

# **Establishing a Unified Optimal Investment Strategy for Defined Contribution Pension Plans Clipped versus Directly Constrained Mean--Variance Controls and a Three-Layer Glide-Path Design**

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Presented to the  
33rd International Congress of Actuaries (ICA2026)  
November 8th – 13th, 2026

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## 1 Introduction

Defined contribution (DC) pension investment is often discussed by comparing exogenously specified target-date, reverse, or U-shaped glide paths. Once member objectives, continuing contributions, the time inconsistency of mean–variance optimisation, and practical investment constraints are considered, however, the glide path implied by a given decision principle is not obvious. This paper develops a unified framework in which glide paths arise endogenously from constrained dynamic optimisation, and compares four solution concepts under a common DC model and the constraint  $0 \leq \pi_t \leq X_t$ .

Because the mean–variance objective is time-inconsistent, the solution depends on which decision date is given authority. Pre-commitment mean–variance (PCMV) commits to the initial plan, whereas dynamically optimal mean–variance (DOMV) replans from the current state. Time-consistent mean–variance with constant variance aversion (cTCMV) and with total-pension-wealth-dependent variance aversion (dTCMV) seek an equilibrium among successive selves. We preserve these distinct decision principles while holding the market, contributions, and investment constraint common.

Even at the same expected retirement wealth, policies can differ in lower-tail risk, upside participation, and the timing of risk-taking. We therefore use two comparison bases: matched expected terminal wealth at inception, and the same date and observed balance at intermediate dates. The latter evaluates the conditional outcomes of existing continuation policies, rather than re-optimising them. The conditional framework applies to all four MV policies and a constant-proportion benchmark; the main text focuses particularly on cTCMV and dTCMV. Based on the law of total variance, a rolling conditional outcome profile summarises expected outcomes, quantiles, tail risk, and current risky allocation from the observed balance onward.

The primary objects of comparison are state-dependent policies. A mean glide path is the risky-asset share averaged over the balance distribution generated by a policy. As a supplementary implementation analysis, we distinguish unconstrained solutions, their ex-post clipped approximations, and directly constrained feedback that internalises future constraint binding in continuation values or equilibrium moments. Clipping is implementable, but its optimality for the constrained problem is not guaranteed; its reproduction accuracy differs across solution concepts.

Numerically, a Markov–Gauss–Hermite (MGH) backward–forward engine uses the same quadrature-induced transition kernel to solve the four concepts and propagate terminal and rolling conditional distributions. Validation combines independent one-million-path Euler–Monte Carlo evaluation, moment consistency, probability-mass conservation, grid/boundary/quadrature sensitivity, and an external benchmark against published results. These checks provide numerical evidence for the adopted finite model; they do not establish general convergence to the continuous-time problem.

The analysis finds that target-date, U-shaped, and intermediate glide paths can arise endogenously from different decision principles and variance-aversion specifications. Under the matched initial mean, PCMV has the smallest standard deviation but the lowest q05 and lower-5% mean; DOMV and cTCMV produce relatively similar distributions; and dTCMV expands both upside participation and downside risk. Together with state dependence and implementation burden, these findings motivate a three-layer design: cTCMV as a candidate for standard default management, DOMV for periodic review, and dTCMV for personalised investment advice based on total pension wealth. PCMV serves as a design benchmark for target setting and mean–variance trade-offs. The practical aim is to connect the decision principle and the member’s current state to how much risk should be taken until retirement.

The main text proceeds through the model, comparison conditions, initial terminal distributions, intermediate conditional outcomes, and practical implications. Section 2 explains the positioning in prior research, and Section 5 describes the common numerical method and validation design. The appendices collect theoretical derivations, detailed procedures, and clipping diagnostics. Mean–variance–skewness (MVS) is treated there as an exploratory extension: a zero skewness coefficient recovers the corresponding MV problem, while positive-coefficient findings require further theoretical research before practical use.

**Keywords:** defined contribution pensions, dynamic mean–variance, investment constraints, endogenous glide paths, rolling conditional evaluation, three-layer design.

## 2 Literature Review and Positioning

### 2.1 The Expansion of DC Pensions and the Scope of the Analysis

The share of DC arrangements in pension assets has increased across countries, shifting part of investment risk from plan providers to members (OECD, 2025; Thinking Ahead Institute, 2026). In DC plans, investment outcomes during accumulation determine the distribution of retirement assets and thereby constrain the range of annuitisation and drawdown choices available after retirement. Accordingly, lower-tail outcomes, the timing of risk-taking, and conditional outcomes from an observed balance, in addition to expected terminal wealth, may lie outside the traditional scope of pension actuarial work but are intrinsically closely related to pension actuarial analysis and plan design. This paper focuses on the accumulation phase and does not address the decumulation phase.

### 2.2 Prior Literature and Three Main Questions

**Question 1 (O1): Comparing four solution concepts under a common DC constraint.** Lifecycle studies treat labour income and human capital as non-tradable background wealth and derive state-dependent allocation from total risk-bearing capacity (Merton, 1969; Bodie et al., 1992; Viceira, 2001; Cocco et al., 2005). DC research likewise shows that a monotonically declining age-only glide path need not always dominate and considers designs incorporating balances, contributions, and targets (Cairns et al., 2006; Basu et al., 2011; Forsyth and Vetzal, 2019; Estrada, 2019). Target-based DC studies connect

replacement rates, downside risk, quadratic loss, and the MV efficient frontier (Vigna and Haberman, 2001; Haberman and Vigna, 2002; Højgaard and Vigna, 2007; Vigna, 2014; Menoncin and Vigna, 2017).

Dynamic MV admits several notions of solution: PCMV commits to the time-zero plan (Li and Ng, 2000; Zhou and Li, 2000); DOMV resolves the remaining-horizon problem (Pedersen and Peskir, 2017; Menoncin and Vigna, 2020); and TCMV seeks an equilibrium among future selves (Basak and Chabakauri, 2010; Björk and Murgoci, 2014; Björk et al., 2017). van Staden et al. (2021) compare terminal distributions for PCMV, DOMV, constant- and wealth-dependent TCMV, and a constant-proportion strategy at a common mean. The O1 contribution of this paper is limited to comparing these four solution concepts in one finite DC model with deterministic contributions, non-tradable background wealth, and the account-level constraint  $0 \leq \pi_t \leq X_t$ , using matched initial expected terminal wealth and conditional outcomes at the same intermediate date and balance. We do not claim novelty for DC mean–variance optimisation or for any individual solution concept; the contribution lies in this combination of solution concepts, constraints, and comparison design. The formulation is given in Sections 3–4, and the scope of theoretical guarantees is stated in Appendices A and G.

**Question 2 (O2): Distinguishing directly constrained and clipped policies through to member outcomes.** Short-sale bans, bankruptcy constraints, and convex portfolio constraints generate constrained value functions, projection structures, and free boundaries (Li et al., 2002; Li and Xu, 2016; Bensoussan et al., 2014; Guan et al., 2022). Because jointly modelling contributions and constraints in a DC setting can be computationally demanding, implementations that clip an unconstrained solution to the feasible range are also used (Di Giacinto et al., 2014; Ferreira Morici and Vigna, 2025). Ex-post clipping projects only the current control while retaining the continuation value of the unconstrained problem, and is therefore a different policy from directly constrained feedback that internalises the constraint in the future value function or equilibrium moments. Its optimality for the constrained problem is not guaranteed. The four-policy comparison is the main analysis; this implementation difference is examined as a supplementary analysis in Section 6, Appendix C, and Appendix E. Through this analysis, we examine when simple clipping is in fact effective in implementation.

**Question 3 (O3): Evaluating conditional outcomes at a fixed intermediate balance.** Simulating and comparing terminal distributions from the initial date combines both cross-state differences created by the evaluation date and investment risk remaining thereafter, and may therefore lead to the unconditional conclusion, by construction, that PCMV is best. This can also be viewed as an interpretation that ignores the time inconsistency of the mean–variance approach. Motivated by DOMV “rolling”, which re-solves the remaining-horizon problem as a decision principle, the conditional evaluation in this paper measures the outcome from continuing any stored policy from the same date and balance. Using the law of total variance, we separate cross-state differences created by the evaluation date from investment risk remaining thereafter, and Section 7 compares conditional distributions under the recalibrated policies. Definitions and computation are given in Appendix B.8.

These three questions constitute the main analytical agenda of the paper. Beyond them, MVS is also considered in the appendix as a promising exploratory finding. Time-consistent formulations of MVS have an established theoretical basis (Mu et al., 2019). In the constant-coefficient unconstrained setting of that paper, the equilibrium risky investment under cTCMV coincides with cTCMV, and the policies obtained by ex-post clipping their common unconstrained solution also coincide. Taking this as a reference, we also attempt MVS extensions for the other solution concepts, and Appendix D presents the simulation results as advanced material. Practical implementation, however, requires additional theoretical research.

Markov-chain approximation, semi-Lagrangian approximation, and Gauss–Hermite quadrature are established numerical tools (Kushner and Dupuis, 2001; Falcone and Ferretti, 2014; Abramowitz and Stegun, 1972; Judd, 1998). This paper builds an MGH engine by applying these tools and reduces computational burden by converting the calculation into a deterministic simulation method, while using the same computational structure consistently across the four constrained MV policies, terminal distributions, conditional outcomes, and implementation diagnostics. Results obtained within the finite model from the MGH engine are distinguished from, and reported separately from, the independent evaluation used to validate those results.

Table 1: Prior literature, three main questions, and the response of this paper

| Research area             | Established results                                                       | Remaining question                                                                    | Response in this paper               |
|---------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------|
| Dynamic MV and DC         | Background wealth, target-based DC, multiple dynamic MV solution concepts | Comparison of four concepts under a common DC constraint                              | O1: Sections 3–4; Appendices A and G |
| Constraint implementation | Projection structures for convex constraints and clipped implementation   | Tracing differences from directly constrained policies through to policy and outcomes | O2: Section 6; Appendices C and E    |
| Conditional outcomes      | Dynamic programming and methods for conditional distributions             | Cross-policy comparison at a fixed observed balance                                   | O3: Section 7; Appendix B.8          |

Taken together, the contribution of this paper lies not in any one numerical technique, but in integrating these elements into a single framework for DC investment design. The four dynamic MV solution concepts are compared under the same market setting, contribution structure, and DC-specific investment constraint; differences between clipped approximations and directly constrained policies are followed beyond the controls themselves to glide paths and member outcomes. At intermediate dates, continuation policies are compared from the same observed balance, making it possible to distinguish remaining-horizon risks and outcomes that are obscured by a time-zero evaluation alone. Within this framework, familiar declining TDF-like paths, U-shaped paths, and intermediate forms can be interpreted as endogenous consequences of the adopted decision principle and member state rather than as shapes imposed in advance.

Numerical reliability is assessed using a common transition kernel, independent Monte Carlo re-evaluation of stored policies, internal consistency checks, grid-sensitivity analysis, and comparison with published benchmark results, thereby confirming the validity of the MGH engine. **The practical aim is then to connect the resulting differences to distinct DC roles: standard default management, periodic investment review, personalised advice, and a design benchmark.**

In this sense, “unified” in the title does not mean that a single glide path is universally optimal. It refers to a common analytical framework in which policies based on different decision principles are compared under the same DC model and evaluation criteria and are then interpreted in relation to distinct practical roles. Likewise, “optimal investment strategy” does not denote one policy that is optimal for every member; it denotes the policy implied by the chosen decision principle, the member’s current state, and the DC-specific constraint. The two parts of the subtitle—“Clipped versus Directly Constrained Mean–Variance Controls” and “Three-Layer Glide-Path Design”—correspond, respectively, to the comparison of constraint implementations and to the translation of the resulting policies into a practical allocation of roles in DC investment management.

### 3 Model and Solution Concepts

#### 3.1 Financial Market and the DC Account

Consider a finite horizon  $[0, T]$ . The risk-free asset  $B_t$  and risky asset  $S_t$  satisfy

$$\frac{dB_t}{B_t} = r dt, \quad (1)$$

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad \sigma > 0, \quad (2)$$

where  $r, \mu, \sigma$  are constants and  $\beta = \mu - r$  is the excess return. Let  $X_t$  denote the DC account balance,  $\pi_t$  the amount invested in the risky asset, and  $c_t \geq 0$  the deterministic contribution flow. Then  $X_t$  satisfies

$$dX_t = (rX_t + c_t + \beta\pi_t) dt + \sigma\pi_t dW_t, \quad X_0 = x_0 \geq 0 \quad (3)$$

For a standard DC default arrangement, it is natural to prohibit both purchasing risky assets in excess of the account balance and short selling the risky asset. Accordingly, this paper imposes

$$0 \leq \pi_t \leq X_t \quad (4)$$

as the basic constraint. This is the definition of the control set itself.

#### 3.2 Background Wealth and an Auxiliary Transformation

Non-tradable pension-related cash flows, such as future contributions and DB benefits, are represented by their present value as deterministic background wealth. In the baseline model, interest rates and cash flows are deterministic, and background wealth  $H_t$  is defined by

$$H_t = \int_t^T e^{-r(s-t)} c_s ds + e^{-r(T-t)} D_T, \quad (5)$$

where  $D_T$  denotes a deterministic terminal benefit received at retirement, with a DB benefit as the leading example. For brevity, we refer to it below as a “DB benefit”, although other deterministic retirement benefits are included in the same formulation. Setting  $D_T = 0$  returns the case with future contributions only. The process  $H_t$  satisfies

$$\dot{H}_t = rH_t - c_t, \quad H_T = D_T \quad (6)$$

For convenience, define

$$Y_t = X_t + H_t \quad (7)$$

Then, from (3) and (6),

$$dY_t = (rY_t + \beta\pi_t) dt + \sigma\pi_t dW_t, \quad Y_T = X_T + D_T \quad (8)$$

so the deterministic contribution flow can be removed formally. This transformation serves two purposes. First,  $X_t + H_t$  is interpreted as total pension wealth and used to define the wealth-dependent variance aversion in dTCMV. Second, and particularly importantly, removing the investment constraint transforms the auxiliary problem into the standard Black–Scholes form and makes it straightforward to construct the unconstrained solution used as the input to the clipped approximation. The amount that can actually be invested is  $X_t$ , whereas  $Y_t$  is total pension wealth including background wealth for valuation purposes. Accordingly, the HJB equations defining directly constrained feedback, the verification theorems, and the baseline numerical calculations are all formulated consistently in  $X$ -space;  $Y$ -space is used to interpret total pension wealth and construct unconstrained benchmarks.

#### 3.3 Admissible Controls and Glide Paths

**Definition 1** (Admissible controls in  $X$ -space). The set of admissible controls starting from  $(t, x)$  is defined by

$$\mathcal{A}_{t,x}^X = \{\pi : \pi_s \text{ is progressively measurable, } 0 \leq \pi_s \leq X_s^\pi, \quad X_s^\pi \geq 0, \quad s \in [t, T]\} \quad (9)$$

The risky-asset share within the DC account is defined as

$$g_t = \frac{\pi_t}{X_t} \quad (10)$$

We call this quantity, viewed as a function of time, a glide path. For state-dependent glide paths, the average is used as the representative glide path. Comparison of these representative glide paths is one of the analytical measures in this paper. By mathematical definition, when  $X_t = 0$  no investment is possible, and we set  $g_t = 0$ . Numerically,  $x = 0$  is the lower boundary and negative discrete transition values are truncated at zero. With contributions, the balance can become positive again in subsequent periods. The baseline calculations use an upper state bound of 600 with grid spacing 0.1, and risky investment is linearly interpolated between grid points.

### 4 Four Decision Principles and Comparison Metrics

Mean–variance optimisation is time-inconsistent, so the preferred policy may depend on the decision date. The notion of optimality must therefore be specified first.

The mathematical definitions and verification theorems are collected in Appendix A. A *directly constrained policy* chooses the control within the admissible interval  $0 \leq \pi \leq x$  using the constrained continuation-value problem. By contrast, a *clipped approximation* takes the risky investment from a separate unconstrained problem and projects it ex post onto  $0 \leq \pi \leq x$ . Both satisfy the current-period investment constraint, but they differ in how future constraints are incorporated into the current decision.

Table 2: Guide to the four mean–variance solution concepts

| Solution concept | Decision principle                                                                                         | Intertemporal property                | Main input                        | Practical role considered in the comparison                        |
|------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------|--------------------------------------------------------------------|
| PCMV             | Commit at time 0 to a terminal target and policy                                                           | Pre-commitment                        | Embedded target $m$               | Design benchmark for organising the target–mean–variance trade-off |
| DOMV             | Re-solve the remaining-horizon problem at each reached date and state                                      | Rolling optimum                       | $\gamma_D$ or local target        | Periodic review                                                    |
| cTCMV            | Seek an intrapersonal equilibrium that cannot be improved by a short-horizon deviation of the current self | Subgame-perfect Nash-type equilibrium | Constant $\gamma_c$               | Standard default candidate                                         |
| dTCMV            | Seek an intrapersonal equilibrium under variance aversion linked to total pension wealth                   | Subgame-perfect Nash-type equilibrium | $\Gamma(t, x) = \rho_d/(x + H_t)$ | Personalised advice based on total pension wealth                  |

## 4.1 Bellman Problems and Equilibrium Problems

The objective function of the mean–variance optimisation problem is

$$J(\pi) = \mathbb{E}[W_T^\pi] - \frac{\gamma}{2} \text{Var}(W_T^\pi)$$

and contains  $(\mathbb{E}[W_T^\pi])^2$ , so it requires treatment different from an ordinary expected-terminal-utility objective  $\mathbb{E}[U(W_T^\pi)]$ .

Below we state only the minimal formulations for convenience; the details are as given in Appendix A. Let  $W_T^\pi := X_T^\pi + D_T$ . PCMV solves at time zero

$$\inf_{\pi \in \mathcal{A}_{0,x_0}^X} \text{Var}(W_T^\pi) \quad \text{subject to} \quad \mathbb{E}[W_T^\pi] = d,$$

commits to that policy, and is computed through the embedded fixed-target loss  $\inf_{\pi \in \mathcal{A}_{t,x}^X} \mathbb{E}_{t,x}[(W_T^\pi - m)^2]$ . DOMV instead re-solves at each reached state  $(t, x)$

$$\sup_{\pi \in \mathcal{A}_{t,x}^X} \left\{ \mathbb{E}_{t,x}[W_T^\pi] - \frac{\gamma_D}{2} \text{Var}_{t,x}(W_T^\pi) \right\}$$

and implements only the current control. Thus PCMV is initial commitment, whereas DOMV is re-optimisation from the current state. More specifically, the actual solution method for PCMV and DOMV transforms the problem into an auxiliary quadratic-loss problem with a fixed target and applies dynamic programming to that auxiliary problem.

By contrast, cTCMV and dTCMV seek intrapersonal equilibria rather than globally re-solving the remaining-horizon problem. Their criteria are

$$J^c = \mathbb{E}_{t,x}[W_T^\pi] - \frac{\gamma_c}{2} \text{Var}_{t,x}(W_T^\pi), \quad J^d = \mathbb{E}_{t,x}[W_T^\pi] - \frac{\Gamma(t, x)}{2} \text{Var}_{t,x}(W_T^\pi), \quad \Gamma(t, x) = \frac{\rho_d}{x + H_t},$$

and a candidate  $\hat{\pi}^j$  ( $j = c, d$ ) is an equilibrium if every admissible short-horizon deviation  $\nu$  satisfies

$$\liminf_{h \downarrow 0} \frac{J^j(t, x; \hat{\pi}^j) - J^j(t, x; \nu \otimes_h \hat{\pi}^j)}{h} \geq 0.$$

Hence cTCMV and dTCMV differ in the specification of variance aversion, not in the equilibrium concept. The HJB and extended-HJB systems and verification conditions are given in Appendix A.

## 4.2 Comparison Design

For the initial comparison, market assumptions, contributions, investment constraints, and the evaluation horizon are held common, and the parameters of each decision principle are adjusted so that  $\mathbb{E}[W_T^j] \simeq \bar{m}$ . The common mean provides a basis for comparing outcome distributions generated by different objectives at the same expected level. We then compare standard deviation, q05, q50, q95, the lower-5% mean, and average allocation.

For the intermediate comparison,  $(t, x)$  is held common and  $\mathbb{E}_{t,x}[W_T^j]$  and the conditional distribution are evaluated for each policy. Parameters remain fixed at their time-zero calibration. Expected outcomes and downside risk are reported together to describe the benefit–risk trade-off from the member’s current state.

The numerical calculations below report policies and their induced distributions on a finite grid. Verification theorems for the continuous-time model apply under the conditions on candidate-solution regularity, admissibility, and related assumptions stated in Appendices A and G.

# 5 Common Solution Method and Numerical Validation Design

## 5.1 Directly Constrained Policies and Clipped Approximations

The treatment of investment constraints is one of the main axes of comparison in this paper. An unconstrained solution can be projected onto the feasible range to construct a computationally inexpensive clipped approximation. By contrast, directly constrained feedback incorporates the constraint into future continuation values when solving for the policy. We assess how closely the clipped approximation reproduces the directly constrained feedback.

For solution concept  $j \in \{\text{PCMV}, \text{DOMV}, \text{cTCMV}, \text{dTCMV}\}$ , the clipped approximation constructed from the unconstrained solution  $\pi^{j,0}$  is

$$\pi^{j,\text{clip}}(t, x) = \text{Proj}_{[0,x]} \{ \pi^{j,0}(t, x + H_t) \}$$

The clipped approximation projects the current control onto the feasible range, whereas the directly constrained policy internalises future constraint binding in the continuation value or equilibrium moments; they therefore need not coincide. In this sense, the unconstrained solution, clipped approximation, and directly constrained policy are distinct objects of comparison. Full HJB derivations, free boundaries, and verification theorems are given in Appendix A.

We make explicit the unconstrained solutions  $\pi^{j,0}$  that form the basis of the clipped approximation. These reproduce the equations in the appendix and therefore use the same equation numbers. Let  $Y_t = X_t + H_t$ ,  $A := \beta^2/\sigma^2$ , and  $y = x + H_t$ . After transformation to  $Y$ -space, the unconstrained solutions are

$$\pi^{p,0}(t, y; m) = \frac{\beta}{\sigma^2} \{m e^{-r(T-t)} - y\} \quad (17),$$

$$\pi^{D,0}(t, y; \gamma_D) = \frac{\beta}{\gamma_D \sigma^2} e^{(A-r)(T-t)} \quad (18),$$

$$\pi^{c,0}(t, y) = \frac{\beta}{\gamma_c \sigma^2} e^{-r(T-t)} \quad (19),$$

$$\pi^{d,0}(t, y) = \theta_{\rho_d}(t) y \quad (20).$$

Under PCMV the unconstrained risky investment is determined by the embedded target  $m$ ; under DOMV and cTCMV by the variance-aversion parameters  $\gamma_D$  and  $\gamma_c$ , respectively; and under dTCMV by multiplying total pension wealth  $y$  by the time-dependent proportion  $\theta_{\rho_d}(t)$ . The definition and computation of  $\theta_{\rho_d}$  and further details are given in Appendix A.

## 5.2 Markov–Gauss–Hermite Backward–Forward Engine

To solve backward and evaluate forward the four constrained MV problems in Section 4 efficiently under common conditions, we use common Gauss–Hermite conditional expectations, grid-based dynamic programming, and probability-mass propagation. The *unified Markov–Gauss–Hermite (MGH) engine* is not a new quadrature formula; it is a computational architecture used consistently for simulations of the four solution concepts, directly constrained solutions and clipped approximations, terminal distributions, rolling conditional distributions, and MV/MVS nesting checks. Its advantage is that conditional expectations with respect to normal shocks can be evaluated using  $K$  deterministic nodes. This makes it possible to perform the backward calculation without generating many random paths for every state and candidate control, thereby reducing computational burden, and, when comparing the four solution concepts under a common transition kernel, avoids introducing Monte Carlo sampling error into the internal calculation. This is why we adopt this method.

The core of the engine is the one-period transition and the common conditional-expectation operator detailed in Appendix B. At date  $t_n$ , balance  $x_i$ , and candidate risky investment  $\pi$ , Euler discretisation and the  $K$ -point Gauss–Hermite nodes  $(\xi_k, w_k)$  give

$$\hat{x}_{n+1}^{(k)}(x_i, \pi) = x_i + (rx_i + c_n + \beta\pi)\Delta t + \sigma\pi\sqrt{\Delta t}\xi_k, \quad \sum_{k=1}^K w_k = 1, \quad (11)$$

where  $\xi_k = \sqrt{2}z_k$  and  $w_k = \omega_k/\sqrt{\pi}$  for the usual Hermite nodes and weights  $(z_k, \omega_k)$ . Off-grid points are evaluated with the linear interpolation operator  $\mathcal{J}_{n+1}$ , so the one-period conditional expectation of any continuation quantity  $f$  is represented by

$$(\mathcal{Q}_n^\pi f)(x_i) := \sum_{k=1}^K w_k (\mathcal{J}_{n+1} f)(\hat{x}_{n+1}^{(k)}(x_i, \pi)). \quad (12)$$

This operator  $\mathcal{Q}_n^\pi$  is the common computational kernel across all four solution concepts. For example, the fixed-target quadratic-loss problem underlying PCMV/DOMV is propagated backward as

$$V_n(x_i; m) = \min_{\pi \in [0, x_i]} (\mathcal{Q}_n^\pi V_{n+1})(x_i), \quad (13)$$

whereas cTCMV/dTCMV apply the same  $\mathcal{Q}_n^\pi$  to the first and second terminal-wealth moments and save the control that maximises the one-step-deviation objective. Thus the objective differs by solution concept, while the treatment of Gaussian shocks, interpolation, and the feasible control set  $0 \leq \pi \leq x_i$  is common.

In the **forward calculation**, the same candidate points in (11) are used under the stored policy  $\pi_n^*(x_i)$ . If  $b_\ell(y)$  denotes the linear weight assigning a point  $y$  to next-period grid point  $x_\ell$ , probability mass evolves as

$$p_{n+1, \ell} = \sum_i \sum_{k=1}^K p_{n,i} w_k b_\ell(\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i))). \quad (14)$$

Hence the backward calculation applies the same discrete transition kernel to future continuation quantities through conditional expectation, while the forward calculation applies the transpose operation to move current probability mass to the next date. This is the sense in which the two directions form one MGH engine. Starting the same forward operation from an intermediate  $(t_n, x_i)$  gives a rolling conditional distribution. The forward calculation therefore differs from Monte Carlo evaluation based on randomly generated individual paths. Detailed interpolation, boundary handling, concept-specific recursions, and mass-conservation diagnostics are given in Appendix B.

## 5.3 Numerical Settings and Independent Monte Carlo Evaluation

Table 3 summarises the common numerical settings used for re-optimisation, recalibration, and independent evaluation in the main text. Policies are computed by monthly-grid MGH backward calculation, using common state grids, candidate controls, and Gauss–Hermite order across the four solution concepts.

The final policy-specific calibration parameters are reported in Table 4. Because these parameters play different roles across the solution concepts, their numerical values are not compared directly; instead, outcome distributions and policy shapes are compared after calibration to the common expected-terminal-wealth criterion defined in Section 6.

The main outcome distributions are obtained by re-evaluating the stored policies from the MGH backward calculation with an independent one-million-path Euler–Monte Carlo simulation. Because the MGH forward calculation uses the same discrete transition kernel as the backward calculation, Monte Carlo is used as a separate stochastic evaluator for validation. Policies are not re-optimised; the stored policies are linearly interpolated in the state dimension, and common random numbers are used for cross-policy comparisons. Consistency of terminal distributions and mean glide paths between MGH forward calculation and Monte Carlo evaluation is examined in Appendix E.3.

Storage locations and regeneration procedures are described in Appendix F. Sensitivity of the mean glide paths to changes in market parameters and contribution design is reported in Appendix C.2, holding the policy coefficients fixed.

Table 3: Common settings for re-optimisation and recalibration

| Item                                         | Symbol            | Value                                                    |
|----------------------------------------------|-------------------|----------------------------------------------------------|
| Accumulation horizon / decision interval     | $(T, \Delta t)$   | (40, 1/12)                                               |
| Initial DC balance / annual contribution     | $(x_0, c)$        | (1/12, 1)                                                |
| Terminal DB benefit                          | $D_T$             | 0                                                        |
| Risk-free rate / expected risky-asset return | $(r, \mu)$        | (0.015, 0.055)                                           |
| Volatility                                   | $\sigma$          | 0.180                                                    |
| State grid for policy calculation            | $(x_{\max}, N_x)$ | (600, 6001), spacing 0.1                                 |
| Candidate control ratios / GH order          | $(N_a, K)$        | (129, 7)                                                 |
| DOMV target grid                             | $\mathcal{M}$     | 0, 2, ..., 650                                           |
| Independent evaluation                       | —                 | Gaussian Euler, one million paths, common random numbers |

Table 4: Final parameters used for the re-optimised and recalibrated policies in the main text

| Policy | Parameter                              | Final value         |
|--------|----------------------------------------|---------------------|
| PCMV   | Embedded target $m_P^*$                | 101.357421875       |
| DOMV   | Variance-aversion parameter $\gamma_D$ | 0.08624755859375    |
| cTCMV  | Variance-aversion parameter $\gamma_c$ | 0.05411865234375    |
| dTCMV  | Scale parameter $\rho_d$               | 1.6142578125        |
| CP     | Constant proportion $\theta_{cp}$      | 0.46888421662151814 |

## 5.4 Theoretical Guarantees and the Scope of the Numerical Results

For PCMV and DOMV, Appendix G gives an existence result based on viscosity solutions together with the scope of classical verification under a smooth-solution assumption. For DOMV, additional conditions are imposed on jointly measurable selection of the target and optimal control and on suitable solvability of the closed-loop equation after diagonal concatenation. The local feedback representation and verification theorem assume a  $C^{1,2}$  candidate function and sufficient integrability.

For the continuous-time cTCMV and dTCMV problems, global existence, uniqueness, and regularity of equilibrium solutions under the state-dependent two-sided constraint  $0 \leq \pi \leq X$  used in this paper remain subjects for future work. Unless otherwise stated by context, references in the main text to an “optimal policy” or “equilibrium policy” mean a policy in the finite-state, finite-control model. Verification theorems for the continuous-time problems apply under the regularity and integrability conditions stated in Appendices A and G.

# 6 Comparison of Four Policies at Matched Initial Expected Terminal Wealth

## 6.1 Comparison Design and Evaluation Scope

The four directly constrained policies and the constant-proportion strategy CP are compared under the same market and plan assumptions. To compare different decision principles at a similar level of expected outcomes, each policy is calibrated so that expected terminal wealth at inception satisfies

$$\mathbb{E}[W_T^j] \simeq \bar{m}, \quad \bar{m} = 84.78, \quad (15)$$

The common expected terminal wealth of 84.78 is the monthly PCMV baseline value 84.777086 rounded to two decimal places. This is a comparison normalisation for terminal-distribution shape, downside risk, upside participation, and the timing of risk-taking, rather than a normative retirement target. The principle of matching expected terminal wealth follows [van Staden et al. \(2021\)](#). The main values below use the independent Euler–Monte Carlo evaluation defined in Section 5.3.

## 6.2 Terminal Distributions

Table 5: Independent one-million-path evaluation of re-optimised and recalibrated policies

| Policy | Mean    | Standard deviation | q05     | Median  | q95      | Lower-5% mean |
|--------|---------|--------------------|---------|---------|----------|---------------|
| PCMV   | 84.7715 | 18.9424            | 38.1937 | 92.3501 | 100.2246 | 28.7985       |
| DOMV   | 84.7831 | 24.4139            | 46.8690 | 83.4760 | 127.0602 | 39.2036       |
| cTCMV  | 84.7913 | 23.1570            | 47.1606 | 84.4674 | 123.4706 | 38.5271       |
| dTCMV  | 84.7863 | 32.3127            | 43.1728 | 79.2581 | 145.2516 | 37.3322       |
| CP     | 84.7854 | 31.6874            | 45.2269 | 78.9299 | 144.1862 | 39.8995       |

PCMV has the smallest standard deviation and the highest median, but also the lowest q05 and lower-5% mean among the five policies. It strongly concentrates terminal wealth around the target while producing weaker left-tail measures. DOMV and cTCMV generate similar distributions: cTCMV has a slightly higher q05, while DOMV has the higher lower-5% mean. dTCMV has the largest standard deviation and q95, increasing both upside participation and downside risk. CP has a standard deviation and q95 close to dTCMV, while achieving the highest lower-5% mean among the five policies. Policy preference therefore depends on whether emphasis is placed on standard deviation, quantiles, or tail means.

## 6.3 Timing of Risk-Taking and State Dependence

Figure 2 shows the mean glide path obtained by averaging the realised risky-asset share across paths in the independent evaluation at each date. PCMV, DOMV, and cTCMV generally decline, although at different rates. The mean risky-asset share under dTCMV initially remains at the upper constraint, then falls rapidly, reaches a trough around the middle of the horizon, and rises again toward retirement. CP is constant by definition. Each curve reflects not only age but also the state-dependent feedback and the balance distribution generated by that policy.

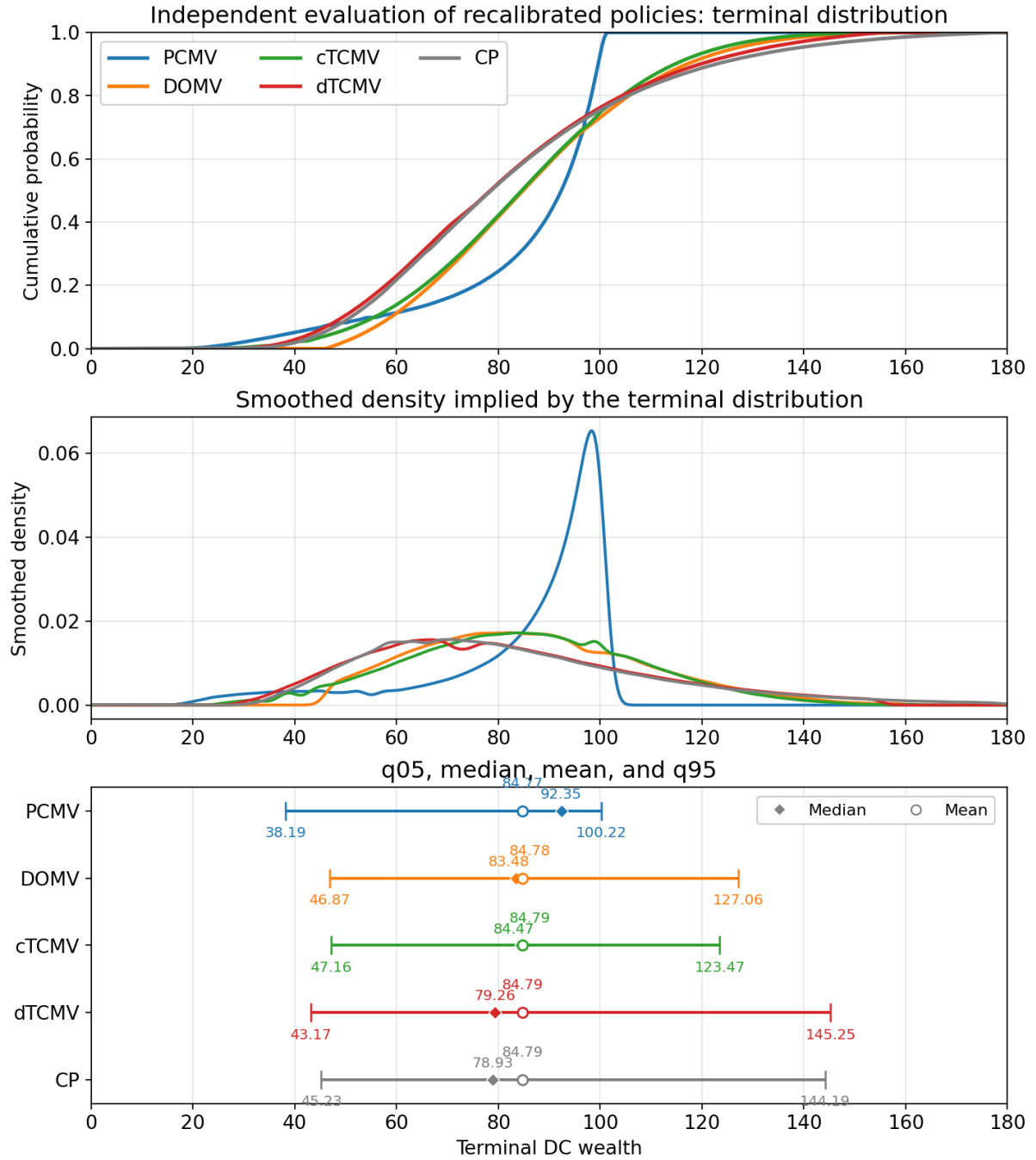


Figure 1: Terminal DC-wealth distributions. The top panel shows cumulative distribution functions, the middle panel shows smoothed densities implied by the terminal distributions, and the bottom panel places q05, median, mean, and q95 directly at their corresponding wealth levels.

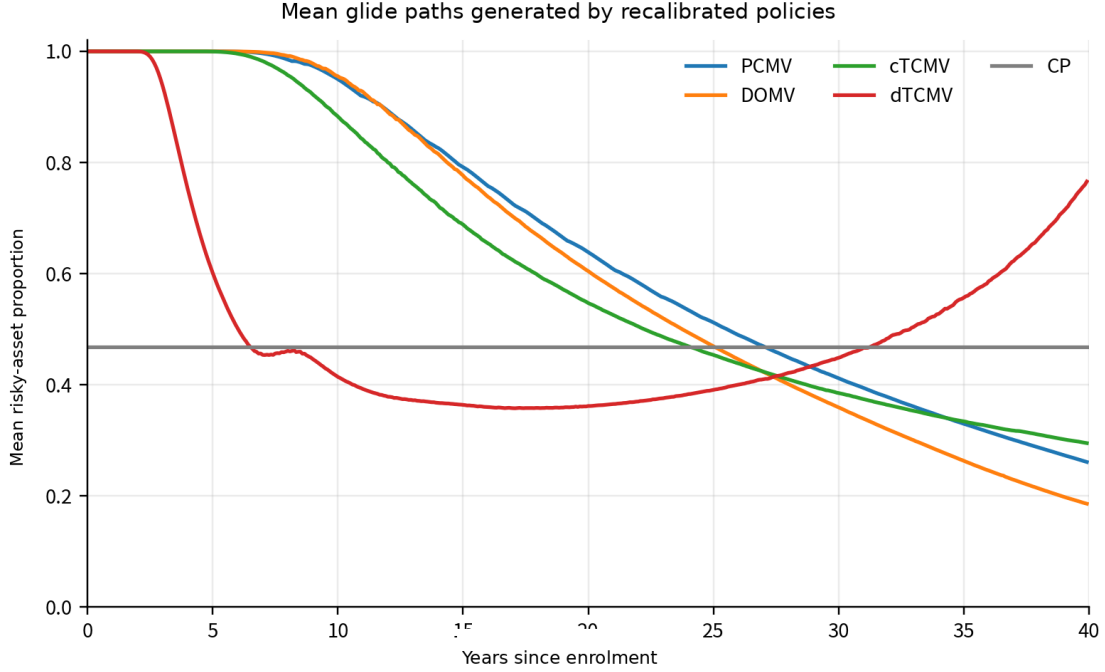


Figure 2: Mean glide paths generated by the re-optimised and recalibrated policies under independent one-million-path Euler–Monte Carlo evaluation

#### 6.4 Role of the Clipped Approximation

The main comparison is conducted using directly constrained policies, which incorporate the constraint inside the backward calculation. Ex-post clipping of an unconstrained solution satisfies the current constraint but does not reflect future constraint binding in the continuation value or equilibrium moments. The difference is therefore treated as a test of implementation approximation in Appendix C.

### 7 Re-evaluating Outcomes from Intermediate Balances

Matching expected terminal wealth at time zero does not imply matching conditional outcomes from the same intermediate state. In this section, the recalibrated policies from Section 6 are continued without changing their coefficients and evaluated independently with one million paths from the same date and balance. This is a re-evaluation of the originally calibrated policy conditional on an observed current state. DOMV “rolling” refers to the decision principle of re-solving the remaining-horizon problem, whereas the conditional evaluation here assesses outcomes from continuing any stored policy from the same date and balance. The definition of the evaluation and the law-of-total-variance decomposition motivating it are given in Appendix B.8.

As examples with different remaining horizons and current balances, we consider three states: year 20 with balance 20, year 30 with balance 45, and year 35 with balance 30. The conditional-evaluation framework applies to all five policies, but the main text compares cTCMV and dTCMV in greater detail because their state dependence is especially informative. Summary values for all five policies are reported in Appendix Table 14.

Table 6: Conditional outcomes from common states under the recalibrated policies

| State               | Policy | Initial share | Mean  | Standard deviation | q05   | Median | q95    | Lower-5% mean | Skewness |
|---------------------|--------|---------------|-------|--------------------|-------|--------|--------|---------------|----------|
| Year 20, balance 20 | cTCMV  | 0.750         | 67.98 | 18.04              | 38.40 | 67.89  | 97.85  | 31.37         | 0.05     |
|                     | dTCMV  | 0.391         | 69.78 | 25.19              | 36.86 | 65.65  | 116.62 | 32.08         | 1.13     |
| Year 30, balance 45 | cTCMV  | 0.438         | 72.19 | 13.00              | 50.83 | 72.19  | 93.59  | 45.40         | 0.00     |
|                     | dTCMV  | 0.453         | 77.95 | 24.63              | 44.70 | 74.36  | 123.43 | 39.53         | 0.97     |
| Year 35, balance 30 | cTCMV  | 0.719         | 42.07 | 9.15               | 26.98 | 42.06  | 57.13  | 23.31         | 0.02     |
|                     | dTCMV  | 0.617         | 42.52 | 10.91              | 27.13 | 41.22  | 62.33  | 24.49         | 0.77     |

At year 30 with balance 45, dTCMV has a conditional mean 5.76 higher and q95 29.84 higher than cTCMV, while its standard deviation is 11.63 larger, q05 is 6.13 lower, and lower-5% mean is 5.87 lower. At year 20 with balance 20 and year 35 with balance 30, the differences in q05 and lower-5% mean are smaller than in the year-30, balance-45 case. Across the three states, the combination of expected outcome, upside participation, and downside tail varies with current balance, remaining horizon, and the chosen evaluation metric.

These results demonstrate concretely that “the same mean at inception” and “the same state at an intermediate date” are different comparison conditions. The time-zero unconditional distribution combines cross-state differences created up to the intermediate date with conditional risk remaining thereafter. When explaining a continuation policy to a member, the observed balance should therefore be treated as given and conditional mean, standard deviation, quantiles, and tail means should be reported together. Extending the comparison to the full state space is included among the future-work items in Section 9.

### 8 Practical Implications of the Comparison

We organise the practical roles of the policies in light of their decision principles, explainability, implementation burden, and suitability for individual members. In the baseline calibration, PCMV has the smallest standard deviation but the lowest q05 and lower-5% mean among the five policies. In light of this distributional profile and its commitment to the initial plan, PCMV

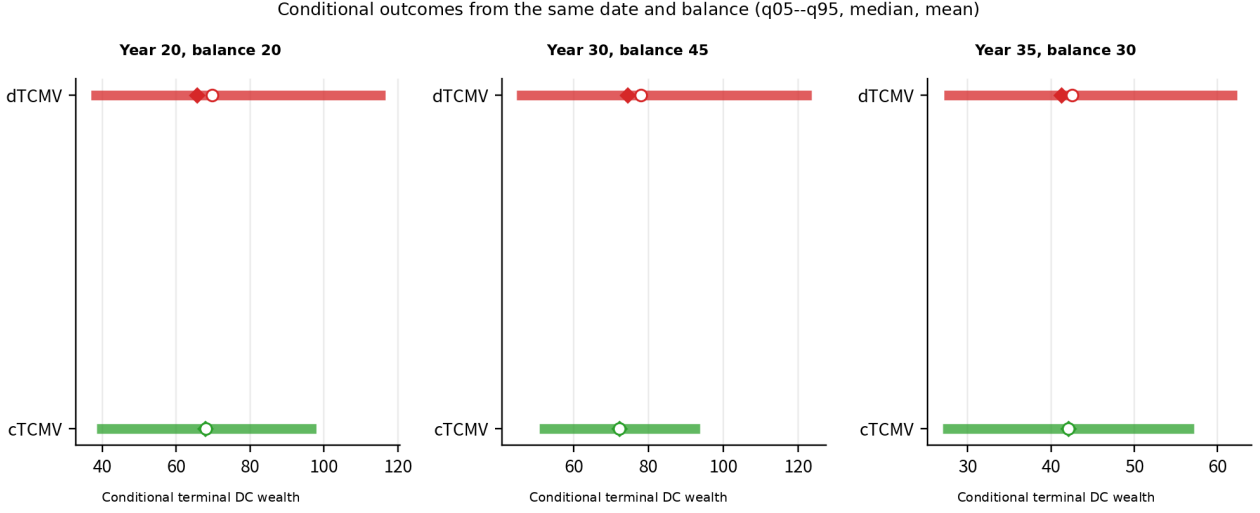


Figure 3: Conditional outcomes from the same date and balance. Horizontal lines indicate q05--q95, diamonds the median, and white circles the mean.

is kept outside the three operating layers and instead serves as a design-benchmark candidate for examining the relationship between terminal targets and mean-variance outcomes. The operating layers are summarised in Figure 4. This three-layer structure is a design hypothesis for members based on each policy’s decision principle and state dependence, not on a numerical ranking.

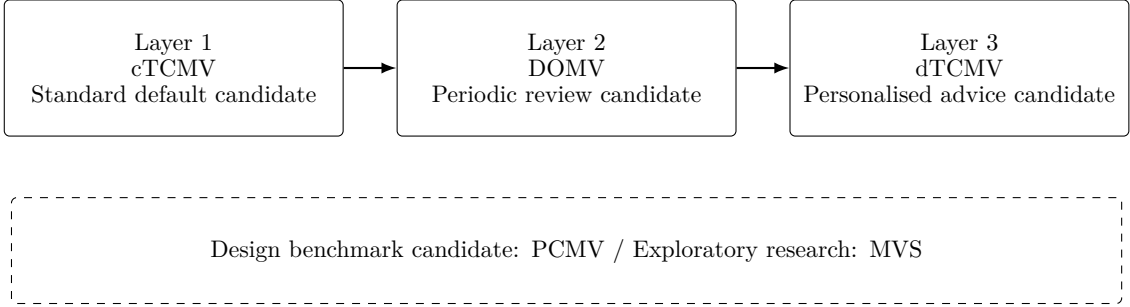


Figure 4: Design hypothesis for practical role allocation across the four policies

cTCMV uses a constant coefficient and has relatively small clipping error, making it comparatively easy to implement; for example, it can serve as a benchmark when defining a default investment option. DOMV re-solves the remaining-horizon problem from the current state and is therefore a candidate for periodic review, although frequent recomputation and target changes increase implementation burden. dTCMV varies risk-taking with total pension wealth and is therefore consistent with personalised management. At the same time, as shown in Section 7, its relationship with downside risk must be checked state by state. In every layer, expected outcomes, downside quantiles, tail means, costs, and coefficient calibration should be reflected in member-facing explanations.

MVS is organised in Appendix D as an exploratory extension requiring further validation, including non-concavity and numerical stability. The U-shaped glide path observed under dTCMV can be reshaped by adding higher moments into forms that may be more conventional and acceptable to members. The three-layer structure in this paper is limited to mean-variance policies for which the theoretical and numerical validation is more developed; connecting MVS to practice remains a subject for future research.

## 9 Limitations and Conclusions

### 9.1 Limitations and Future Work

The market contains one risk-free and one risky asset, and contributions are deterministic. Background wealth  $H_t$  represents the value of deterministic cash flows including future contributions and retirement benefits from other arrangements. Re-optimisation is performed in a monthly finite-state, finite-control model; the independent evaluation holds the resulting stored policies fixed and re-evaluates outcome distributions under a monthly Euler process. A general convergence theorem to the continuous-time model and uniform discretisation-error bounds over the full state space remain future work, so the principal numerical results should be interpreted as results for the adopted finite model. The common expected terminal wealth of 84.78 is a comparison calibration, and choosing a different common mean may change quantitative policy shapes and distributional comparisons. Mindful of these limitations, the validation combines internal consistency, independent Monte Carlo evaluation, grid/boundary/quadrature sensitivity, and an external benchmark based on the continuous-time problem of van Staden et al. (2021). The resulting maximum relative errors at the evaluated policy points are 0.080% for PCMV, 0.231% for DOMV, and 0.879% for cTCMV; for PCMV and DOMV, terminal-CDF errors also decrease along the refinement path. For dTCMV, the sign of the squared term differs between the equation printed in the original paper and the equilibrium-condition equation derived in this paper, so the dTCMV results are reported only for reference. These checks do not prove general continuous-time convergence, but they provide numerical evidence that, in externally verifiable cases, the common MGH engine can reproduce continuous-time reference solutions closely. For PCMV and DOMV, Appendix G gives existence results based on viscosity solutions and relaxed controls and the scope of verification under a classical-solution assumption. The conditional evaluation considers three illustrative states. Future work should extend it to the full state space and examine how state- and

time-dependent risk choices can be visualised and connected to practical decisions. Model extensions to stochastic interest rates, stochastic contributions, multiple assets, and decumulation with mortality risk are also important. For MVS, further work is required on global solutions to non-concave objectives, grid convergence, and economic calibration of coefficients before practical applicability can be assessed with confidence in existence, stability, and numerical robustness.

## 9.2 Conclusions

This paper develops a framework in which DC pension glide paths can be derived endogenously from dynamic optimisation and compared on a common basis once member objectives, current state, and investment constraints are specified. Reflecting the time inconsistency of mean–variance problems, PCMV, DOMV, cTCMV, and dTCMV are compared under common market, contribution, and investment constraints. Even after calibration to the same expected terminal wealth, the policies generate different terminal distributions, timing of risk-taking, and mean glide paths. The optimal DC glide path should therefore be interpreted as arising endogenously from the adopted decision principle and member state.

For implementation of investment constraints, we compare the clipped approximation, which projects an unconstrained solution onto the feasible range, with a directly constrained policy that internalises the constraint in the continuation value. Their closeness differs by solution concept. Where reproduction accuracy is high, clipping can simplify implementation; where the gap is large, directly constrained optimisation matters for both policy shape and outcome distribution. Practical implementation should therefore be chosen jointly on the basis of computational simplicity and accuracy in reproducing member outcomes and the timing of risk-taking.

At intermediate dates, conditional outcomes are compared after fixing the date and current balance. Even from the same current balance, cTCMV and dTCMV generate different combinations of expected outcomes, upper and lower quantiles, and tail risk, and these differences vary with remaining horizon and balance state. By separating cross-state dispersion created before the evaluation date from investment risk remaining beyond the current state, the rolling conditional outcome profile provides a member-level evaluation axis for explaining state-dependent investment management.

As a design hypothesis connecting these findings to practical roles, we propose a three-layer structure: cTCMV as a candidate for standard default management, DOMV as a candidate for periodic investment review, and dTCMV as a candidate for personalised advice based on total pension wealth. PCMV supports these layers as a benchmark for organising the terminal-target and mean–variance trade-off. MVS is retained as an extension in which higher moments can alter the timing of risk-taking and is left for further theoretical and numerical validation.

Taken together, the “unified optimal investment strategy” developed in this paper does not mean applying one glide path to every member. It means comparing different decision principles under the same DC model and constraint, evaluating future risk and outcomes conditional on the member’s current state, and then assigning the resulting policies to appropriate practical roles. The clip–direct comparison indicates when a simpler implementation can adequately reproduce the directly constrained policy, while the three-layer structure provides a design hypothesis linking the resulting policies to standard default management, periodic investment review, and personalised advice. PCMV serves as a design benchmark for these comparisons.

Accordingly, the central conclusion is to reframe DC investment design from “which glide-path shape should be chosen?” to “under which decision principle, and from the member’s current state, how much risk should be taken until retirement?” Starting from this question allows glide paths to be derived endogenously rather than fixed in advance, their implementation accuracy and outcome distributions to be examined, and the resulting policies to be connected to distinct practical roles. In addition, although this evidence is obtained in a deliberately simplified model, the analysis shows that the familiar monotonically declining glide path used in conventional target-date funds need not be imposed as an exogenous rule: under some decision principles, it can emerge endogenously as an outcome of optimisation for retirement-asset accumulation, implicitly suggesting that conventional industry practice is mathematically justified.

## A Constrained Mean–Variance Theory

### A.1 Three Objects of Comparison: Unconstrained Solution, Clipped Approximation, and Directly Constrained Feedback

#### A.1.1 Why distinguish the three objects?

The central implementation issue in this paper is how to handle the investment constraint. Because the unconstrained solution may violate the DC constraint, it is not itself necessarily implementable. Its role is to provide the basis for a computationally inexpensive clipped approximation. The practically important question is then how closely that clipped approximation reproduces the directly constrained feedback that internalises future constraint binding in the continuation value.

Let the set of four solution concepts be  $\mathcal{J} := \{\text{PCMV}, \text{DOMV}, \text{cTCMV}, \text{dTCMV}\}$ . For each  $j \in \mathcal{J}$ , define the comparison objects as

$$\begin{cases} \pi^{j,\text{clip}}(t, x) := \text{Proj}_{[0, x]} \{\pi^{j,0}(t, x + H_t)\} & (\text{clipped approximation}), \\ \pi^{j,\text{str}}(t, x) & (\text{directly constrained feedback}). \end{cases} \quad (16)$$

The clipped approximation is constructed by projecting the unconstrained solution onto the feasible range. The directly constrained feedback instead reflects future constraint binding in the continuation value. Thus the two objects in (16) incorporate the constraint in fundamentally different ways.

#### A.1.2 Unconstrained solutions in $Y$ -space

Because  $Y_t = X_t + H_t$  satisfies

$$dY_t = (rY_t + \beta\pi_t)dt + \sigma\pi_t dW_t$$

the auxiliary problem without the investment constraint can be implemented as a standard Black–Scholes-type problem despite DC contributions. Let  $A := \beta^2/\sigma^2$  and  $y = x + H_t$ . The unconstrained solutions used to construct the clipped approximations are

$$\pi^{p,0}(t, y; m) = \frac{\beta}{\sigma^2} \{me^{-r(T-t)} - y\}, \quad (17)$$

$$\pi^{D,0}(t, y; \gamma_D) = \frac{\beta}{\gamma_D \sigma^2} e^{(A-r)(T-t)}, \quad (18)$$

$$\pi^{c,0}(t, y) = \frac{\beta}{\gamma_c \sigma^2} e^{-r(T-t)}, \quad (19)$$

$$\pi^{d,0}(t, y) = \theta_{\rho_d}(t)y. \quad (20)$$

where the PCMV expression corresponds to the embedded target  $m$ , whereas the DOMV expression corresponds to the variance-aversion parameter  $\gamma_D$ . Throughout this paper, DOMV is written consistently as  $\mathbb{E}[W] - (\gamma_D/2) \text{Var}(W)$ . Because DOMV re-solves the remaining-horizon problem at each date and state, the numerical comparison calibrates  $\gamma_D$  or the embedded target consistently with the objective. The DOMV parameter  $\gamma_D$  and the dTCMV scale parameter  $\rho_d$  play different roles and are denoted separately. The time-dependent dTCMV proportion  $\theta_{\rho_d}$  is obtained from a Volterra-type integral equation corresponding to the Björk–Murgoci formulation with variance aversion dependent on total pension wealth. Numerically, the two auxiliary integrals are converted into a system of ordinary differential equations and integrated backward from the terminal conditions. The unconstrained representation is then applied to  $Y$ , which includes background wealth. To make the theoretical structure explicit, Appendix A presents the continuous-time HJB and extended-HJB representations. The baseline numerical calculations, however, do not solve these PDEs directly by finite differences; instead they use the Markov–Gauss–Hermite backward calculations in Appendix B.

**Definition 2** (Clipped approximation). Given the corresponding unconstrained solution, define

$$\pi^{j,\text{clip}}(t, x) = \text{Proj}_{[0, x]} \{\pi^{j,0}(t, x + H_t)\} \quad (21)$$

and call it the *clipped approximation*. It is an implementable policy satisfying the DC constraint and is an ex-post projection based on the continuation value of the unconstrained problem. It is compared with the directly constrained policy obtained from the corresponding constrained HJB equation or constrained extended-HJB system.

**Definition 3** (Directly constrained feedback). Incorporate the constraint  $0 \leq \pi \leq x$  directly into the optimisation set of the HJB equation, extended-HJB system, or Markov–Gauss–Hermite backward calculation, and denote the feedback obtained from the corresponding constrained HJB equation or constrained extended-HJB system by

$$\pi^{j,\text{str}}(t, x)$$

Under this policy, the possibility that the constraint binds at future dates is reflected in the value function or equilibrium moments.

**Remark 1** (Local projection versus ex-post clipping). Directly constrained feedback also has a projection form onto  $[0, x]$  when the local Hamiltonian is maximised or minimised. Its projection centre, however, is computed from the *constrained* value function or equilibrium moments. By contrast, the clipped approximation projects ex post the solution of a *different unconstrained problem*. Therefore, in general,

$$\pi^{j,\text{str}}(t, x) \neq \pi^{j,\text{clip}}(t, x).$$

The two coincide only when the unconstrained policy remains inside the constraint throughout the reachable region, or when the clipped approximation can separately be shown to satisfy the corresponding constrained HJB equation or constrained extended-HJB system. The numerical analysis below measures this difference as a projection gap.

### A.2 PCMV and DOMV: Constrained Quadratic-Loss HJB Problems

#### A.2.1 Embedded quadratic-loss problem

This subsection considers a member who receives a deterministic terminal benefit  $D_T \geq 0$ , such as a DB pension benefit, and evaluates total terminal pension wealth  $X_T + D_T$ , combining the DC account balance and DB benefit, under a mean–variance criterion. To align the evaluation object with the cTCMV and dTCMV formulations below and preserve comparability across solution concepts, PCMV and DOMV are also formulated in terms of terminal wealth  $X_T + D_T$ .

For PCMV and DOMV, however,  $D_T$  is deterministic and unaffected by the control. Hence,

$$\text{Var}_{t,x}(X_T^\pi + D_T) = \text{Var}_{t,x}(X_T^\pi), \quad \mathbb{E}_{t,x}[X_T^\pi + D_T] = \mathbb{E}_{t,x}[X_T^\pi] + D_T \quad (22)$$

Thus a mean constraint expressed in terms of total pension wealth,  $\mathbb{E}[X_T^\pi + D_T] = d$ , is equivalent to a DC-balance constraint  $\mathbb{E}[X_T^\pi] = d - D_T$ . Likewise, for the embedded quadratic loss,

$$(X_T^\pi + D_T - m)^2 = (X_T^\pi - (m - D_T))^2 \quad (23)$$

Therefore, after shifting the target mean or embedded target by  $D_T$ , the optimal controls, feedback rules, and glide paths for PCMV and DOMV are identical between the cases  $D_T > 0$  and  $D_T = 0$ . In other words,  $D_T$  effectively drops out of the optimisation for PCMV and DOMV and is neither a state variable nor a source of risk. It is retained explicitly only to keep the unit of comparison common across the other solution concepts.

**Definition 4** (PCMV (pre-commitment mean-variance)). Following van Staden et al. (2021), Li and Ng (2000), Zhou and Li (2000), and Højgaard and Vigna (2007); Vigna (2014); Menoncin and Vigna (2017), who establish the target-based representation for DC pensions, PCMV denotes the solution concept that commits at the initial state  $(0, x_0)$  to the selected mean-variance efficient strategy through maturity and does not re-solve the original mean-variance problem at intermediate dates. We use the Li-Ng-Zhou embedding employed by van Staden et al. (2021), and denote by  $m$  the target corresponding to the embedding parameter  $\gamma/2$  in that paper. Thus, for fixed

$m \in \mathbb{R}$ , solve

$$L^m(t, x) = \inf_{\pi \in \mathcal{A}_{t,x}^X} \mathbb{E}_{t,x}[(X_T^\pi + D_T - m)^2] \quad (24)$$

and continue to implement the continuation part of the optimal feedback chosen at time zero. The relation to the mean-constrained PCMV problem is given below in Theorem 2.

The corresponding constrained HJB equation is

$$v_t^m + \inf_{0 \leq \pi \leq x} \left\{ (rx + c_t + \beta\pi)v_x^m + \frac{1}{2}\sigma^2\pi^2 v_{xx}^m \right\} = 0, \quad v^m(T, x) = (x + D_T - m)^2. \quad (25)$$

**Assumption 1** (Regularity required for verification). *For the verification theorems, the candidate function is assumed to be of class  $C^{1,2}$ , to satisfy the terminal condition continuously, and to have polynomial growth. We further assume sufficient square integrability of the associated control process to apply Itô's formula and localisation so that the stopped stochastic integral is a true martingale with zero expectation, together with uniform integrability or dominated-convergence conditions sufficient to pass to the limit in the localisation stopping times. These technical conditions justify removing the stochastic-integral term in expectation and taking the stopping-time limit in the verification proof. Numerical solutions may in general need to be understood as viscosity rather than classical solutions; the theoretical presentation here uses the classical-solution framework to make the projection structure explicit.*

The existence discussion differs between PCMV/DOMV and cTCMV/dTCMV. For the fixed-target PCMV and DOMV problems, using the risky-asset proportion  $u_t = \pi_t/X_t \in [0, 1]$  as the control converts the state-dependent interval  $[0, X_t]$  into a fixed compact control set. The deterministic contribution  $c_t$  is an inhomogeneous drift term in a linear state equation, so strong existence, uniform moment estimates, the dynamic programming principle, and viscosity-solution existence can be treated within a standard framework. Proposition 3 in Appendix G summarises existence, in the viscosity-solution/relaxed-control sense, for the fixed-target value function and optimal relaxed Markov control, the outer target choice in scalarised PCMV, and measurable rolling-target selection in DOMV (Li et al., 2002; Li and Xu, 2016; Fleming and Soner, 2006). Additional assumptions are required for  $C^{1,2}$  classical solutions, regular free boundaries, and unique continuous strict feedback controls.

For cTCMV and dTCMV, construction of a global equilibrium solution becomes a fixed-point problem specific to time-inconsistent control. Bensoussan et al. (2014) construct an equilibrium under a one-sided short-sale constraint. The state-dependent two-sided constraint  $0 \leq \pi_t \leq X_t$  used here requires additional theoretical work. Accordingly, Theorems 5 and 7 are positioned as local verification results conditional on a sufficiently regular global equilibrium candidate. Existence, uniqueness, and regularity of equilibrium solutions under this constraint remain subjects for future research.

**Theorem 1** (Verification for the constrained PCMV embedding problem). *For fixed  $m \in \mathbb{R}$ , suppose  $v^m$  satisfies Assumption 1 and the HJB equation (25), and suppose further that  $v_{xx}^m > 0$ . Then  $v^m = L^m$ . Moreover,*

$$\pi^{m,*}(t, x) = \text{Proj}_{[0,x]} \left( -\frac{\beta v_x^m(t, x)}{\sigma^2 v_{xx}^m(t, x)} \right) \quad (26)$$

*if this feedback is admissible and generates a unique state process, then  $\pi^{m,*}$  is optimal for (24).*

*Proof.* Apply Itô's formula to any admissible control  $\pi$  up to a localisation stopping time  $\tau_k$ . This yields

$$\mathbb{E}[v^m(\tau_k, X_{\tau_k}^\pi)] - v^m(t, x) = \mathbb{E} \int_t^{\tau_k} (v_s^m + \mathcal{L}^{\pi_s} v^m)(s, X_s^\pi) ds \geq 0$$

where the inequality follows from the minimisation condition in the HJB equation. Letting  $k \rightarrow \infty$  and then taking the terminal limit shows that  $v^m(t, x)$  is no greater than the quadratic loss under any admissible control. When  $v_{xx}^m > 0$ , the part of the Hamiltonian depending on  $\pi$  is a convex quadratic; projecting its unconstrained minimiser onto  $[0, x]$  as in (26) attains the pointwise minimum. Under this control, the preceding inequality becomes an equality, proving  $v^m = L^m$  and optimality. Details of the localisation limit are given in Appendix G.2.  $\square$

**Corollary 1** (Free-boundary regions for PCMV). *Define the local unconstrained candidate by*

$$u^m(t, x) = -\frac{\beta v_x^m(t, x)}{\sigma^2 v_{xx}^m(t, x)}$$

*Then the state space is partitioned into*

$$\mathcal{R}_0^m = \{u^m \leq 0\}, \quad \mathcal{R}_I^m = \{0 < u^m < x\}, \quad \mathcal{R}_U^m = \{u^m \geq x\}$$

and, in the respective regions,

$$\begin{aligned}\mathcal{R}_0^m : \quad & v_t^m + (rx + c_t)v_x^m = 0, \\ \mathcal{R}_I^m : \quad & v_t^m + (rx + c_t)v_x^m - \frac{\beta^2}{2\sigma^2} \frac{(v_x^m)^2}{v_{xx}^m} = 0, \\ \mathcal{R}_U^m : \quad & v_t^m + \{(r + \beta)x + c_t\}v_x^m + \frac{1}{2}\sigma^2 x^2 v_{xx}^m = 0.\end{aligned}$$

The boundaries separating these three regions are free boundaries.

*Proof.* Let  $u^m$  denote the projection centre in (26). When  $u^m \leq 0$ ,  $0 < u^m < x$ , and  $u^m \geq x$ , the minimising controls are respectively  $0, u^m, x$ . Substituting these controls into the HJB equation (25), and using the minimum value of the convex quadratic  $-(\beta^2/2\sigma^2)(v_x^m)^2/v_{xx}^m$  only in the interior region, yields the three PDEs displayed above. Because the switching conditions depend on derivatives of the unknown candidate function, the separating boundaries are free boundaries.  $\square$

### A.2.2 From the embedded quadratic-loss problem to PCMV

This embedding follows the line of Højgaard and Vigna (2007); Vigna (2014); Menoncin and Vigna (2017), which relates the MV efficient frontier in DC pensions to target-based quadratic loss. We formulate that relation under a state-dependent upper investment bound and establish optimality through the constrained HJB equation and its verification theorem.

Define the mean-constrained PCMV problem by

$$\inf_{\pi \in \mathcal{A}_{0,x_0}^X} \text{Var}(X_T^\pi + D_T) \quad \text{subject to} \quad \mathbb{E}[X_T^\pi + D_T] = d \quad (27)$$

Introduce a Lagrange multiplier  $\lambda$  and set  $m = d + \lambda$ . Then, for any  $\pi$  satisfying the mean constraint,

$$\mathbb{E}[(X_T^\pi + D_T - m)^2] = \text{Var}(X_T^\pi + D_T) + \lambda^2$$

**Theorem 2** (Verification for constrained PCMV). *Suppose that for some  $m = d + \lambda$  there exist  $v^m$  and an optimal control  $\pi^{m,*}$  satisfying the conditions of Theorem 1, and that*

$$\mathbb{E}_{0,x_0}[X_T^{\pi^{m,*}} + D_T] = d$$

*holds. Then  $\pi^{m,*}$  is optimal for the mean-constrained PCMV problem (27).*

*Proof.* For any admissible control  $\pi$  satisfying the mean constraint, setting  $m = d + \lambda$  gives

$$\mathbb{E}[(X_T^\pi + D_T - m)^2] = \text{Var}(X_T^\pi + D_T) + \lambda^2$$

The control  $\pi^{m,*}$  minimises the embedded quadratic loss and, by assumption, satisfies the same mean constraint. Hence, for every feasible  $\pi$ ,

$$\text{Var}(X_T^{\pi^{m,*}} + D_T) + \lambda^2 \leq \text{Var}(X_T^\pi + D_T) + \lambda^2$$

Cancelling the common term  $\lambda^2$  yields optimality for the mean-constrained PCMV problem.  $\square$

### A.2.3 DOMV as a rolling local PCMV problem

**Definition 5** (DOMV (dynamically optimal mean–variance)). Following Pedersen and Peskir (2017), Menoncin and Vigna (2020), and van Staden et al. (2021), DOMV treats each reached date and state  $(t, x)$  as a new initial condition, re-solves the remaining-horizon mean–variance problem, and implements only the current control of the resulting optimal continuation strategy. In the standard representation of van Staden et al. (2021) with variance-aversion parameter  $\gamma_D > 0$ , at each  $(t, x)$  one solves

$$\pi_{t,x}^{D,*} \in \arg \max_{\pi \in \mathcal{A}_{t,x}^X} \left\{ \mathbb{E}_{t,x}[X_T^\pi + D_T] - \frac{\gamma_D}{2} \text{Var}_{t,x}(X_T^\pi + D_T) \right\}$$

and defines the DOMV feedback by  $\pi^{\text{DOMV}}(t, x) = \pi_{t,x}^{D,*}(t, x)$ . At the next date, the same operation is repeated from the new state. DOMV therefore does not commit to the time-zero strategy as PCMV does, and it also differs from the game-theoretic equilibrium underlying TCMV.

In the terminology of Menoncin and Vigna (2020), pre-commitment corresponds to a fixed target, whereas the dynamically optimal strategy uses a target that is updated with time and state. We implement the same rolling optimality through a local target mean  $d(t, x)$  and embedded target  $m(t, x) = d(t, x) + \lambda(t, x)$ .

**Theorem 3** (Verification for constrained DOMV). *For each  $(t, x)$ , suppose the embedded problem associated with target  $m(t, x)$  satisfies the conditions of Theorem 1, with optimal control  $\pi_{t,x}^{D,*}(s, z)$ . Suppose further that*

$$\mathbb{E}_{t,x}[X_T^{\pi_{t,x}^{D,*}} + D_T] = d(t, x)$$

*holds. Then  $\pi_{t,x}^{D,*}$  is optimal for the local constrained MV problem at  $(t, x)$ , and the DOMV feedback is defined by*

$$\pi^{\text{DOMV}}(t, x) = \pi_{t,x}^{D,*}(t, x) \quad (28)$$

*Proof.* Treat each fixed state  $(t, x)$  as a new initial condition. The DOMV remaining-horizon problem is then exactly an embedded PCMV problem with target  $m(t, x)$ . Applying Theorem 1 and the mean-consistency condition to this local problem shows that  $\pi_{t,x}^{D,*}$  solves the local MV problem on  $[t, T]$ . DOMV does not commit to the entire continuation strategy; it implements only the current component  $\pi_{t,x}^{D,*}(t, x)$  and repeats the same local optimisation at the next date. Equation (28) therefore defines the rolling-optimal feedback.  $\square$

**Remark 2** (DOMV re-optimisation and conditional evaluation). DOMV is a decision principle that re-solves the remaining-horizon problem at each date and state. Conditional evaluation in Appendix B.8, by contrast, measures outcomes from continuing a stored policy from the current state and applies to PCMV, DOMV, cTCMV, dTCMV, and CP. The two operations serve different purposes.

### A.3 cTCMV: Time-Consistent Equilibrium with Constant Variance Aversion

#### A.3.1 Equilibrium criterion and moment functions

cTCMV is the time-consistent mean-variance (TCMV) equilibrium concept in which every successive self uses the same constant variance-aversion coefficient  $\gamma_c > 0$  (Basak and Chabakauri, 2010; Björk et al., 2017; van Staden et al., 2021). Throughout this paper, the coefficient multiplying variance in  $\mathbb{E}[W] - (\gamma/2) \text{Var}(W)$  is referred to as the “variance-aversion coefficient”.

**Definition 6** (cTCMV (constant-variance-aversion time-consistent MV)). For the self at state  $(t, x)$ , define the criterion

$$J^c(t, x; \pi) = \mathbb{E}_{t,x}[X_T^\pi + D_T] - \frac{\gamma_c}{2} \text{Var}_{t,x}(X_T^\pi + D_T)$$

In van Staden et al. (2021), time consistency is expressed by requiring that, for the same future date  $v$  and future state  $z$ , the continuation control calculated from the initial date  $t_0$  and the control recalculated at a later date  $t'$  satisfy

$$u_c^*(t_0; z, v) = u_c^*(t'; z, v), \quad v \geq t' \geq t_0,$$

We formulate this time-consistent solution concept as a continuous-time Markov equilibrium in the sense of Björk et al. (2017), treating the decision maker at each date and state as a distinct “self”. A candidate policy is an equilibrium if the current self cannot produce a positive first-order improvement in the criterion by switching unilaterally to another admissible control over an infinitesimal time interval and then returning to the candidate policy. Formally, let  $\hat{\pi}^c$  be a candidate equilibrium feedback rule and consider a short-horizon deviation  $\nu \otimes_h \hat{\pi}^c$  that uses another admissible control  $\nu \in [0, x]$  only on  $[t, t+h]$  and then returns to  $\hat{\pi}^c$ <sup>1</sup>. The rule  $\hat{\pi}^c$  is a constrained cTCMV equilibrium if, for every  $(t, x)$  and every admissible deviation  $\nu$ ,

$$\liminf_{h \downarrow 0} \frac{J^c(t, x; \hat{\pi}^c) - J^c(t, x; \nu \otimes_h \hat{\pi}^c)}{h} \geq 0 \quad (29)$$

holds. Thus cTCMV is not DOMV, which globally re-solves the remaining-horizon problem at every date. It is instead a subgame-perfect Nash-type intrapersonal equilibrium that is locally stable against short-horizon deviations by each current self.

Under the candidate equilibrium rule, define the conditional terminal mean and equilibrium value function by

$$F^c(t, x) = \mathbb{E}_{t,x}^{\hat{\pi}^c}[X_T + D_T], \quad V^c(t, x) = \mathbb{E}_{t,x}^{\hat{\pi}^c}[X_T + D_T] - \frac{\gamma_c}{2} \text{Var}_{t,x}^{\hat{\pi}^c}(X_T + D_T) \quad (30)$$

and define the generator associated with control  $\pi$  by

$$\mathcal{L}^\pi \phi = (rx + c_t + \beta\pi)\phi_x + \frac{1}{2}\sigma^2\pi^2\phi_{xx} \quad (31)$$

Specialising the equilibrium-control definition of Björk and co-authors to the present problem yields the following extended HJB system for cTCMV.

$$F_t^c + \mathcal{L}^{\hat{\pi}^c} F^c = 0, \quad F^c(T, x) = x + D_T, \quad (32)$$

$$V_t^c + \sup_{0 \leq \pi \leq x} \left\{ (rx + c_t + \beta\pi)V_x^c + \frac{1}{2}\sigma^2\pi^2V_{xx}^c - \frac{\gamma_c}{2}\sigma^2\pi^2(F_x^c)^2 \right\} = 0, \quad V^c(T, x) = x + D_T. \quad (33)$$

#### A.3.2 Local feedback representation

**Theorem 4** (Local feedback representation for constrained cTCMV). Suppose  $(F^c, V^c, \hat{\pi}^c)$  satisfies (32)–(33), and suppose further that, at each  $(t, x)$ ,

$$D^c(t, x) = V_{xx}^c(t, x) - \gamma_c(F_x^c(t, x))^2 < 0 \quad (34)$$

holds. Then the equilibrium feedback rule that maximises the constrained Hamiltonian is

$$\hat{\pi}^c(t, x) = \text{Proj}_{[0,x]} \left( -\frac{\beta V_x^c(t, x)}{\sigma^2 D^c(t, x)} \right) \quad (35)$$

*Proof.* Removing from the Hamiltonian in (33) the terms independent of  $\pi$  leaves

$$h^c(\pi) = \beta V_x^c \pi + \frac{1}{2}\sigma^2 D^c \pi^2$$

Because  $D^c < 0$ ,  $h^c$  is strictly concave and its unconstrained maximiser is  $-\beta V_x^c / (\sigma^2 D^c)$ . Maximising a concave function over the closed interval  $[0, x]$  is achieved by projecting this unconstrained maximiser onto the interval, which gives (35).  $\square$

#### A.3.3 Verification theorem

**Theorem 5** (Verification theorem for constrained cTCMV). Suppose  $(F^c, V^c, \hat{\pi}^c)$  satisfies (32)–(33), and that  $\hat{\pi}^c$  attains the supremum in (33) pointwise. Suppose also that  $F^c, V^c$  and the associated controlled process satisfy the regularity and integrability conditions needed to justify the short-horizon expansion. Then  $\hat{\pi}^c$  is a time-consistent equilibrium for constrained cTCMV. In particular, when (34) holds, the directly constrained feedback given by the local representation in Theorem 4 is an equilibrium.

*Proof.* At state  $(t, x)$ , change the control to  $\nu$  only on  $[t, t+h]$  and return thereafter to  $\hat{\pi}^c$ . Using an Itô expansion and the backward equation for  $F^c$ , the first-order change in the criterion due to this deviation can be written as

$$J^c(t, x; \nu \otimes_h \hat{\pi}^c) - J^c(t, x; \hat{\pi}^c) = h \{ \mathcal{H}^c(t, x, \nu) - \mathcal{H}^c(t, x, \hat{\pi}^c) \} + o(h)$$

where  $\mathcal{H}^c$  is the local Hamiltonian inside the braces in (33). Because  $\hat{\pi}^c$  attains its maximum pointwise, the first-order term on the right-hand side is non-positive, which gives (29). In this short-horizon expansion, combining the first-order changes in the conditional mean and conditional second moment produces the local Hamiltonian difference including the variance-correction term.  $\square$

<sup>1</sup>This type of local perturbation is also called a spike variation in control theory. We use the term “short-horizon deviation” because it directly describes the equilibrium definition of Björk and co-authors.

**Corollary 2** (Free-boundary regions for cTCMV). *Define the local unconstrained candidate by*

$$u^c(t, x) = -\frac{\beta V_x^c(t, x)}{\sigma^2 \{V_{xx}^c(t, x) - \gamma_c (F_x^c(t, x))^2\}}$$

*The state space is partitioned into*

$$\mathcal{R}_0^c = \{u^c \leq 0\}, \quad \mathcal{R}_I^c = \{0 < u^c < x\}, \quad \mathcal{R}_U^c = \{u^c \geq x\}$$

*In the respective regions,  $F^c$  satisfies the linear PDE obtained by substituting  $\hat{\pi}^c = 0, u^c, x$ , while  $V^c$  satisfies the corresponding extended HJB equation.*

*Proof.* By Theorem 4, the maximising control is  $0, u^c, x$  according as the local candidate  $u^c$  lies below zero, inside the interval, or above  $x$ . Substituting this control into (32) and (33) yields the linear backward equation for  $F^c$  and the corresponding extended HJB equation in each region. Because the switching conditions are determined endogenously by derivatives of  $F^c$  and  $V^c$ , the boundaries are free boundaries.  $\square$

## A.4 dTCMV: Variance Aversion Dependent on Total Pension Wealth

### A.4.1 Variance aversion based on total pension wealth

dTCMV is a time-consistent mean–variance equilibrium concept that uses a variance-aversion coefficient depending on current total pension wealth. van Staden et al. (2021) refer to the case in which variance aversion is inversely proportional to current wealth as dTCMV. In this paper, “current wealth” is interpreted not only as the tradable DC balance but as total pension wealth  $Y_t = X_t + H_t$ , including future contributions and DB benefits, thereby fixing the specification to make explicit the interpretation of the third layer as personalised advice based on total pension wealth. Thus,

$$\Gamma(t, x) = \frac{\rho_d}{x + H_t}, \quad \rho_d > 0 \quad (36)$$

where  $\rho_d$  is a scale parameter determined once at calibration and then held fixed across time and states, while the variance-aversion coefficient actually used by the self at each date and state is the state-dependent  $\Gamma(t, x)$ . We therefore distinguish clearly between  $\rho_d$  and  $\Gamma(t, x)$  rather than using a time-zero subscript on the coefficient. Under this specification, even for the same DC account balance  $x$ , a member with a larger present value  $H_t$  of future contributions or DB benefits has lower variance aversion. dTCMV can thus be interpreted as personalised advice that reflects total pension wealth in the allocation of risk.

**Remark 3** (Roles of  $\Gamma$  and the investment ceiling).  $\Gamma(t, x)$  determines the scale of variance aversion, whereas the investable amount is determined by  $0 \leq \pi_t \leq X_t$ . Background wealth affects risk-bearing capacity, while actual risky-asset investment remains within the DC account balance. Separating these roles captures both risk tolerance based on total pension wealth and implementability inside the DC account.

### A.4.2 Moment formulation

For the self at state  $(t, x)$ , consider the criterion under the fixed variance-aversion coefficient  $\Gamma(t, x) > 0$ ,

$$J^d(t, x; \pi) = \mathbb{E}_{t,x}[X_T + D_T] - \frac{\Gamma(t, x)}{2} \text{Var}_{t,x}(X_T + D_T) \quad (37)$$

**Definition 7** (dTCMV (time-consistent MV with total-pension-wealth-dependent variance aversion)). dTCMV preserves the same time-consistency requirement as cTCMV while allowing the variance-aversion coefficient used by each current self to depend on total pension wealth. In van Staden et al. (2021), the cTCMV requirement that continuation controls coincide for the same future date and future state is retained, while the variance coefficient in the objective is set to  $\rho_d/(2w)$ . Replacing  $w$  with total pension wealth  $Y_t = X_t + H_t$  yields  $\rho_d/(2Y_t) = \Gamma(t, x)/2$ . In the Markov-equilibrium representation, the self at  $(t, x)$  treats the contemporaneous value  $\Gamma(t, x)$  as fixed while evaluating a short-horizon deviation and imposes the same local equilibrium condition as under cTCMV. Thus the difference from cTCMV lies not in the equilibrium concept but in the preference specification: constant variance aversion  $\gamma_c$  versus  $\Gamma(t, x)$  inversely proportional to total pension wealth. The formal short-horizon deviation condition is given in (42).

Under the candidate equilibrium feedback rule  $\hat{\pi}^d$ , define the conditional first and second moments by

$$F^d(t, x) = \mathbb{E}_{t,x}^{\hat{\pi}^d}[X_T + D_T], \quad G^d(t, x) = \mathbb{E}_{t,x}^{\hat{\pi}^d}[(X_T + D_T)^2] \quad (38)$$

Then,

$$F_t^d + \mathcal{L}^{\hat{\pi}^d} F^d = 0, \quad F^d(T, x) = x + D_T, \quad (39)$$

$$G_t^d + \mathcal{L}^{\hat{\pi}^d} G^d = 0, \quad G^d(T, x) = (x + D_T)^2 \quad (40)$$

holds.

Because  $\Gamma(t, x)$  is state-dependent in dTCMV, writing the Hamiltonian only in terms of derivatives of the equilibrium value function

$$V^d(t, x) = F^d(t, x) - \frac{\Gamma(t, x)}{2} \{G^d(t, x) - (F^d(t, x))^2\}$$

would introduce  $\Gamma_x$  and  $\Gamma_{xx}$ . However, when the self at  $(t, x)$  evaluates a short-horizon deviation, its current variance-aversion coefficient  $\Gamma(t, x)$  is held fixed. Consequently, an extended-HJB representation consistent with the equilibrium definition of Björk and co-authors must be constructed using  $F^d$  and  $G^d$ , rather than derivatives of  $V^d$  alone.

We follow the game-theoretic framework of Björk et al. (2014) and Björk et al. (2017). The former studies a mean–variance portfolio problem with state-dependent variance aversion, while the latter formulates equilibrium controls, equilibrium value functions, and extended HJB systems for continuous-time time-inconsistent stochastic control in general. We consider a short-horizon deviation in which the self at  $(t, x)$  uses another admissible control only over the infinitesimal interval  $[t, t + h)$  and then returns to the candidate equilibrium feedback rule. The self’s variance-aversion coefficient  $\Gamma = \Gamma(t, x)$  is held fixed during the deviation evaluation. The infinitesimal changes in the first and second moments can formally be written as

$$F^{d,\varepsilon} = F^d + \varepsilon(F_t^d + \mathcal{L}^\pi F^d) + o(\varepsilon), \quad G^{d,\varepsilon} = G^d + \varepsilon(G_t^d + \mathcal{L}^\pi G^d) + o(\varepsilon)$$

Substituting these expressions into  $F^d - (\Gamma/2)\{G^d - (F^d)^2\}$  and expanding to first order yields the short-horizon-deviation Hamiltonian

$$\mathcal{H}^d(t, x, \pi) = (1 + \Gamma F^d)(F_t^d + \mathcal{L}^\pi F^d) - \frac{\Gamma}{2}(G_t^d + \mathcal{L}^\pi G^d) \quad (41)$$

Equation (41) is therefore the extended-HJB theory of Björk and co-authors specialised to the first- and second-moment notation used here. The contribution of this paper is to solve this local equilibrium condition under the DC investable-amount constraint  $0 \leq \pi \leq x$  and derive the associated constrained projection feedback and free-boundary structure.

The short-horizon deviation condition for the equilibrium feedback rule in Definition 7 is, for every  $\nu \in [0, x]$ ,

$$\liminf_{h \downarrow 0} \frac{J^d(t, x; \hat{\pi}^d) - J^d(t, x; \nu \otimes_h \hat{\pi}^d)}{h} \geq 0 \quad (42)$$

#### A.4.3 Local feedback representation and verification theorem

**Theorem 6** (Local feedback representation for constrained dTCMV). *Suppose  $(F^d, G^d, \hat{\pi}^d)$  satisfies (39)–(40) and the equilibrium condition*

$$\hat{\pi}^d(t, x) \in \arg \max_{0 \leq \pi \leq x} \mathcal{H}^d(t, x, \pi) \quad (43)$$

Define

$$A^d(t, x) = (1 + \Gamma F^d)F_x^d - \frac{\Gamma}{2}G_x^d, \quad (44)$$

$$B^d(t, x) = (1 + \Gamma F^d)F_{xx}^d - \frac{\Gamma}{2}G_{xx}^d \quad (45)$$

and assume  $B^d(t, x) < 0$ . Then

$$\boxed{\hat{\pi}^d(t, x) = \text{Proj}_{[0, x]} \left( -\frac{\beta A^d(t, x)}{\sigma^2 B^d(t, x)} \right)} \quad (46)$$

*Proof.* Expanding (41) using the generator (31), the terms depending on  $\pi$  are

$$h^d(\pi) = \beta A^d \pi + \frac{1}{2} \sigma^2 B^d \pi^2$$

Because the current  $\Gamma(t, x)$  is held fixed during evaluation of the deviation, neither  $\Gamma_x$  nor  $\Gamma_{xx}$  appears in this local maximisation. Under  $B^d < 0$ ,  $h^d$  is strictly concave, and projecting the unconstrained maximiser  $-\beta A^d/(\sigma^2 B^d)$  onto  $[0, x]$  gives (46).  $\square$

**Theorem 7** (Verification theorem for constrained dTCMV). *Suppose  $(F^d, G^d, \hat{\pi}^d)$  satisfies (39)–(40), and that  $\hat{\pi}^d$  maximises (41) pointwise over  $[0, x]$ . Suppose also that  $F^d, G^d$  and the associated controlled process satisfy the regularity and integrability conditions required to justify the short-horizon expansion. Then  $\hat{\pi}^d$  is a time-consistent equilibrium for constrained dTCMV. In particular, when  $B^d < 0$ , the directly constrained feedback given by the local representation in Theorem 6 is an equilibrium.*

*Proof.* Fix the current self's  $\Gamma(t, x)$  and consider a short-horizon deviation that uses  $\nu$  only on  $[t, t+h)$  and returns thereafter to the candidate equilibrium rule. Substituting the Itô expansions of the first and second conditional moments into the objective (37) yields

$$J^d(t, x; \nu \otimes_h \hat{\pi}^d) - J^d(t, x; \hat{\pi}^d) = h\{\mathcal{H}^d(t, x, \nu) - \mathcal{H}^d(t, x, \hat{\pi}^d)\} + o(h)$$

By assumption,  $\hat{\pi}^d$  maximises  $\mathcal{H}^d$  over  $[0, x]$ , so the first-order term is non-positive and (42) follows. When  $B^d < 0$ , the maximiser is the directly constrained feedback in Theorem 6. Here again, combining the short-horizon changes in the first and second conditional moments with state-dependent variance aversion represents the loss from deviation as a local Hamiltonian difference.  $\square$

**Corollary 3** (Free-boundary regions for dTCMV). *Define the local unconstrained candidate by*

$$u^d(t, x) = -\frac{\beta A^d(t, x)}{\sigma^2 B^d(t, x)}$$

The state space is partitioned into

$$\mathcal{R}_0^d = \{u^d \leq 0\}, \quad \mathcal{R}_I^d = \{0 < u^d < x\}, \quad \mathcal{R}_U^d = \{u^d \geq x\}$$

In these regions,  $F^d$  and  $G^d$  satisfy the linear PDEs obtained by substituting  $\hat{\pi}^d = 0, u^d, x$ , respectively. In this sense, dTCMV is also characterised by an extended-HJB system with free boundaries.

*Proof.* Let  $u^d$  denote the projection centre in Theorem 6. In the three regions  $u^d \leq 0$ ,  $0 < u^d < x$ , and  $u^d \geq x$ , the controls are respectively  $0, u^d, x$ . Substitution into the moment equations (39)–(40) yields the displayed linear PDEs in each region. Because the switching sets depend on  $A^d, B^d$  and therefore on derivatives of the unknown moment functions, the boundaries are free boundaries.  $\square$

**Remark 4** (Why dTCMV cannot be reduced to  $V^d$  alone). If  $\Gamma$  were constant, derivatives of  $V^d = F^d - (\Gamma/2)\{G^d - (F^d)^2\}$  could be used to recover a cTCMV-type representation. When  $\Gamma = \Gamma(t, x)$  is state-dependent, however,  $V_x^d$  and  $V_{xx}^d$  contain  $\Gamma_x$  and  $\Gamma_{xx}$ . Because the current self's  $\Gamma(t, x)$  is held fixed when evaluating a short-horizon deviation, the Hamiltonian consistent with the equilibrium definition must be written in the form (41).

## A.5 CP: Constant-Proportion Benchmark

PCMV, DOMV, cTCMV, and dTCMV are each derived from some optimality or equilibrium condition for a mean–variance problem. By contrast, the constant-proportion (CP) strategy is not the solution of an optimisation problem but a practical benchmark that keeps the risky-asset share of the DC account fixed in advance. Following [van Staden et al. \(2021\)](#), define it as follows.

**Definition 8** (DC constant-proportion strategy). Let  $\theta_{cp} \in [0, 1]$  be constant. Define the CP strategy by

$$\pi^{cp}(t, x) = \theta_{cp}x, \quad g^{cp}(t, x) = \frac{\pi^{cp}(t, x)}{x} = \theta_{cp} \quad (47)$$

and set  $\pi^{cp}(t, 0) = 0$  when  $x = 0$ .

CP is an implementable constant-proportion benchmark given by  $\pi^{cp}(t, x) = \theta_{cp}x$  and plays a different role from the four MV policies derived from HJB or equilibrium conditions. We calibrate  $\theta_{cp}$  so that expected terminal wealth matches the four MV strategies and compare distributional shape at the same expected value. This design follows the common-mean comparison principle in [van Staden et al. \(2021\)](#).

## A.6 Unified Constrained Projection and Free-Boundary Structure

This subsection organises the directly constrained feedback derived above into a common structure. Although the solution concepts differ in objective and time consistency, incorporating the DC constraint inside the optimisation leads in each case to a local candidate computed from the constrained continuation value and projected onto  $[0, x]$ .

**Theorem 8** (Unified projection structure). *Suppose candidate functions exist for the corresponding constrained HJB equations for PCMV/DOMV and for the constrained extended-HJB systems for cTCMV and dTCMV, and that their effective curvatures satisfy*

$$v_{xx}^m > 0, \quad D^c = V_{xx}^c - \gamma_c(F_x^c)^2 < 0, \quad B^d < 0$$

*Then the directly constrained feedback for each solution concept has the form*

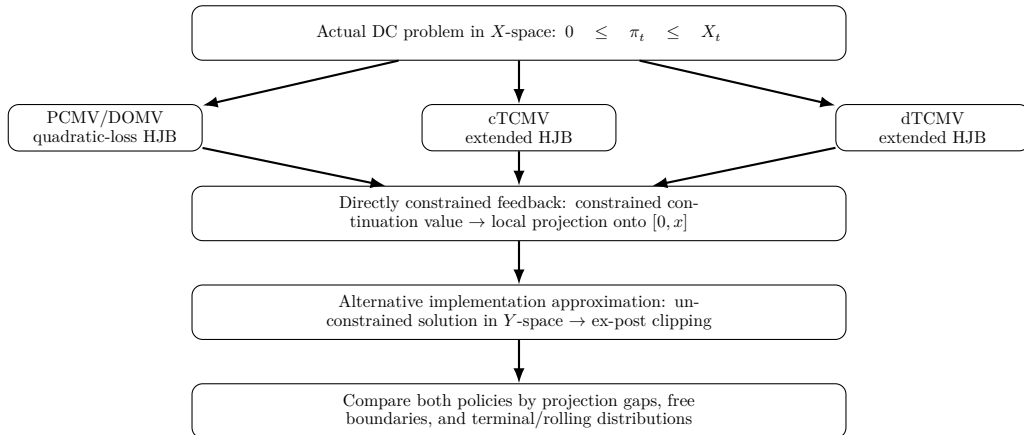
$$\begin{aligned} \pi^{p, \text{str}}(t, x) &= \text{Proj}_{[0, x]} \left( -\frac{\beta v_x^m}{\sigma^2 v_{xx}^m} \right), \\ \pi^{c, \text{str}}(t, x) &= \text{Proj}_{[0, x]} \left( -\frac{\beta V_x^c}{\sigma^2 D^c} \right), \\ \pi^{d, \text{str}}(t, x) &= \text{Proj}_{[0, x]} \left( -\frac{\beta A^d}{\sigma^2 B^d} \right) \end{aligned}$$

DOMV has the same form as the local re-solution version of PCMV.

*Proof.* Applying Theorem 1 for PCMV/DOMV, Theorem 4 for cTCMV, and Theorem 6 for dTCMV gives the respective projection representations. Their common feature is that, under the sign condition on effective curvature, the local Hamiltonian is a convex or concave quadratic. The functions defining the projection centres differ across solution concepts and are computed from the solutions of the corresponding constrained HJB equation or constrained extended-HJB system. Thus what is common is the projection form, while the projection centre is determined by the constrained value function or equilibrium moments of each solution concept.  $\square$

**Remark 5** (Constrained projection versus clipping). The  $v^m$  appearing in the projection centre in Theorem 8 solves the constrained HJB equation, while  $F^c, V^c, F^d, G^d$  solve the corresponding constrained extended-HJB systems. This projection therefore differs from the ex-post clipping of an unconstrained solution defined in Section A.1. The analytical solutions in  $Y$ -space are used to construct the clipped approximation at low cost, while Theorem 8 characterises the actual constrained solution. The implementable policies induced by these two objects are the objects of comparison.

The state space is divided into a lower-bound region, an interior region, and an upper-bound region according as the constrained local candidate is negative, lies between 0 and  $x$ , or exceeds  $x$ . Because these boundaries are determined endogenously from the value function or equilibrium moments, they are free boundaries.



## B A Unified Markov–Gauss–Hermite Engine for Constrained MV Control

### B.1 Positioning and Contribution of the Computational Engine

To solve and forward-evaluate the four constrained MV problems formulated in Appendix A under common conditions, we combine Gauss–Hermite conditional expectations, grid-based dynamic programming, and probability-mass propagation

and call this computational architecture the *unified Markov–Gauss–Hermite engine*. Each component rests on established numerical theory: Gauss–Hermite quadrature is a standard deterministic integration method (Abramowitz and Stegun, 1972; Judd, 1998); Markov approximation and semi-Lagrangian-type backward calculations for controlled diffusions have a general theory (Kushner and Dupuis, 2001; Falcone and Ferretti, 2014); and numerical HJB methods for MV asset allocation are well established (Wang and Forsyth, 2010, 2012). The contribution here is to apply these components consistently across four solution concepts, directly constrained and clipped policies, terminal and rolling conditional distributions, and MV/MVS nesting checks.

The numerical contribution is the *reuse of these numerical components across multiple solution concepts and evaluation stages*. Table 7 distinguishes established components from the integration specific to this paper.

Table 7: Positioning of the unified Markov–Gauss–Hermite engine

| Component                                                                       | Methodological status                                                                                                     | Function and contribution in this paper                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gauss–Hermite conditional expectations                                          | Standard deterministic quadrature for Gaussian shocks                                                                     | Uses the same one-step-ahead expectation evaluator for PCMV, DOMV, cTCMV, dTCMV, and MVS extensions, holding shock approximation and interpolation rules common across different objectives.                                                      |
| Markov / semi-Lagrangian-type backward calculation                              | Established general method for solving stochastic-control problems on a grid                                              | Incorporates the state-dependent interval $[0, x]$ directly into the admissible set at each grid point and handles ordinary Bellman recursion and moment recursion for extended HJB systems within one computational structure.                   |
| Parallel evaluation of directly constrained feedback and clipped approximations | Clipping itself is an easily implemented ex-post projection and is a different problem from directly constrained feedback | Compares directly constrained feedback and clipped approximations on the same state and control grids, with the same quadrature nodes, interpolation, and forward evaluator, isolating the projection gap as a difference in policy construction. |
| Forward distribution propagation using the same kernel                          | Probability-mass propagation under a Markov transition is standard                                                        | Reuses the transition kernel from the backward calculation for both terminal and rolling conditional distributions, evaluating densities, CDFs, quantiles, lower tails, and moment consistency within one framework.                              |
| Common implementation across four concepts and MV/MVS                           | An integrative design of this paper rather than a standalone numerical method                                             | Places four solution concepts with different intertemporal decision principles and the nested MVS extension on common inputs, outputs, and diagnostics, thereby connecting mathematical characterisation to three-layer implementation decisions. |

This common architecture serves four purposes. First, it *controls comparison conditions*: when comparing solution concepts or directly constrained feedback with clipped approximations, the conditional-expectation method, state and control grids, and forward-distribution evaluator are held common, so differences can be attributed primarily to the objective, time-consistency concept, and policy construction. Second, it *links theory to member outcomes*: the local feedback characterised by the verification theorems can be propagated through the same kernel to terminal-wealth distributions, lower tails, and rolling conditional distributions. Third, it provides *reproducible diagnostics of free boundaries and projection gaps*: because no random Monte Carlo simulation is placed in the inner loop, small differences between candidate controls and the locations of constraint boundaries are not distorted by sampling noise. Fourth, it provides *numerical governance*: backward moments can be checked against forward distributions, probability-mass conservation can be monitored, and nesting regression tests such as the return to MV at  $\eta = 0$  can be executed within the common implementation.

**Scope of the numerical contribution.** The numerical contribution is to integrate standard numerical components into one backward–forward architecture and reuse them across four constrained MV solution concepts, the comparison of directly constrained feedback and clipped approximations, MV/MVS nesting, backward policy calculation, forward evaluation of terminal and rolling distributions, and practical governance. In the forward calculation, probability mass is propagated deterministically along the discrete transition kernel constructed from the quadrature nodes. Independent Monte Carlo evaluation is used as a separate evaluator from this internal calculation.

**Why Gauss–Hermite quadrature is used.** Because the numerical procedure repeatedly evaluates conditional expectations for each date, state, and candidate control, Gauss–Hermite quadrature is used to avoid introducing sampling noise into the inner loop. Reusing the discrete transition kernel generated from the same quadrature nodes in forward propagation makes policy, free boundaries, terminal distributions, and rolling distributions reproducibly evaluable.

## B.2 Numerical State Space and Controls

**Notation for discrete indices.** We retain the continuous-time ordering of the state variables  $(t, x)$  after discretisation. Thus, for example, we write  $V_n(x_i) := V(t_n, x_i)$ ,  $M_n(x_i) := M(t_n, x_i)$ , and  $\pi_n(x_i) := \pi(t_n, x_i)$ , placing the time index  $n$  on the function and using  $x_i$  for the current-balance grid point. Probability mass is denoted by  $p_{n,i}$ , the control-grid index by  $a$ , the Gauss–Hermite node index by  $k$ , and the strategy index in rolling evaluation by  $j$ . The candidate next-period state generated from  $(t_n, x_i)$  by candidate control  $\pi$  and the  $k$ th Gauss–Hermite node is written  $\hat{x}_{n+1}^{(k)}(x_i, \pi)$ .

### B.2.1 Time grid

Partition the interval  $[0, T]$  into  $N_t$  steps and define

$$t_n = n\Delta t, \quad \Delta t = \frac{T}{N_t}, \quad n = 0, 1, \dots, N_t \quad (48)$$

The baseline calculation uses  $N_t = 480$  ( $\Delta t = 1/12$ ), corresponding to a monthly discrete-time model over 40 years of contributions and investment review. This is a model specification matching the institutional decision frequency; convergence to the continuous-time HJB is treated as a separate validation issue. For sensitivity analysis spanning many parameter combinations, a screening specification with  $N_t = 80$  is used to control computational cost and is distinguished in purpose from the baseline calculation with  $N_t = 480$ . Backward calculations proceed for  $n = N_t - 1, N_t - 2, \dots, 0$ , while distribution propagation proceeds for  $n = 0, 1, \dots, N_t - 1$ .

### B.2.2 Wealth grid

Define the grid for the DC account balance  $X_t$  by

$$0 = x_0 < x_1 < \dots < x_I = x_{\max} \quad (49)$$

and let the number of wealth grid points be  $N_x = I + 1$ . On an equally spaced grid,  $x_i = i\Delta x$ . The baseline calculation uses upper state bound  $x_{\max} = 600$ ,  $N_x = 6001$  grid points, and spacing  $\Delta x = 0.1$ . Since the initial DC balance  $X_0 = 1/12$  is not itself a grid point, the first transition is taken from the exact initial state and subsequent values are linearly interpolated between neighbouring grid points. In the low-balance region where the constraint binds strongly, the invested amount  $\pi = xg$  rather than the ratio  $g = \pi/x$  is interpolated, and  $\pi = 0$  is imposed at  $x = 0$ . The effects of the upper boundary, state spacing, control grid, and quadrature order are checked separately in the robustness analysis in Appendix E.4.

All PDE calculations and distribution propagation are performed on the  $X$ -space grid. Only for dTCMV, background wealth enters the criterion to affect risk tolerance through

$$\Gamma(t_n, x_i) = \frac{\rho_d}{x_i + H_{t_n}} \quad (50)$$

Since  $H_{t_n}$  is an exogenous deterministic quantity calculated backward, no additional state dimension or  $Y$ -space grid is introduced. We also do not generalise to partial background wealth through an  $\alpha$  parameter.

### B.2.3 Control grid and admissible interval

At each grid point  $(t_n, x_i)$ , the admissible controls satisfy

$$\Pi_{n,i} = [0, x_i] \quad (51)$$

Numerically, candidate controls are discretised as

$$\pi_i^{(a)} = \frac{a}{N_a - 1} x_i, \quad a = 0, 1, \dots, N_a - 1 \quad (52)$$

For higher accuracy, after a coarse candidate search one can exploit the near-quadratic form of the local Hamiltonian or discrete objective and use a one-dimensional golden-section search or Brent method to maximise or minimise over  $[0, x_i]$ .

**Remark 6** (Control grid and projection formula). The theory above shows that, when smooth value or moment functions are available, directly constrained feedback can be represented by projecting a local unconstrained candidate onto  $[0, x]$ . In the numerical implementation, candidate controls are searched directly over  $[0, x_i]$ , so binding and interior regions are treated by the same procedure. The projection formula is used to diagnose the computed solution and interpret free boundaries and projection gaps.

## B.3 Gauss–Hermite Conditional Expectations

At this point, the explicit controls for the unconstrained problems and the structure of state-dependent feedback and binding regions for the directly constrained problems are largely characterised. The remaining task in the numerical calculation is to evaluate continuation values in the HJB or extended-HJB systems. In the main computation, rather than discretising the HJB PDEs directly by finite differences, we construct a discrete transition kernel from one-period-ahead conditional expectations. This stabilises evaluation of local controls near switching boundaries and allows the effects of time, state and control grids, interpolation, and quadrature order to be diagnosed separately.

Specifically, we evaluate one-period-ahead conditional expectations directly and use the resulting transition kernel to construct the dynamic backward calculation. Gaussian shocks are integrated by Gauss–Hermite quadrature, and the same transition kernel is used for both backward calculation and forward probability-mass propagation. Numerical solutions are further checked through grid refinement, boundary-mass diagnostics, and consistency between backward moments and forward distributions. This makes it possible to monitor approximation error separately by time, state and control discretisation, interpolation, and quadrature order.

### B.3.1 One-period transition

At date  $t_n$ , state  $x_i$ , and candidate control  $\pi$ , the next-period DC account balance is approximated by Euler discretisation as

$$X_{n+1}^\pi = x_i + (rx_i + c_n + \beta\pi)\Delta t + \sigma\pi\sqrt{\Delta t}Z, \quad Z \sim N(0, 1) \quad (53)$$

where  $c_n = c_{t_n}$ . Rather than estimating the expectation over  $Z$  by random sampling, we use  $K$ -point Gauss–Hermite quadrature.

Let  $(z_k, \omega_k)_{k=1}^K$  be the nodes and weights of the standard Hermite polynomial rule, and define

$$\int_{-\infty}^{\infty} e^{-z^2} f(z) dz \approx \sum_{k=1}^K \omega_k f(z_k) \quad (54)$$

Then the expectation with respect to a standard normal  $Z$  is approximated by

$$\mathbb{E}[f(Z)] = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} f(\sqrt{2}z) dz \approx \sum_{k=1}^K w_k f(\xi_k), \quad (55)$$

where

$$\xi_k = \sqrt{2}z_k, \quad w_k = \frac{\omega_k}{\sqrt{\pi}}, \quad \sum_{k=1}^K w_k = 1 \quad (56)$$

Hence the one-step-ahead candidate states are

$$\hat{x}_{n+1}^{(k)}(x_i, \pi) = x_i + (rx_i + c_n + \beta\pi)\Delta t + \sigma\pi\sqrt{\Delta t}\xi_k, \quad k = 1, \dots, K \quad (57)$$

### B.3.2 Interpolation operator

When the grid values of  $V_{n+1}(x)$  are known,  $\hat{x}_{n+1}^{(k)}(x_i, \pi)$  will generally not lie exactly on the grid. We therefore use a linear interpolation operator  $\mathcal{I}_{n+1}$  and evaluate

$$\mathbb{E}[V_{n+1}(X_{n+1}^\pi) | X_n = x_i] \approx \sum_{k=1}^K w_k (\mathcal{I}_{n+1} V_{n+1})(\hat{x}_{n+1}^{(k)}(x_i, \pi)) \quad (58)$$

The same linear interpolation between neighbouring grid points is used on a non-uniform grid. Shape-preserving monotone interpolation can be used when needed, but higher-order interpolation may destroy HJB monotonicity, so the baseline implementation uses linear interpolation.

## B.4 Embedded Quadratic-Loss HJB for PCMV and DOMV

### B.4.1 Continuous-time partial differential equation

The numerical calculations for PCMV and DOMV start from an embedded quadratic-loss problem with a mean constraint or target parameter. For a representative terminal target  $m$ , solve

$$V(t, x; m) = \inf_{\pi \in \mathcal{A}_{t,x}^X} \mathbb{E}_{t,x} [(X_T + D_T - m)^2] \quad (59)$$

Here  $D_T$  denotes a deterministic terminal benefit at retirement. If  $D_T$  is deterministic, the terminal condition is

$$V(T, x; m) = (x + D_T - m)^2 \quad (60)$$

The formal HJB equation is

$$V_t + \inf_{0 \leq \pi \leq x} \left\{ (rx + c_t + \beta\pi)V_x + \frac{1}{2}\sigma^2\pi^2V_{xx} \right\} = 0, \quad V(T, x) = (x + D_T - m)^2. \quad (61)$$

Define the local Hamiltonian by

$$\mathcal{H}^Q(t, x, \pi; V) = (rx + c_t + \beta\pi)V_x + \frac{1}{2}\sigma^2\pi^2V_{xx} \quad (62)$$

In regions where  $V_{xx} > 0$ , the unconstrained candidate is

$$\pi^{\text{loc}}(t, x) = -\frac{\beta V_x(t, x)}{\sigma^2 V_{xx}(t, x)} \quad (63)$$

and the directly constrained feedback can be diagnosed as

$$\pi^{\text{str}}(t, x) = \text{Proj}_{[0,x]}(\pi^{\text{loc}}(t, x)) \quad (64)$$

In the actual numerical calculation, however, (61) is solved directly by finite differences or the Markov–GH backward calculation.

### B.4.2 Markov–Gauss–Hermite backward calculation

The Gauss–Hermite backward calculation can be written as a semi-Lagrangian approximation to the PDE (61). Set the terminal condition to

$$V_{N_t}(x_i; m) = (x_i + D_T - m)^2 \quad (65)$$

and, for  $n = N_t - 1, \dots, 0$ , compute

$$V_n(x_i; m) = \min_{\pi \in [0, x_i]} \sum_{k=1}^K w_k(\mathcal{J}_{n+1} V_{n+1}(\cdot; m)) \left( \hat{x}_{n+1}^{(k)}(x_i, \pi) \right), \quad n = N_t - 1, \dots, 0 \quad (66)$$

At the same time, store the control attaining the minimum as

$$\pi_n^{\text{QL}}(x_i; m) = \arg \min_{\pi \in [0, x_i]} \sum_{k=1}^K w_k(\mathcal{J}_{n+1} V_{n+1}(\cdot; m)) \left( \hat{x}_{n+1}^{(k)}(x_i, \pi) \right) \quad (67)$$

This  $\pi_n^{\text{QL}}(x_i; m)$  is the directly constrained feedback for the embedded quadratic-loss problem. The superscript QL denotes a quadratic-loss policy and distinguishes it from the second-moment function  $Q$ .

**Remark 7** (Semi-Lagrangian interpretation). Equation (66) is a semi-Lagrangian-type approximation: instead of approximating the diffusion term directly by finite differences, it interpolates the value function along the one-step-ahead transition distribution and takes the conditional expectation. Unlike an implicit finite-difference method that discretises the HJB as a differential operator, this representation makes transitions under each candidate control transparent and incorporates the constraint  $0 \leq \pi \leq x$  naturally. We use this Markov–GH form as the baseline implementation and reuse the same discrete transition kernel for backward calculation and forward distribution propagation.

### B.4.3 PCMV target consistency under the scalarised MV objective

If PCMV is defined by a target mean  $m_0^{\text{tar}}$ , compute under the stored feedback  $\pi^{\text{QL}}(m)$

$$\mu^m(t, x) := \mathbb{E}_{t,x}^{\pi^{\text{QL}}(m)} [X_T + D_T]$$

and search for  $m$  satisfying  $\mu^m(0, x_0) = m_0^{\text{tar}}$ . For numerical experiments that compare PCMV with DOMV under the same variance-aversion coefficient, instead embed the scalarised objective

$$\mathbb{E}[W_T] - \frac{\gamma}{2} \text{Var}(W_T), \quad W_T = X_T + D_T,$$

into quadratic loss. Let the value function for fixed target  $m$  be given by (59); then

$$\sup_{\pi} \left\{ \mathbb{E}[W_T] - \frac{\gamma}{2} \text{Var}(W_T) \right\} = \sup_{m \in \mathbb{R}} \left\{ m - \frac{1}{2\gamma} - \frac{\gamma}{2} V(t, x; m) \right\}. \quad (68)$$

and, at an interior solution,

$$m^*(t, x) = \mu^{m^*}(t, x) + \frac{1}{\gamma} \quad (69)$$

holds. PCMV maximises (68) once at  $(0, x_0)$  and commits through maturity to the fixed-target feedback associated with the resulting  $m_P^*$ .

#### B.4.4 DOMV as rolling target selection

Under DOMV, at each date and state we re-solve the remaining-horizon MV problem

$$\sup_{\pi \in \mathcal{A}_{t,x}^X} \left\{ \mathbb{E}_{t,x}[W_T] - \frac{\gamma_D}{2} \text{Var}_{t,x}(W_T) \right\} \quad (70)$$

Numerically, for each candidate target  $m_\ell$  in  $\mathcal{M} = \{m_1, \dots, m_{N_m}\}$ , we solve the quadratic-loss problem (66) while holding  $m_\ell$  fixed over the entire remaining horizon. Then, at each grid point, select

$$m_D^*(t_n, x_i) \in \arg \max_{m_\ell \in \mathcal{M}} \left\{ m_\ell - \frac{1}{2\gamma_D} - \frac{\gamma_D}{2} V_n(x_i; m_\ell) \right\} \quad (71)$$

and adopt only the first control from the corresponding fixed-target problem,

$$\pi_n^{\text{DOMV}}(x_i) = \pi_n^{\text{QL}}(x_i; m_D^*(t_n, x_i)) \quad (72)$$

At the next date, (71) is re-evaluated from the new state.

The important point is that, while evaluating a candidate  $m_\ell$  in the backward calculation, the target is not switched to another  $m_k$  midway. Switching to a state-dependent target during the backward recursion would no longer evaluate the fixed-target problem  $\mathbb{E}[(W_T - m_\ell)^2]$  and would produce a heuristic different from rolling DOMV. The *target-grid approximation* used here means solving multiple fixed-target problems to completion and only then selecting the first control at the current date and state.

### B.5 Extended-HJB Solver for cTCMV and dTCMV

#### B.5.1 Moment functions

For time-consistent MV, the objective takes the form

$$J(t, x) = M(t, x) - \frac{\Gamma(t, x)}{2} (Q(t, x) - M(t, x)^2) \quad (73)$$

where

$$M(t, x) = \mathbb{E}_{t,x}^{\pi^*}[X_T + D_T], \quad (74)$$

$$Q(t, x) = \mathbb{E}_{t,x}^{\pi^*}[(X_T + D_T)^2] \quad (75)$$

and  $\pi^*$  is the equilibrium feedback rule. For cTCMV,

$$\Gamma(t, x) = \gamma_c \quad (76)$$

is constant, whereas for dTCMV,

$$\Gamma(t, x) = \frac{\rho_d}{x + H_t} \quad (77)$$

Here  $H_t$  is the present value of future contributions and deterministic background pension wealth.

#### B.5.2 Discrete one-step equilibrium condition

At date  $t_n$  and state  $x_i$ , suppose candidate control  $\pi$  is used only from  $t_n$  to  $t_{n+1}$  and the already-computed equilibrium feedback rule is followed thereafter. The one-step-ahead quantities under candidate control  $\pi$  are

$$M_n(x_i; \pi) = \sum_{k=1}^K w_k (J_{n+1} M_{n+1}) (\hat{x}_{n+1}^{(k)}(x_i, \pi)), \quad (78)$$

$$Q_n(x_i; \pi) = \sum_{k=1}^K w_k (J_{n+1} Q_{n+1}) (\hat{x}_{n+1}^{(k)}(x_i, \pi)) \quad (79)$$

Define the discrete-time objective for a one-step deviation as follows.

$$\Psi_n(x_i; \pi) = M_n(x_i; \pi) - \frac{\Gamma(t_n, x_i)}{2} (Q_n(x_i; \pi) - M_n(x_i; \pi)^2) \quad (80)$$

The equilibrium feedback rule is defined by

$$\pi_n^{\text{TC}}(x_i) \in \arg \max_{\pi \in [0, x_i]} \Psi_n(x_i; \pi) \quad (81)$$

After the feedback is selected, update

$$M_n(x_i) = M_n(x_i; \pi_n^{\text{TC}}(x_i)), \quad (82)$$

$$Q_n(x_i) = Q_n(x_i; \pi_n^{\text{TC}}(x_i)), \quad (83)$$

$$J_n(x_i) = M_n(x_i) - \frac{\Gamma(t_n, x_i)}{2} (Q_n(x_i) - M_n(x_i)^2) \quad (84)$$

The terminal conditions are

$$M_{N_t}(x_i) = x_i + D_T, \quad Q_{N_t}(x_i) = (x_i + D_T)^2, \quad J_{N_t}(x_i) = x_i + D_T \quad (85)$$

**Remark 8** (Relation to the continuous-time equilibrium condition). Equation (81) changes only the current control and follows the stored equilibrium policy thereafter. This finite model replaces the continuous-time short-horizon deviation condition with a one-period condition. Gauss–Hermite quadrature evaluates next-period conditional moments instead of constructing the Hamiltonian from derivatives. This correspondence explains the construction of the method; it does not prove general convergence to a continuous-time equilibrium under grid refinement.

#### B.5.3 Equilibrium algorithm for cTCMV and dTCMV

For both equilibrium concepts, set the terminal moments to  $M_{N_t}(x_i) = x_i + D_T$  and  $Q_{N_t}(x_i) = (x_i + D_T)^2$  and work backward. At each node and for each admissible control, use Gauss–Hermite quadrature to evaluate the one-period conditional moments,

maximise the one-step deviation objective, and store both the maximising control and the induced moments. The only structural difference is the variance-aversion coefficient:

$$\Gamma_{n,i} = \gamma_c \quad (\text{cTCMV}), \quad \Gamma_{n,i} = \frac{\rho_d}{x_i + H_{t_n}} \quad (\text{dTCMV}). \quad (86)$$

The two policies use the same backward calculation with different specifications of variance aversion. In dTCMV, the coefficient depends on time and current total pension wealth. Section 8 discusses their practical roles.

#### B.5.4 CP benchmark within the common numerical engine

For CP, no backward HJB problem is solved. Instead, (47) is used directly at each grid date and probability mass is propagated forward with the same Gauss–Hermite transition kernel as for the optimal strategies. Differences between CP and the four MV strategies therefore arise from the feedback rule rather than from random seeds or distribution-evaluation methods. In the common-mean comparison,  $\theta_{cp}$  is calibrated numerically so that expected terminal wealth matches the four MV strategies.

**Remark 9** (No policy iteration in the baseline calculation). The baseline calculation uses Markov–Gauss–Hermite backward recursion and direct search over the admissible interval  $[0, x]$ . Each fixed-target PCMV/DOMV problem is solved by ordinary dynamic programming recursion, while cTCMV/dTCMV directly maximise the one-step-deviation criterion in (81). Howard-type policy iteration for implicit finite-difference equations is therefore not used for the baseline results.

### B.6 Numerical Implementation of Directly Constrained Feedback and Clipped Approximations

The two implementable policies distinguished in Section A.1 are evaluated on the same grid and with the same forward-distribution engine. Directly constrained feedback is obtained from the solution-concept-specific backward problem defined above: PCMV by minimising quadratic loss for the fixed target selected at time zero; DOMV by reselecting the target at each state; and cTCMV/dTCMV by maximising the one-step equilibrium objective. In common notation,

$$\pi_n^{\text{str}}(x_i) = \begin{cases} \pi_n^{\text{QL}}(x_i; m_P^*), & \text{PCMV,} \\ \pi_n^{\text{DOMV}}(x_i), & \text{DOMV,} \\ \pi_n^{\text{TC}}(x_i), & \text{cTCMV and dTCMV,} \end{cases} \quad (87)$$

The difference between cTCMV and dTCMV is represented by their respective  $\Gamma(t_n, x_i)$ . By contrast, the clipped approximation is constructed from the unconstrained  $Y$ -space solution in Section A.1 as

$$\pi_n^{\text{clip}}(x_i) = \text{Proj}_{[0, x_i]}(\pi^{\text{unc}}(t_n, x_i + H_{t_n})) \quad (88)$$

The former recursively evaluates constrained continuation values, whereas the latter truncates only the current control while leaving the unconstrained continuation value unchanged.

Probability mass is then propagated forward independently under each policy. Hence, the projection gap affects not only the control difference at the same state point but also future terminal and rolling distributions through the different balance distributions induced by the two policies.

#### B.6.1 Projection-gap metrics

We measure the difference between directly constrained feedback and the clipped approximation using the following metrics.

$$\Delta\pi_n(x_i) = \pi_n^{\text{str}}(x_i) - \pi_n^{\text{clip}}(x_i), \quad (89)$$

$$\Delta g_n(x_i) = \frac{\pi_n^{\text{str}}(x_i)}{x_i} - \frac{\pi_n^{\text{clip}}(x_i)}{x_i}, \quad x_i > 0 \quad (90)$$

In addition, define the forward-distribution-weighted mean absolute error by

$$\text{PG}_{\text{WMAE}} = \frac{1}{N_t} \sum_{n=0}^{N_t-1} \sum_{i=1}^I p_{n,i}^{\text{str}} |\Delta g_n(x_i)| \quad (91)$$

where  $p_{n,i}^{\text{str}}$  is the forward distribution induced by the directly constrained feedback itself. Thus  $\text{PG}_{\text{WMAE}}$  measures the average projection gap in regions that are actually likely to be reached under the directly constrained benchmark policy.

#### B.6.2 Classification of free-boundary regions

Each node is classified into one of the following three regions.

$$\mathcal{R}_0 = \{(n, i) : \pi_n^{\text{str}}(x_i) = 0\}, \quad (92)$$

$$\mathcal{R}_I = \{(n, i) : 0 < \pi_n^{\text{str}}(x_i) < x_i\}, \quad (93)$$

$$\mathcal{R}_U = \{(n, i) : \pi_n^{\text{str}}(x_i) = x_i\}. \quad (94)$$

$\mathcal{R}_0$  is the no-risky-asset region,  $\mathcal{R}_I$  the interior region, and  $\mathcal{R}_U$  the upper-bound region in which the entire DC balance is invested in the risky asset. Numerically, a free boundary is defined as the boundary between  $\mathcal{R}_I$  and either  $\mathcal{R}_0$  or  $\mathcal{R}_U$ . The numerical analysis distinguishes the free boundaries of the directly constrained feedback from binding states under the clipped approximation.

### B.7 Forward Distribution Propagation without Random Monte Carlo

#### B.7.1 Probability-mass propagation on the wealth grid

Once the feedback  $\pi_n^*(x_i)$  has been obtained by backward calculation, we evaluate the terminal-wealth distribution. Rather than generating a large number of random Monte Carlo paths, we propagate probability mass forward using the Gauss–Hermite transitions.

For a grid-point starting state, set the initial distribution to

$$p_{0,i} = \mathbf{1}_{\{x_i = x_0\}} \quad (95)$$

An off-grid starting point can also be initialised by splitting mass between neighbouring nodes. The baseline results and Appendix E.3, however, use equation (106) to start the first transition from the exact initial balance. After initialisation, given mass  $p_{n,i}$  at date  $n$ , compute for each state  $x_i$  and Gauss–Hermite node  $k$

$$\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) = x_i + (rx_i + c_n + \beta\pi_n^*(x_i))\Delta t + \sigma\pi_n^*(x_i)\sqrt{\Delta t}\xi_k \quad (96)$$

Allocate this point linearly to the next-period grid and add weight  $p_{n,i}w_k$ . Specifically, if  $x_\ell \leq \hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) \leq x_{\ell+1}$ , then

$$p_{n+1,\ell} \leftarrow p_{n+1,\ell} + p_{n,i}w_k \frac{x_{\ell+1} - \hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i))}{x_{\ell+1} - x_\ell}, \quad (97)$$

$$p_{n+1,\ell+1} \leftarrow p_{n+1,\ell+1} + p_{n,i}w_k \frac{\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) - x_\ell}{x_{\ell+1} - x_\ell} \quad (98)$$

Transitions with  $\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) < 0$  are assigned to the lower boundary  $x_0 = 0$ , while transitions with  $\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) > x_{\max}$  are assigned to the upper boundary. Probability mass leaving the grid before boundary assignment is recorded separately in Section B.9. This yields a discrete distribution of  $X_T$  without random sampling. Total terminal pension wealth is  $X_T + D_T$ .

### B.7.2 Moment and tail-risk measures

From the terminal distribution  $p_{N_t,i}$  we compute the following measures.

$$\hat{\mu}_T = \sum_i p_{N_t,i}(x_i + D_T), \quad (99)$$

$$\hat{\sigma}_T^2 = \sum_i p_{N_t,i}\{(x_i + D_T) - \hat{\mu}_T\}^2, \quad (100)$$

$$\hat{m}_{3,T} = \sum_i p_{N_t,i}\{(x_i + D_T) - \hat{\mu}_T\}^3. \quad (101)$$

Quantiles are obtained from the cumulative mass

$$F_{N_t}(x_\ell) = \sum_{i \leq \ell} p_{N_t,i} \quad (102)$$

Let  $W_i = x_i + D_T$ , define  $F_T^W(w) := \sum_{i: W_i \leq w} p_{N_t,i}$  and  $q_\tau := \inf\{w : F_T^W(w) \geq \tau\}$ . Because the forward-propagated distribution is discrete, only the fraction of probability mass at the quantile required to make the tail probability exactly  $\alpha$  is included. The lower-tail LCVaR is approximated by

$$\text{LCVaR}_\alpha = \frac{1}{\alpha} \left[ \sum_{W_i < q_\alpha} p_{N_t,i}W_i + \left( \alpha - \sum_{W_i < q_\alpha} p_{N_t,i} \right) q_\alpha \right] \quad (103)$$

To measure upside participation, we also report the conditional mean of the upper  $\alpha$  tail,

$$\text{UCVaR}_\alpha = \frac{1}{\alpha} \left[ \sum_{W_i > q_{1-\alpha}} p_{N_t,i}W_i + \left( \alpha - \sum_{W_i > q_{1-\alpha}} p_{N_t,i} \right) q_{1-\alpha} \right] \quad (104)$$

### B.7.3 Summary of path-dependent glide paths

The mean glide path is the distributional average of members' investment proportions at each decision date. It can be computed without generating individual trajectories: from the state distribution  $p_{n,i}$  and feedback  $g_{n,i} = \pi_{n,i}^*/x_i$ , for  $n \geq 1$ ,

$$\bar{g}_n = \sum_{i: x_i > 0} p_{n,i} g_n(x_i) \quad (105)$$

The initial balance  $x_0 = 1/12$ , however, is deterministic. If its initial probability mass were split between two state-grid points and the ratios averaged, the convenient convention  $g(0,0) = 0$  at the zero-balance point would artificially depress the initial risky share. We therefore define the initial investment amount as

$$\pi_0 = \text{Interp}_x\{x_i g_0(x_i)\}(x_0), \quad \bar{g}_0 = \frac{\pi_0}{x_0} \quad (106)$$

and propagate the first period directly from the exact initial state  $x_0$ , allocating the post-transition probability mass to the state grid. In other words, we interpolate the amount invested  $\pi = xg$ , rather than the ratio  $g$ , preventing the convenient zero-balance convention from contaminating the initial glide path.

Controls are defined only on intervals  $[t_n, t_{n+1})$  for  $n = 0, \dots, N_t - 1$ ; there is no new investment decision at the terminal date  $T = t_{N_t}$ . Glide-path figures therefore show only the decision dates  $t_0, \dots, t_{N_t-1} = T - \Delta t$  and do not fill the undefined terminal control with zero. Mechanically appending a zero terminal control would create a discontinuity that could be mistaken for de-risking immediately before retirement. The distribution  $p_{n,i}$  also gives the median, 10th percentile, and 90th percentile of the investment proportion at each date. These are pointwise cross-sectional quantiles, not representative trajectories followed by a single member throughout the horizon.

## B.8 Rolling Conditional Evaluation

### B.8.1 Definition of the rolling conditional outcome profile

Let  $W_T^{j;t,x}$  denote terminal wealth obtained by continuing the stored feedback for strategy  $j$  from date  $t$  and current balance  $x$  to maturity. Rather than reducing the remaining-horizon outcome to a single mean–variance value, we represent it by the following vector of measures:

$$\begin{aligned} \mathbf{O}_j(t, x; \alpha) = & (\mu_j(t, x), s_j(t, x), q_{j,\alpha}(t, x), q_{j,0.50}(t, x), q_{j,1-\alpha}(t, x), \\ & \text{LCVaR}_{j,\alpha}(t, x), \text{UCVaR}_{j,\alpha}(t, x), \text{Skew}_j(t, x), g_j(t, x), \pi_j(t, x)), \end{aligned} \quad (107)$$

where  $\mu_j$  and  $s_j$  are the conditional mean and standard deviation of  $W_T^{j;t,x}$ ,  $q_{j,\alpha}$  is a conditional quantile, and  $g_j$  and  $\pi_j = xg_j$  are the current risky-asset share and investment amount. We refer to (107) as the *rolling conditional outcome profile*. It jointly

presents expected outcomes, upper and lower tail risk, distributional asymmetry, and current investment behaviour from the same conditional distribution.

### B.8.2 Positioning through the laws of total expectation and total variance

Let  $X_t^j$  denote the balance at date  $t$  generated under the stored feedback for strategy  $j$ , and let terminal wealth be  $W_T^j = X_T^j + D_T$ . Define the state-fixed conditional mean and conditional variance by

$$\mu_j(t, x) = \mathbb{E}[W_T^j | X_t^j = x], \quad v_j(t, x) = \text{Var}(W_T^j | X_t^j = x) \quad (108)$$

Under Markov feedback,  $\mathbb{E}[W_T^j | \mathcal{F}_t] = \mu_j(t, X_t^j)$ .

**Proposition 1** (Decomposition of time-zero and rolling conditional evaluation). *For any  $t \in [0, T]$ , if the required second moments are finite,*

$$\mathbb{E}[W_T^j] = \mathbb{E}[\mu_j(t, X_t^j)], \quad (109)$$

$$\text{Var}(W_T^j) = \mathbb{E}[v_j(t, X_t^j)] + \text{Var}(\mu_j(t, X_t^j)) \quad (110)$$

hold.

*Proof.* Equation (109) follows from the law of iterated expectations. Further, decompose

$$W_T^j - \mathbb{E}[W_T^j] = \{W_T^j - \mu_j(t, X_t^j)\} + \{\mu_j(t, X_t^j) - \mathbb{E}[\mu_j(t, X_t^j)]\}$$

and condition on  $X_t^j$ ; the conditional expectation of the cross term is zero, which yields (110).  $\square$

The first term in (110) is the average conditional variance remaining from intermediate balances; the second is dispersion in expected terminal wealth across those balances. With a deterministic initial balance the second term is zero at time zero, but total variance at an intermediate date includes both terms. Conditional evaluation fixes the current balance and extracts the distribution from that state onward. This relationship applies to PCMV, DOMV, cTCMV, dTCMV, and CP; Section 8 discusses their practical roles.

We distinguish two visualisations. The first uses the quantiles  $q_{t,k}^j$  of the balance distribution generated by each strategy itself as starting states,

$$\mathbf{O}_j(t_n, q_{n,\tau}^j; \alpha), \quad \tau \in \{0.10, 0.50, 0.90\} \quad (111)$$

and shows outcomes from low, median, and high balances generated by each policy itself. The second design continues each policy from a common date and balance. It fixes the current state, rather than the entire past trajectory.

### B.8.3 Conditional starting states

Starting states can be specified as common dates and balances or selected as quantiles of each policy's intermediate balance distribution. Representative own-path starting states at date  $t_n$  are

$$q_{n,0.10}^j, \quad q_{n,0.50}^j, \quad q_{n,0.90}^j \quad (112)$$

These correspond to low-, median-, and high-balance members.

### B.8.4 Conditional remaining-horizon distributions

Treat date  $t_n$  and state  $x_i$  as a new initial condition and propagate the distribution from  $t_n$  to  $T$  using the stored feedback. Define the conditional terminal distribution function for strategy  $j$  by

$$F_j(z | t_n, x_i) := \Pr(X_T^j + D_T \leq z | X_{t_n}^j = x_i), \quad z \in \mathbb{R} \quad (113)$$

From this conditional distribution we compute the conditional mean, conditional standard deviation, conditional quantiles, conditional LCVaR, conditional UCVaR, and conditional third central moment.

Common-state evaluation holds each policy's initial calibration coefficients fixed and continues the stored policy from the same date and balance. Expected terminal wealth is not rematched at intermediate states. The results therefore describe outcome combinations under different preference specifications and decision principles; they do not isolate a welfare effect caused solely by adding total-wealth dependence.

### B.8.5 Starting states used in the main text

Section 7 and Table 14 specify three common states: 20 years/balance 20, 30 years/balance 45, and 35 years/balance 30. All five policies are evaluated with independent one-million-path Euler–Monte Carlo simulation. These states are presented as specified common states, not as dTCMV quantiles selected by equation (112). The main text examines cTCMV and dTCMV in detail; Appendix F gives the all-policy summary and reproduction settings.

## B.9 Numerical Diagnostics

### B.9.1 Moment consistency

For cTCMV/dTCMV, the backward moments  $M_0, Q_0$  must agree with the terminal mean and second moment obtained from forward distribution propagation. We therefore compute

$$\varepsilon_M = \left| M_0(x_0) - \sum_i p_{N_t, i}(x_i + D_T) \right|, \quad (114)$$

$$\varepsilon_Q = \left| Q_0(x_0) - \sum_i p_{N_t, i}(x_i + D_T)^2 \right| \quad (115)$$

These are basic diagnostics for consistency between backward and forward evaluation. Current backward–forward moment and mass-preservation diagnostics are stored at `results/validation/recalibration/checks.json`.

### B.9.2 Probability-mass conservation and boundary aggregation

To avoid masking errors through normalisation during forward distribution propagation, we record the pre-normalisation mass at each period. Let  $\tilde{p}_{n+1,\ell}$  be the next-period pre-normalisation mass obtained from normalised mass  $p_{n,i}$  at date  $t_n$ , and set

$$P_{n+1}^{\text{tot}} = \sum_{\ell} \tilde{p}_{n+1,\ell} \quad (116)$$

As a numerical diagnostic, only after storing

$$\varepsilon_p(n+1) = |1 - P_{n+1}^{\text{tot}}| \quad (117)$$

do we normalise by  $p_{n+1,\ell} = \tilde{p}_{n+1,\ell} / P_{n+1}^{\text{tot}}$ . This checks whether the transition kernel and mass-allocation procedure conserve probability before normalisation, rather than merely verifying that the normalised total equals one.

In addition, using the candidate point before boundary assignment,  $\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i))$ , define the upper- and lower-bound overflow masses by

$$e_n^{\text{up}} = \sum_{i,k} p_{n,i} w_k \mathbf{1}_{\{\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) > x_{\max}\}}, \quad (118)$$

$$e_n^{\text{low}} = \sum_{i,k} p_{n,i} w_k \mathbf{1}_{\{\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) < 0\}} \quad (119)$$

We separately record the maximum by date, cumulative flow over the horizon, and terminal boundary mass. The cumulative flow  $\sum_n e_n^{\text{up}}$  is not itself an overflow probability because the same mass can reach the boundary in multiple periods. The current baseline ceiling is 600. Boundary-arrival mass and ceiling sensitivity are examined separately; Appendix E.4 reports dTCMV recalibration with a ceiling of 900. Current mass and boundary diagnostics are stored at `results/validation/recalibration/checks.json`, and sensitivity to the state ceiling, grids, and quadrature order at `results/validation/recalibration/sensitivity.csv`.

## C Supplementary Sensitivity Analysis of Glide Paths

### C.1 Positioning of the Sensitivity Analysis

This appendix screens changes in mean-glide-path shape under alternative market and contribution assumptions. It uses 80 periods over 40 years, a non-uniform 101-point state grid, nine candidate controls, and three-point Gauss–Hermite quadrature, holding variance-aversion and other coefficients at their baseline values. Expected terminal wealth is not rematched for each scenario, so both numerical resolution and comparison conditions differ from the monthly recalibrated comparison in the main text. The figures describe sensitivity of policy shape; they do not establish outcome rankings at a common mean or the accuracy of tail-risk reproduction.

Each colour represents the same assumption scenario. Solid lines are directly constrained policies with the constraint  $0 \leq \pi \leq x$  incorporated into the backward calculation; dotted lines are clipped approximations obtained by ex-post projection of the unconstrained policy onto the same interval. Mean glide paths weight risky-asset shares by the balance distribution generated by each policy itself. Differences between the solid and dotted curves therefore reflect both state-by-state investment decisions and the balance distributions accumulated up to that date. The dTCMV clipped approximation uses the current unconstrained coefficients adopted in this paper.

### C.2 Sensitivity to Market Parameters and Contribution Assumptions

**Deterministic terminal benefit.** Figure 5 compares  $D_T = 0$  with a deterministic benefit equal to the terminal value obtained by accumulating contributions in the risk-free asset. The curves for PCMV, DOMV, and cTCMV are almost unchanged, whereas the shape of dTCMV changes because its variance aversion depends on total pension wealth. The effect of deterministic benefits therefore differs across solution concepts.

**Risk-free rate.** Figure 6 compares  $r = 0.005, 0.015, 0.025$ . Higher risk-free rates generally reduce the mean risky-asset share under all four policies. Under dTCMV, however, the mid-horizon trough and the increase before retirement shift at the same time, so the change is not a uniform parallel movement.

**Expected risky-asset return.** Figure 7 compares  $\mu = 0.045, 0.055, 0.065$ . For PCMV, DOMV, and cTCMV, the cases with higher expected returns generally have lower mean risky-asset shares from the middle of the horizon onward. These are averages under the balance distribution generated by each policy and should be distinguished from comparisons of investment decisions at the same balance. Under dTCMV, the response from the beginning through the middle of the horizon is not monotone, and the pre-retirement increase also changes. A change in expected excess return therefore affects both the average amount of risk and the timing of risk-taking.

**Risky-asset volatility.** Figure 8 compares  $\sigma = 0.14, 0.18, 0.22$ . Under PCMV, DOMV, and cTCMV, higher volatility lowers the mean risky-asset share. dTCMV also moves broadly toward lower risky-asset exposure, but at low volatility the curve develops numerical kinks and its shape is less stable than for the other three policies. Interpreting such local shapes as economic thresholds would require recomputation on finer time, state, and control grids.

**Contribution profile.** Figure 9 holds total contributions fixed and compares constant, linearly increasing, and quadratically increasing contributions. Shifting contributions toward later dates changes the relation between future contributions and the current balance and generally raises the mean risky-asset share from the beginning through the middle of the horizon under all four policies. Glide paths therefore depend not only on the total contribution amount but also on the timing of contributions relative to the current balance.

Mean glide paths change endogenously with market assumptions, future contributions, background wealth and benefits, and current state. Solid and dotted lines are generally close under cTCMV, while the gap can widen under other solution concepts depending on the assumption and the date. The extent to which the clipped approximation reproduces the directly constrained policy therefore depends on both the solution concept and the state distribution generated by each policy.

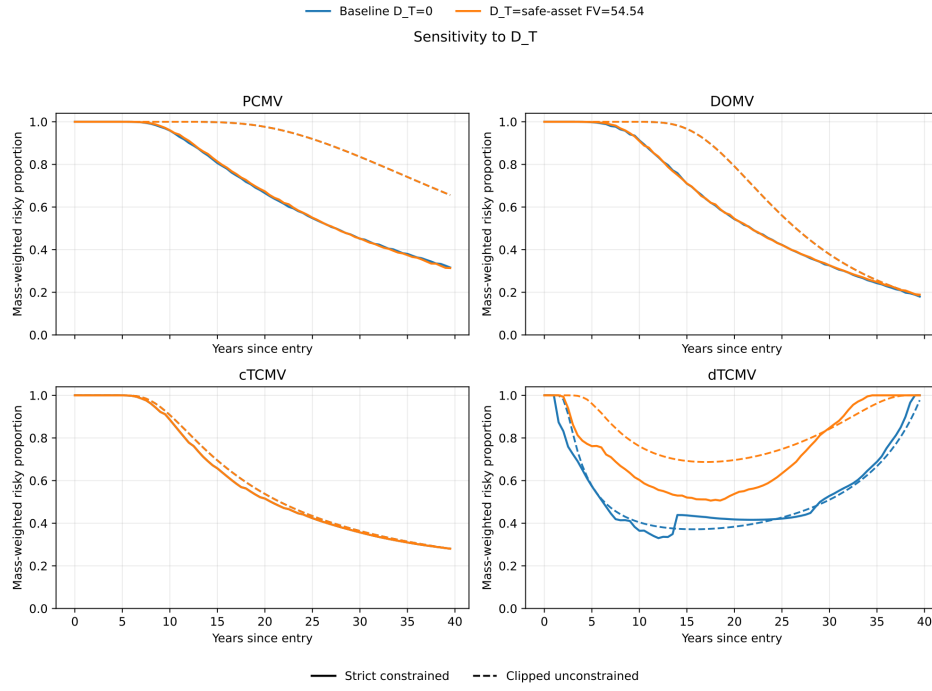


Figure 5: Sensitivity of mean glide paths to the deterministic terminal benefit  $D_T$ . Solid lines show directly constrained policies; dotted lines show clipped approximations.

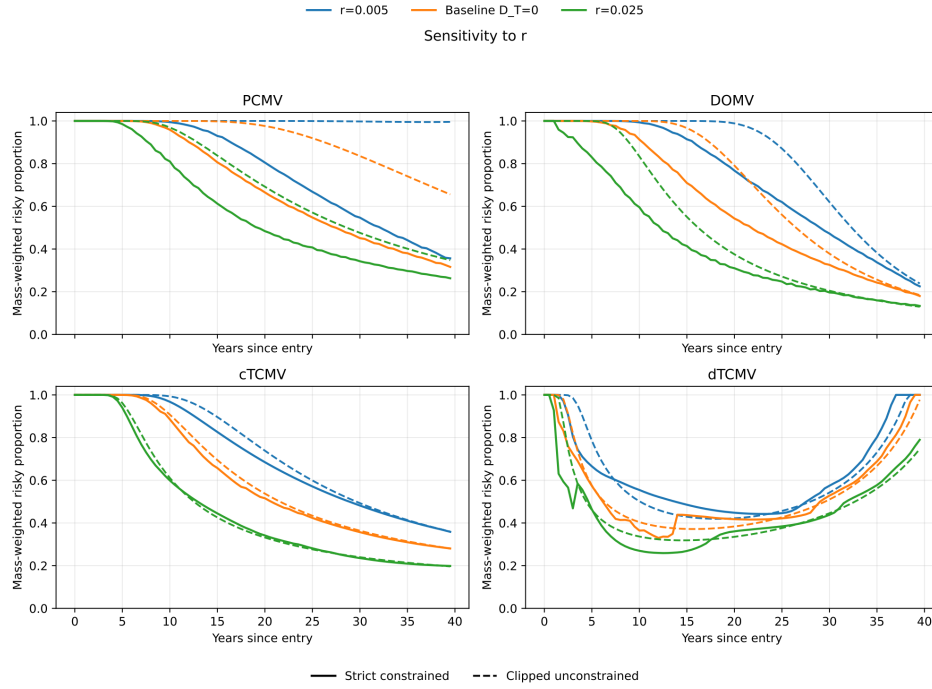


Figure 6: Sensitivity of mean glide paths to the risk-free rate  $r$ . Solid lines show directly constrained policies; dotted lines show clipped approximations.

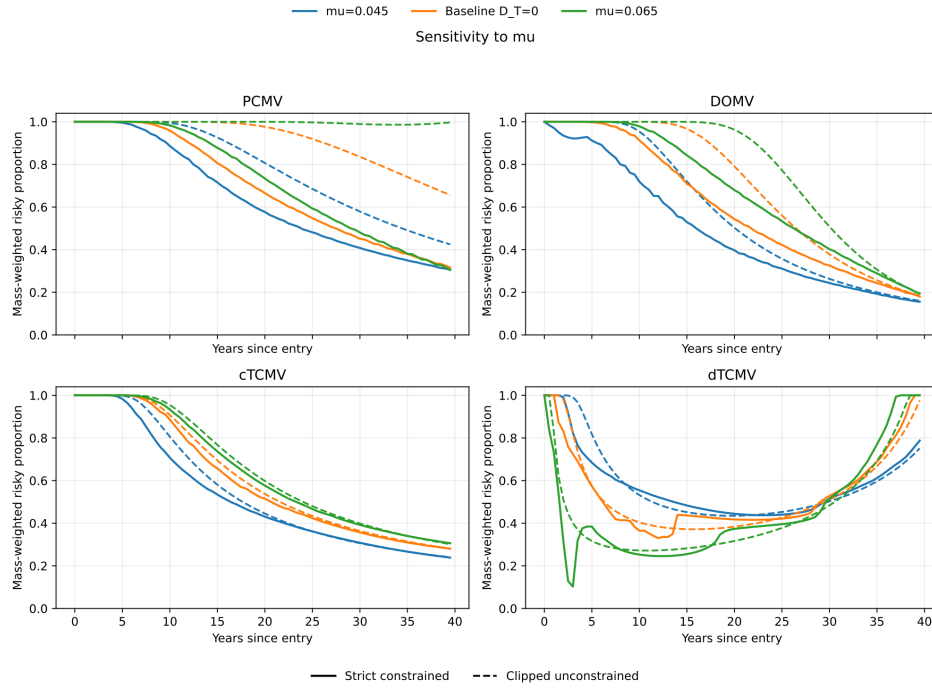


Figure 7: Sensitivity of mean glide paths to the expected risky-asset return  $\mu$ . Solid lines show directly constrained policies; dotted lines show clipped approximations.

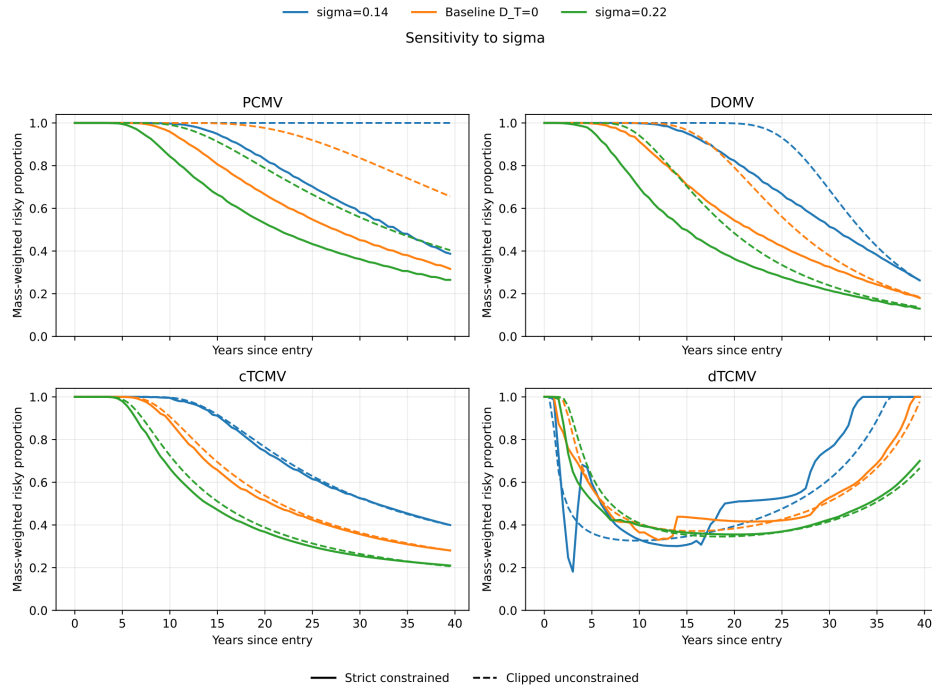


Figure 8: Sensitivity of mean glide paths to risky-asset volatility  $\sigma$ . Solid lines show directly constrained policies; dotted lines show clipped approximations.

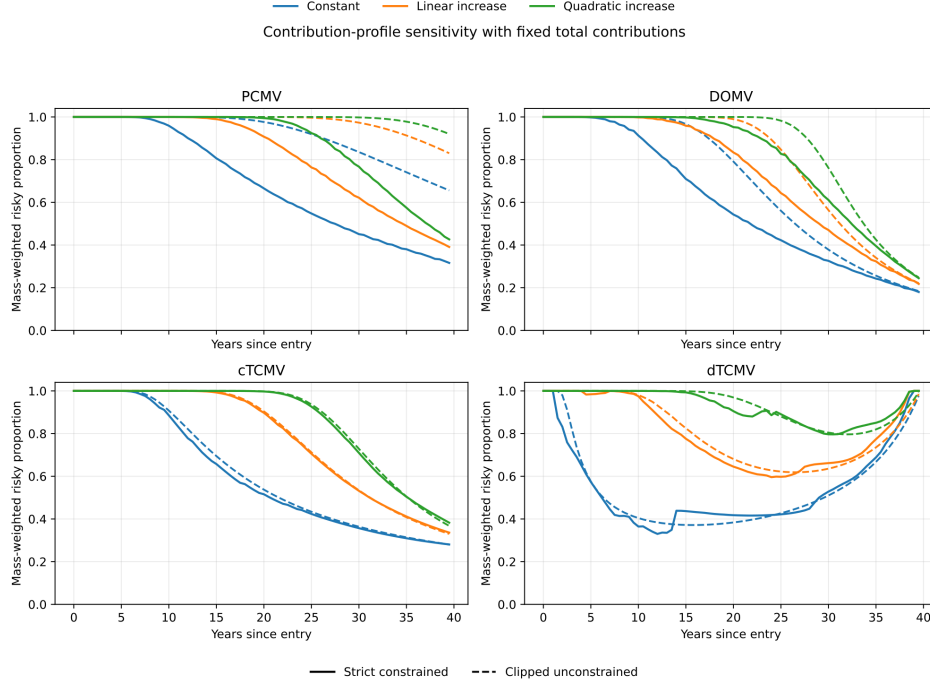


Figure 9: Sensitivity of mean glide paths to the contribution profile with total contributions held fixed. Solid lines show directly constrained policies; dotted lines show clipped approximations.

## D Extension to Mean–Variance–Skewness

This appendix presents exploratory MVS analysis and supplementary figures. Because the objective uses the third central moment rather than standardised skewness, the effect of a positive coefficient depends on scale and discretisation. We therefore report coefficient sensitivity and interpret it separately from the main-text MV comparison calibrated to a common mean.

### D.1 Criterion, Nesting within MV, and the Position of cTCMVS

To incorporate upside asymmetry of terminal wealth explicitly into the objective, consider

$$J^{\text{MVS}} = \mathbb{E}[W_T] - \frac{\gamma}{2} \text{Var}(W_T) + \frac{\eta}{3} \mathbb{E}[(W_T - \mathbb{E}[W_T])^3] \quad (120)$$

PCMVS commits at time zero to the mean, variance, and third central moment, while DOMVS re-solves the remaining-horizon problem at each state. dTCMVS treats total-pension-wealth-dependent variance aversion and skewness preference within an equilibrium among future selves. Thus MVS is not a single common model simply appended to all four solution concepts; it is a higher-moment extension that preserves each concept's original intertemporal decision principle.

**Proposition 2** (Reduction to MV when skewness preference is zero). *Under the same market model, constraints, variance-aversion coefficient, initial condition, and solution concept, setting  $\eta = 0$  makes (120) identical to the corresponding MV criterion. The sets of optimal/equilibrium policies therefore coincide; under uniqueness or the same tie-breaking rule, the feedback, glide path, and terminal distribution coincide as well.*

In an independently implemented small validation calculation with  $(N_t, N_x, N_a, K) = (8, 31, 7, 3)$ , the maximum policy difference between MV and MVS at  $\eta = 0$  was zero to machine precision for PCMV, DOMV, cTCMV, and dTCMV. This regression test confirms that the MVS code reduces to MV when the skewness coefficient is zero. Positive skewness coefficients are considered in the exploratory calculations below.

Applying the closed-form result of Mu et al. (2019) for a constant-coefficient unconstrained Black–Scholes problem to the constant-variance-aversion specification used as the first-layer benchmark here, the unconstrained equilibrium risky investment under cTCMVS coincides with the corresponding cTCMV policy. Hence the clipped approximations obtained by ex-post projection of this common unconstrained solution onto the same interval  $[0, x]$  also coincide. In the unconstrained and clipped cases, adding a positive skewness preference therefore does not change cTCMVS risky investment relative to cTCMV. In the directly constrained problem, the probability of constraint binding and conditional higher moments depend on current wealth, so this cancellation cannot be carried over directly. Characterising the extended-HJB system, equality conditions, and free boundaries for directly constrained cTCMVS with a positive skewness coefficient is left for future work.

### D.2 Interpretation of MVS by Solution Concept and Coefficient Sensitivity of dTCMVS

The meaning of the MVS extension differs substantially by solution concept. For cTCMVS, the equality in the previous subsection holds for the unconstrained solution and clipped approximation, while equality in the directly constrained problem remains unproved. By contrast, in dTCMVS, where variance aversion and skewness preference vary with total pension wealth, the coefficients can alter not only the level of risky-asset exposure but also the *timing of risk-taking*. Because this relates to allocation timing in the four-policy comparison, the coefficient-sensitivity figure is reproduced here. The same image is also retained as historical material at [archive/papers/v14\\_7/figs/fig\\_dtcvm\\_mvs\\_fixed\\_gamma\\_glidepaths\\_refined\\_ja.png](#); material under [archive/](#) is not part of the current authoritative results.

In Figure 10,  $\eta_0 = 0$  corresponds to dTCMV. Increasing the skewness coefficient does not uniformly raise the risky-asset share over the entire horizon. At low coefficient values, the policy leaves the upper constraint later in the early accumulation phase and the mid-horizon trough becomes shallower; at higher values, a high risky-asset share is maintained for a longer period before converging towards its pre-retirement level. Thus, the effect of skewness preference appears not as a uniform increase in risk-taking, but as a reorganisation of the glide path that reallocates the risk budget across time in order to create

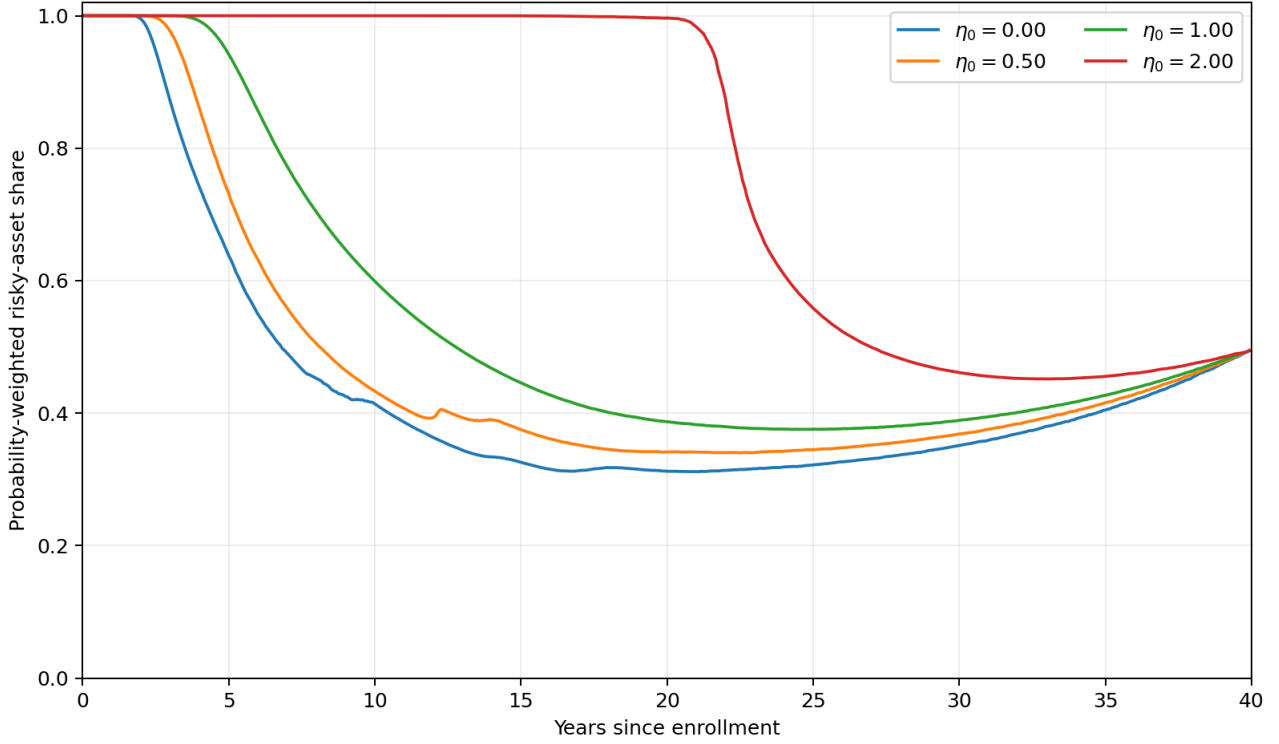


Figure 10: dTCMVS glide paths by coefficient. Variance aversion is fixed at  $\gamma_0 = 2.5$ , while the skewness coefficient is  $\eta_0 = 0, 0.5, 1.0, 2.0$ . Each curve is the risky-asset share weighted by the forward distribution generated by the corresponding policy.

a longer right tail. Because variance aversion is held fixed, the figure directly shows the sensitivity of policy shape to  $\eta_0$  alone. Mean terminal wealth is not matched across coefficients, however, so the figure should not be interpreted as a ranking of outcomes.

For each of these fixed- $\gamma_0$  policies, the terminal distribution is constructed using the same Gauss–Hermite forward mass propagation. The coefficient-sensitivity calculations use monthly time steps, five-point Gauss–Hermite quadrature,  $N_x = 3601$ ,  $N_a = 97$ , and  $x_{\max} = 900$ . Table 8 and Figure 11 report the resulting terminal distributions. The density panel is a smoothed visualisation of the discrete terminal probability mass, whereas the CDF and the statistics in the table are calculated from the unsmoothed probability mass.

Table 8: Terminal-distribution statistics generated by dTCMVS glide paths with fixed  $\gamma_0 = 2.5$  ( $N_x = 3601$ ,  $N_a = 97$ ,  $x_{\max} = 900$ )

| $\eta_0$ | Mean   | Standard deviation | Skewness | q01   | q05   | Median | q95    | LCVaR <sub>5%</sub> | UCVaR <sub>5%</sub> |
|----------|--------|--------------------|----------|-------|-------|--------|--------|---------------------|---------------------|
| 0.00     | 77.65  | 22.05              | 0.865    | 38.98 | 47.16 | 74.72  | 118.15 | 42.21               | 133.31              |
| 0.50     | 79.41  | 23.69              | 0.914    | 38.48 | 47.03 | 76.14  | 123.05 | 41.84               | 139.75              |
| 1.00     | 82.82  | 26.89              | 1.003    | 37.75 | 46.76 | 78.86  | 132.67 | 41.24               | 152.38              |
| 2.00     | 104.24 | 60.08              | 2.294    | 31.62 | 41.98 | 89.21  | 217.53 | 35.64               | 286.49              |

The q05 values for  $\eta_0 = 0, 0.5, 1, 2$  are 47.16, 47.03, 46.76, and 41.98, respectively. As  $\eta_0$  increases from 0 to 2, the mean rises from 77.65 to 104.24 and q95 from 118.15 to 217.53, while q01 falls from 38.98 to 31.62 and LCVaR<sub>5%</sub> from 42.21 to 35.64. Upper-tail UCVaR<sub>5%</sub> rises from 133.31 to 286.49. Positive skewness preference therefore substantially expands upside participation, but at high coefficient values it also worsens the lower tail. Because this non-concave higher-moment extension requires further checks of sensitivity to the state and control searches, the fixed- $\gamma_0$  coefficient sensitivity is not interpreted as a welfare ranking at a common mean or as a recommended calibration.

Because of page constraints, detailed numerical results for positive skewness are reported only for dTCMVS; the other MVS solution concepts are retained at the level of positioning and exploratory summaries. Even this limited experiment is informative. For  $\eta_0 = 0.5$  and 1.0, q05 remains close to its dTCMV value (47.16, 47.03, and 46.76 for  $\eta_0 = 0, 0.5, 1.0$ ), while the median rises from 74.72 to 76.14 and 78.86 and q95 from 118.15 to 123.05 and 132.67. At the same time, the mid-horizon trough in the U-shaped dTCMV glide path becomes shallower, suggesting that skewness preference can shift the policy toward a more conventional TDF-like profile that may be easier to communicate and accept in practice. The means are not matched across coefficients and non-concavity and grid sensitivity remain, so this is neither a welfare ranking nor a recommended calibration. Nevertheless, linking higher moments to member preferences may allow upside participation and the timing of risk-taking to be designed jointly, indicating a promising research direction for future personalised investment.

Some of the small fluctuations visible in the figure may reflect switches in the selected control caused by global search over the control grid, state interpolation, and small changes in the forward distribution under a non-concave local objective. The values  $\eta_0 = 0.5, 1.0, 2.0$  are exploratory points chosen to examine coefficient sensitivity of the policy shape.

PCMVS and DOMVS were also computed and are treated at this stage as exploratory results, separate from the operational proposal in the main text. At small skewness coefficients they remain close to their MV counterparts, whereas above a certain coefficient range numerical branches appear, particularly for PCMVS, with abrupt shifts in the timing of risk-taking. Changes in DOMVS are comparatively smaller, but the late-horizon policy is sensitive to target search, the Lagrange multiplier, the control grid, and interpolation. Related exploratory assets are retained as historical material under [archive/legacy\\_v5/resu](#)

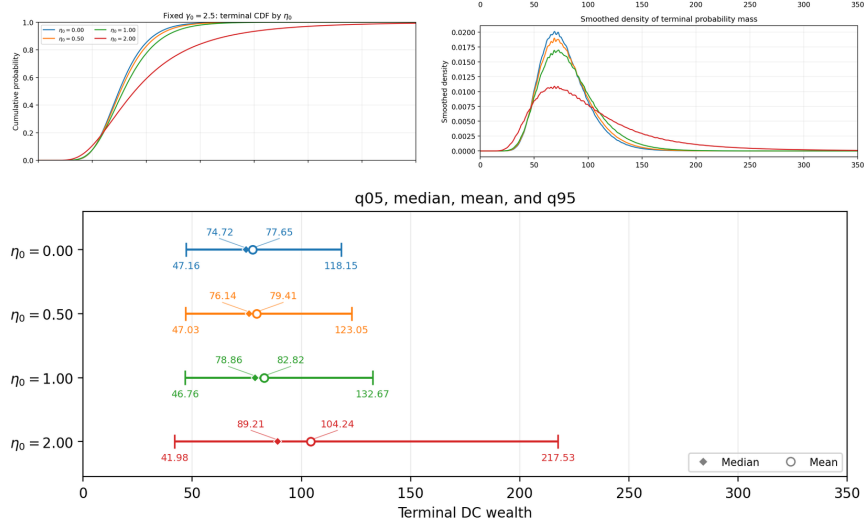


Figure 11: Terminal distributions generated by dTCMVS glide paths with fixed  $\gamma_0 = 2.5$ . The upper-left panel shows the CDF, the upper-right panel a smoothed representation of terminal probability mass, and the bottom panel places q05, median, mean, and q95 directly at their wealth locations.

lts/ and [archive/legacy\\_v5/supplementary/figures/](#), separate from the current principal results; additional validation would be required before member-facing use.

Table 9: Positioning of the MVS extensions by solution concept

| Extension | What is established in this paper                                                                                                                                                                                                                                                                                             | Practical positioning                                                                                                                                          |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cTCMVS    | In the constant-coefficient unconstrained Black–Scholes setting of <a href="#">Mu et al. (2019)</a> , it gives the same equilibrium risky investment as cTCMV, and the clipped approximation constructed from the common unconstrained solution also coincides. The directly constrained problem is left for future research. | The equality is treated as established for the unconstrained solution and clipped approximation; directly constrained cTCMVS remains an exploratory extension. |
| dTCMVS    | Even with $\gamma_0$ fixed, the timing structure of risk-taking changes materially with $\eta_0$ .                                                                                                                                                                                                                            | Reported in this appendix as a glide-path study, but treated as an exploratory result with substantial coefficient sensitivity.                                |
| PCMVS     | Close to MV at small coefficients, but numerical branching and abrupt policy changes emerge as the coefficient increases.                                                                                                                                                                                                     | Related historical exploratory material is retained under <a href="#">archive/legacy_v5/</a> .                                                                 |
| DOMVS     | Changes are smaller than for PCMVS, but the late-horizon policy is sensitive to search and discretisation.                                                                                                                                                                                                                    | Related historical exploratory material is retained under <a href="#">archive/legacy_v5/</a> .                                                                 |

Taken together, the principal research issue raised by MVS is that *higher-moment objectives can alter the timing structure of a glide path nonlinearly and discontinuously*. This paper records, as exploratory evidence, the equality of the cTCMVS unconstrained solution and clipped approximation together with the unresolved directly constrained problem, coefficient-specific dTCMVS glide paths, and the sensitivity of PCMVS and DOMVS. Practical application would require a method for mapping coefficients to retirement-income objectives or preferences, cross-validation of global solutions to the non-concave problem, grid convergence, and confirmation of benefits through conditional evaluation and downside-risk constraints.

## E Independent Validation and Robustness of the Numerical Results

### E.1 Validation design: internal MGH evaluation and independent evaluation

This appendix validates the policies and evaluators used in the current analysis along four dimensions: internal consistency within the finite Markov model, agreement with an independent evaluator, robustness to grid, boundary, and discretisation choices, and consistency with an external benchmark. Optimisation and equilibrium calculations are performed by MGH backward induction, while MGH forward propagation produces terminal and conditional distributions within the same finite Markov model. In the independent Euler–Monte Carlo evaluation, the saved policy is interpolated in the state dimension and applied to a separate monthly Euler process using independent standard-normal shocks; the outcome distribution is evaluated without re-optimising the policy.

First, we verify that the saved policy satisfies  $0 \leq \pi \leq x$  at every grid point. We then check whether the Bellman minimisation for a fixed target, the DOMV comparison across candidate targets, or the one-step-deviation objective for TCMV is consistent with the optimisation or equilibrium condition within the adopted candidate set. Finally, evaluation on alternative state grids or under independent normal shocks is used to examine the sensitivity of the attained mean and distributional statistics. Feasibility, optimality or equilibrium within the finite model, and approximation accuracy relative to

an independent evaluator are therefore treated as separate diagnostics.

The common transition kernel establishes internal consistency within the finite Markov model, while approximation accuracy relative to a monthly Euler process is assessed independently. When the same transition kernel is used for both backward and forward calculations, internal consistency alone cannot independently detect approximation error arising from the state grid, interpolation, boundary treatment, or related numerical choices.

For an off-grid value  $z \in [\ell, u]$ , probability mass is allocated to the two endpoints so as to preserve the mean. Let the post-allocation random value be  $Z$ . Then

$$\Pr(Z = u) = \frac{z - \ell}{u - \ell}, \quad \mathbb{E}[Z] = z, \quad \text{Var}(Z) = (z - \ell)(u - z). \quad (121)$$

Even when the first moment is preserved, the second moment contains an additional term. When the same kernel is used in forward and backward calculations, both remain internally consistent with this added variance. Internal consistency alone therefore cannot reveal grid error. Because the upper boundary may also affect the tail, we combine independent Monte Carlo evaluation with sensitivity checks on the state ceiling, state grid, control grid, and quadrature order.

## E.2 Re-optimisation and recalibration of the current policies

The principal policies in the main text are obtained under fixed market parameters and monthly decision frequency on the state and control grids in Table 3, and are calibrated to the common expected terminal wealth of 84.78.

Policy computation uses a state ceiling of 600, 6001 state points (spacing 0.1), 129 candidate risky-asset proportions, and seven-point GH quadrature. The DOMV target grid runs from 0 to 650 in increments of 2. The horizon comprises 480 monthly periods, with  $r = 0.015$ ,  $\beta = 0.04$ ,  $\sigma = 0.18$ , annual contribution 1, initial balance  $1/12$ , and  $D_T = 0$ . Calibration is evaluated on a separate state spacing of 0.025 with 31-point GH quadrature; the maximum deviation from expected terminal wealth 84.78 is 0.003547. Numerical checks of optimality and equilibrium are carried out on the adopted finite model.

## E.3 MGH forward evaluation versus independent Monte Carlo evaluation

### E.3.1 Purpose of the evaluator comparison

This subsection takes the saved risky-investment arrays of the five current policies as common inputs and compares terminal distributions and mean glide paths obtained from MGH forward mass propagation and an independent one-million-path Euler–Monte Carlo evaluation. No policy is re-optimised, no coefficient is recalibrated, and no market or contribution assumption is changed. MGH uses the same state ceiling of 600, 6001 state points, state spacing 0.1, seven-point GH quadrature, and 480 monthly periods as the policy calculation. This is distinct from the separate calibration-evaluation grid (spacing 0.025, GH31). The initial state follows equation (106), with the first one-period transition starting from the exact  $x_0 = 1/12$ . Because `results/validation/recalibration/checks.json` split the initial mass across two grid points, its initialisation differs from the forward values reported here.

For MC, we use `results/current/independent_mc.npz`, which is the source of Table 5 and Figure 2 in the main text. The random seed is 20260912; there are 100 blocks, with each block seed equal to 20260912 plus a multiple of 1009; there are one million paths, common random numbers across policies, and no upper truncation. Saved risky-investment amounts are linearly interpolated, and beyond the upper state boundary the policy continues at the risky-asset proportion at the boundary. The principal statistics are cross-checked against `results/current/independent_mc.csv`. Regeneration with the same seeds yields a maximum pathwise difference of  $5.90 \times 10^{-9}$  from the saved terminal values. The principal values in the main text use the saved independent-MC evaluations, and SHA-256 hashes of the policies, coefficients, and MC data are retained in the validation records. Current evaluator-comparison outputs are under `results/validation/evaluator_comparison/`: terminal statistics in `results/validation/evaluator_comparison/evaluator_distribution_comparison.csv`, mean-glide-path differences in `results/validation/evaluator_comparison/evaluator_glidepath_comparison.csv`, and the overall summary in `results/validation/evaluator_comparison/evaluator_report.md`.

### E.3.2 Comparison of retirement-wealth distributions

Figure 12 compares the terminal CDFs obtained from MGH forward propagation and an independent one-million-path Euler–Monte Carlo evaluation under the same saved policies. The difference between the full distributions is measured by

$$D_{\text{KS}} = \sup_z |F_{\text{MGH}}(z) - F_{\text{MC}}(z)|,$$

which is used here as a distance between evaluators rather than as a statistical test.  $D_{\text{KS}}$  is largest for PCMV at 0.008514 and ranges from 0.001257 to 0.002079 for the other four policies, indicating broad agreement between the two evaluators. Because the PCMV terminal distribution is concentrated in a narrow region, the stepwise shape of the MGH discrete CDF on a grid of width 0.1 may contribute to local CDF differences. Figure 12 therefore also includes an enlarged view of the upper concentration region. Summary statistics are reported in Table 10.

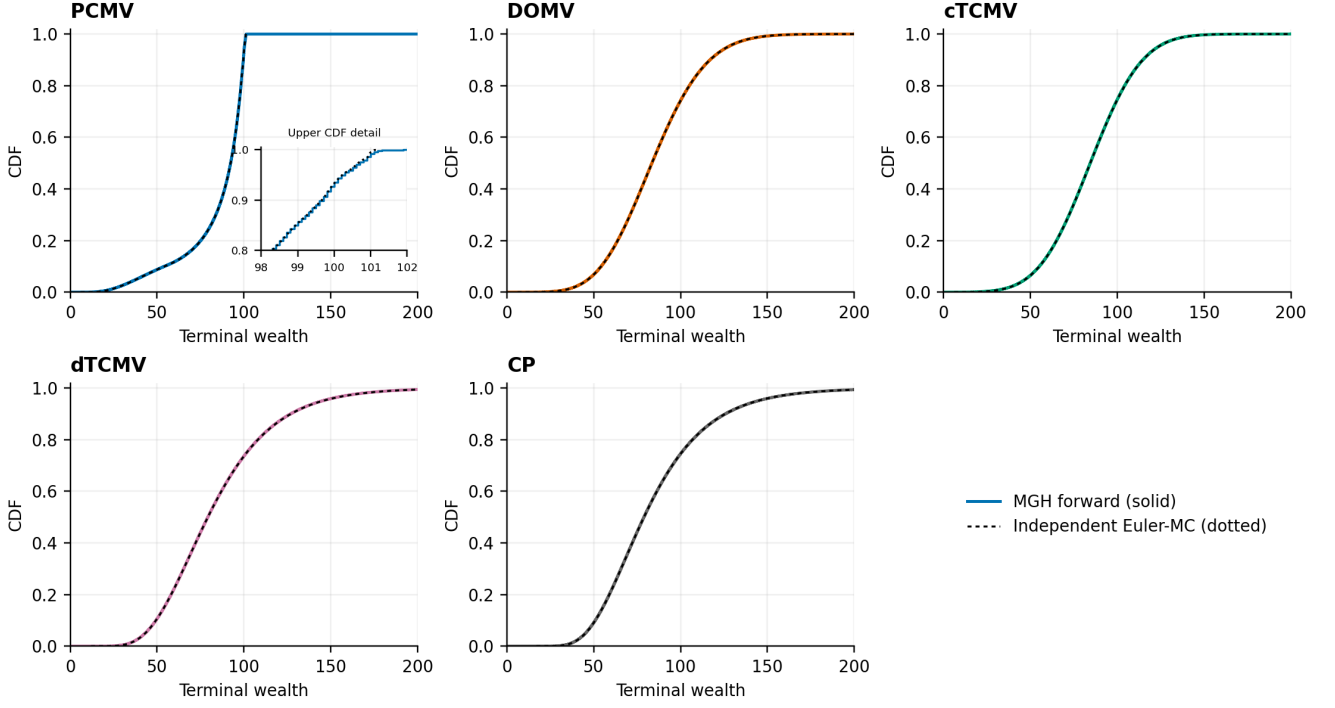


Figure 12: Terminal CDFs under the same saved policies: MGH forward propagation (solid) and independent Euler-MC (dotted). The displayed range is 0–200; summary statistics and maximum CDF differences are calculated over the full support.

Table 10: Terminal statistics by evaluator under the same saved policies

| Policy | Evaluator | Mean    | Standard deviation | q05     | Median  | q95      | Lower 5% mean |
|--------|-----------|---------|--------------------|---------|---------|----------|---------------|
| PCMV   | MGH       | 84.7673 | 18.9593            | 38.1000 | 92.3000 | 100.3000 | 28.7258       |
| PCMV   | MC        | 84.7715 | 18.9424            | 38.1937 | 92.3501 | 100.2246 | 28.7985       |
| DOMV   | MGH       | 84.7861 | 24.4663            | 46.8000 | 83.5000 | 127.2000 | 39.0772       |
| DOMV   | MC        | 84.7831 | 24.4139            | 46.8690 | 83.4760 | 127.0602 | 39.2036       |
| cTCMV  | MGH       | 84.7542 | 23.1470            | 47.1000 | 84.4000 | 123.4000 | 38.3925       |
| cTCMV  | MC        | 84.7913 | 23.1570            | 47.1606 | 84.4674 | 123.4706 | 38.5271       |
| dTCMV  | MGH       | 84.7345 | 32.2180            | 43.1000 | 79.3000 | 144.9000 | 37.2531       |
| dTCMV  | MC        | 84.7863 | 32.3127            | 43.1728 | 79.2581 | 145.2516 | 37.3322       |
| CP     | MGH       | 84.7800 | 31.7442            | 45.2000 | 78.9000 | 144.3000 | 39.8139       |
| CP     | MC        | 84.7854 | 31.6874            | 45.2269 | 78.9299 | 144.1862 | 39.8995       |

Note: MGH quantiles are defined as the smallest grid value at which cumulative probability reaches the specified level. For the lower-5% mean, only the required fraction of probability mass at the boundary grid point is allocated so that total tail probability is exactly 5%. MC uses empirical quantiles and the arithmetic mean of the lowest 5% of observations. The MGH standard deviation is calculated from probability mass, whereas the MC standard deviation uses the sample variance (999999 degrees of freedom for one million observations).

The MGH-minus-MC differences range from  $-0.0947$  to  $0.0568$  for standard deviation, from  $-0.0937$  to  $-0.0269$  for q05, from  $-0.3516$  to  $0.1398$  for q95, and from  $-0.1346$  to  $-0.0727$  for the lower-5% mean. dTCMV has the largest absolute difference in standard deviation and q95, while its maximum CDF difference is the smallest, illustrating that evaluator discrepancies depend on the metric considered. q01, the upper-5% mean, and skewness are also recorded in the CSV file.

### E.3.3 Comparison of mean glide paths

Mean glide paths are calculated according to equations (105) and (106). MGH uses forward probability mass at each month, whereas MC uses the realised risky-asset proportions in the saved one-million-path sample at each month. The comparison covers the 480 decision dates and excludes the terminal date, at which the control is undefined. For  $\Delta G_n = G_{\text{MGH}}(t_n) - G_{\text{MC}}(t_n)$ , we compute the maximum absolute difference, the time-mean absolute error (MAE), and the root mean square error (RMSE). The units in Table 11 are percentage points (pp); a risky-asset-proportion difference of 0.01 corresponds to 1 pp.

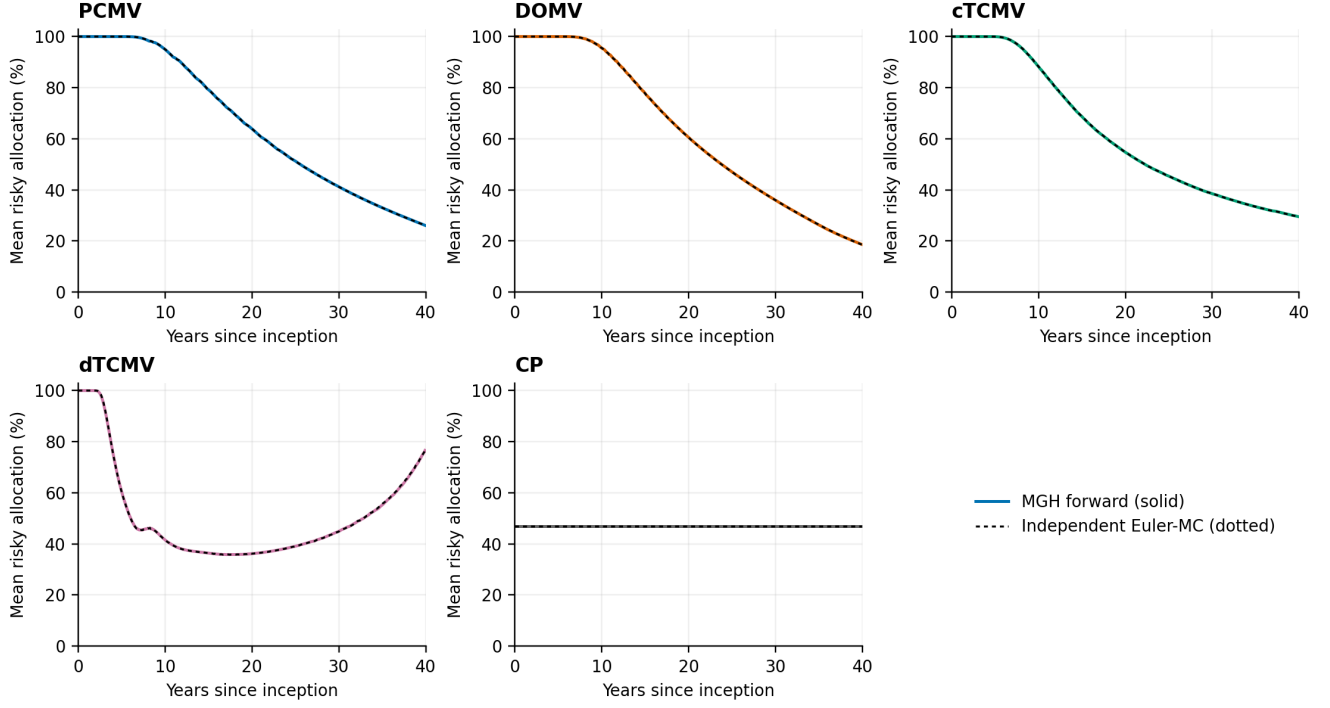


Figure 13: Mean glide paths under the same saved policies: MGH forward propagation (solid) and independent Euler-MC (dotted). The vertical axis is the mean risky-asset proportion (%).

Table 11: Distributional and mean-glide-path differences (MGH versus MC)

| Policy | $D_{KS}$ | Maximum difference (pp) | MAE (pp) | RMSE (pp) | Time of maximum difference (years) |
|--------|----------|-------------------------|----------|-----------|------------------------------------|
| PCMV   | 0.008514 | 0.0684                  | 0.0184   | 0.0239    | 9.92                               |
| DOMV   | 0.002079 | 0.0948                  | 0.0353   | 0.0455    | 39.50                              |
| cTCMV  | 0.001819 | 0.1688                  | 0.0278   | 0.0470    | 13.50                              |
| dTCMV  | 0.001257 | 0.5889                  | 0.0704   | 0.1224    | 34.75                              |
| CP     | 0.001434 | 0.0000                  | 0.0000   | 0.0000    | —                                  |

The mean-glide-path discrepancy is largest for dTCMV, with a maximum of 0.5889 pp at 34.75 years, an MAE of 0.0704 pp, and an RMSE of 0.1224 pp. For CP, the constant proportion  $\theta_{CP} = 0.4688842166$  and the mean proportions under both evaluators agree to rounding error, with a maximum evaluator difference below  $8.6 \times 10^{-12}$  pp. The saved state-dependent feedback policies are common to both evaluators. Differences in the mean glide path arise because the same policy is weighted by the state distributions generated by the two evaluators.

### E.3.4 Summary of validation results

In this evaluator comparison, the principal shapes of the terminal distributions and mean glide paths are broadly consistent across the two evaluators, supporting the independent-MC results in the main text and the design conclusions based on the underlying decision principles. The mean-glide-path difference is largest for dTCMV, whereas the maximum CDF difference is largest for PCMV, so evaluator discrepancies manifest differently across metrics. CP provides a useful reference for isolating evaluator effects such as state discretisation and mass allocation: even when its mean risky-asset proportion coincides across evaluators, its terminal distributions are not identical.

The total probability-mass error in each month is below  $2.2 \times 10^{-14}$ , and both negative mass and lower-boundary mass are zero. The largest upper-boundary mass is  $6.43 \times 10^{-8}$  for CP and  $3.16 \times 10^{-9}$  for dTCMV. Forward first and second moments are consistent with the backward evaluation of the same saved policies, including a check that the one-step second moments incorporate the additional variance induced by mass allocation. Upper-boundary probability is small, and other approximation sources such as state discretisation and mass allocation also contribute to the observed distributional differences.

For the current CP policy, the standard deviation is 31.7442 under MGH and 31.6874 under MC, a relative difference of approximately 0.18%. This difference is treated as a diagnostic of consistency when the same saved policy is supplied to two evaluators. The present validation concerns the finite model and an independent evaluator; convergence to the continuous-time problem and uniform error bounds remain subjects for future validation.

## E.4 Robustness to grids, boundaries, and discretisation

As a sensitivity check on the state ceiling, recalibrating dTCMV with an upper bound of 900 gives a standard deviation of approximately 32.29, close to 32.31 in the main text. The baseline calculation uses a state ceiling of 600, state spacing 0.1, 129 candidate risky-asset proportions, and seven-point GH quadrature; calibration evaluation uses state spacing 0.025 and 31-point GH quadrature. In addition, boundary-arrival mass and differences in terminal distributions and mean glide paths between MGH forward propagation and independent Monte Carlo are examined to diagnose the effects of the state grid, control grid, quadrature order, and boundary treatment separately. Current grid/recalibration sensitivity is stored at `results/validation/recalibration/sensitivity.csv`, the upper-900 summary at `results/validation/recalibration/cap900_rec calibration.json`, and quadrature convergence at `results/validation/recalibration/quadrature_convergence.json`.

In the external benchmark, we vary the time step, state-grid width, GH order (3, 7, 15), state domain, control domain, and optimisation tolerance, and also examine a refinement path that simultaneously refines the time partition and state grid. The

convergence series is stored at `results/validation/van_staden_2021/convergence.csv`. These checks support robustness within the finite model but do not provide a uniform error bound over the entire state domain or a general convergence theorem to the continuous-time model.

## E.5 Scope of the numerical results and theoretical claims

Numerical internal consistency shows that backward and forward calculations agree under the adopted discrete transition kernel. Optimality or equilibrium in the finite model and existence and uniqueness of classical solutions in the continuous-time model are treated as separate validation questions.

The projection representation requires suitable convexity of the value function for PCMV and concavity of the local objective for cTCMV and dTCMV. If the curvature is zero or has the opposite sign, the problem is solved by direct comparison including the endpoints. Moreover, because  $\Gamma = \rho_d/(x + H_t)$  is undefined when  $x + H_t = 0$ , that point must either be excluded from the domain of dTCMV or assigned a separate boundary convention. In the baseline case there are positive future contributions immediately before the terminal date, and no investment decision is made at the terminal date itself.

Verification by classical solutions applies when candidate solutions exist and possess the required regularity. Existence of PCMV and DOMV based on viscosity solutions and relaxed controls is distinguished from the global search over the outer target in the numerical implementation. A deterministic DB benefit can be treated as a parallel shift of terminal wealth for PCMV, DOMV, and constant-coefficient cTCMV, but it can affect dTCMV through the preference coefficient.

In the external validation, the problem setting of van Staden et al. (2021) is solved independently by MGH backward induction, and the resulting policies are evaluated forward and compared with known policies and terminal distributions. Values reverse-engineered from published terminal means and standard deviations are not used as validation inputs. For dTCMV, however, the printed formula and the equilibrium condition derived from the stated objective give different signs for the squared term; the exercise is therefore treated as partial rather than complete external validation. For MVS, the regression test that a zero coefficient recovers MV is separated from checks of accuracy and optimality at positive coefficients.

Existence of PCMV and DOMV based on viscosity solutions and relaxed controls is established under the assumptions and additional conditions in Appendix G. For cTCMV and dTCMV, this paper does not generally prove global existence, uniqueness, or regularity of an equilibrium solution under the state-dependent two-sided constraint  $0 \leq \pi \leq X$ .

## E.6 External benchmark validation of the MGH backward–forward engine

This subsection independently solves the published problem of van Staden et al. (2021) by MGH backward calculation and compares the resulting policies and forward-evaluated terminal distributions with published values. Its specification is separate from the constrained DC baseline. We present inputs, replication methods, comparison results, and refinement sensitivity in that order.

### E.6.1 Benchmark specification and primary sources

We verify equations (2.1) and (5.1), Assumption 3.1, and Table 5.1 of the original paper (pp. 570, 574, 597, and 599), and set  $w_0 = 100$ ,  $T = 10$ ,  $r = 0.00623$ ,  $\mu = 0.0816$ , and  $\sigma = 0.1863$ . There are no contributions or terminal benefits. The risky-asset amount  $u$  is an unconstrained control that permits short selling and borrowing, and trading continues after insolvency. Target means 125 and 250 are explicit input conditions in the original paper. The DC constraint  $0 \leq \pi \leq X$ , the 40-year horizon, contributions, and the common expected terminal wealth of 84.78 used elsewhere in this paper are not imposed on the benchmark.

The PCMV embedded target  $\gamma/2$  and the variance-aversion coefficients for DOMV and cTCMV are obtained from equations (4.2)–(4.4) of the original paper and fixed at their unrounded values across all refinement conditions. DOMV resolves the remaining-horizon problem at each date and state under the fixed variance-aversion coefficient and connects the first control. dTCMV uses  $\rho/(2w)$  from equation (2.10), with  $\rho$  determined from the target-mean condition. Published standard deviations, medians, and other output statistics are not used as inputs; the published table is used only for ex-post comparison. The specification, equation numbers, and input/output distinction are recorded at `recalibration/external_vanstaden_2021/BENCHMARK_SPECIFICATION.md`.

### E.6.2 Replication by MGH backward and forward calculations

The external benchmark uses the same conditional-moment calculations with linear interpolation and the same normalised GH nodes as the DC calculation. For the external problem we use  $Y_t = e^{r(T-t)}W_t$ : a signed geometric grid for centred wealth in PCMV, an equally spaced grid for DOMV and cTCMV, and a positive geometric grid for dTCMV. Translation and scaling symmetries of the grids reduce storage of backward moments, but every candidate control is still evaluated by the common MGH function. The reduction and direct interpolation of moment arrays on the grid are cross-checked at 54 points.

PCMV numerically minimises squared loss; DOMV numerically maximises the scalarised objective over the family of remaining-horizon embedded-target problems. cTCMV and dTCMV numerically search for the current equilibrium control given future backward moments. Analytical policies are not inserted into the backward calculation. A 257-point search over the full control range is combined with interval optimisation near the best candidate, and the search tolerance is varied separately. Forward propagation uses non-negative mass allocation corresponding to the transpose of linear interpolation. First and second moments are cross-checked, including the case in which initial wealth is off-grid.

For dTCMV, the positive proportional-policy branch uses a GBM transition with the proportion fixed over each interval. Thus, the external calculation is not simply the DC-specific constraint and transition code applied unchanged; it is an implementation for the unconstrained problem that shares the common moment kernel. This exercise alone therefore does not validate every DC-specific code path.

**Qualification regarding the dTCMV formula in the original paper.** Equation (3.5) of the original paper gives the integrand in the exponent as  $r + (\mu - r)\theta - \sigma^2\theta^2$ . By contrast, deriving the equilibrium condition from the objective in equation (2.10), with  $E[W_T | w] = aw$  and  $E[W_T^2 | w] = cw^2$ , gives

$$\theta = \frac{\mu - r}{\rho\sigma^2} \left\{ \frac{a}{c} + \rho \left( \frac{a^2}{c} - 1 \right) \right\}, \quad \frac{a}{c} = \exp \left[ - \int_t^T \{ r + (\mu - r)\theta(s) + \sigma^2\theta(s)^2 \} ds \right]$$

and the sign of the squared term differs. Both variants are therefore calculated and recorded separately. Solving the latter integral as a reverse-time ODE with zero initial value and determining  $\rho$  from the target-mean condition gives 8.228651 for target 125 and 0.112272 for target 250. This ODE policy is not supplied to MGH; the MGH backward calculation independently determines the control from the objective function. References below to the dTCMV theoretical solution use this equilibrium derivation. A literal implementation of the printed formula gives a standard deviation of 204.478 at target 250, whereas the equilibrium derivation gives 446.751, which still leaves a residual relative to the published value 444. Values reverse-engineered

as lognormal distributions from the published mean and standard deviation are not used as validation inputs. This difference in the documented formulations remains unresolved, so dTCMV is not described as a complete replication of the original paper.

### E.6.3 Comparison of terminal distributions and policies for the four strategies

Table 12 reports the pre-specified finest condition:  $N_t = 1280$ , seven-point GH quadrature, log spacing 0.00125 for geometric grids, and width 0.0625 for equally spaced grids. Quantiles of each discrete distribution are left inverses of cumulative probability, and the lower-5% mean uses only the required fraction of the boundary atom so that total tail probability equals exactly 5%. Figure 14 compares the unsmoothed discrete MGH CDF with the continuous-time reference CDF. The maximum difference,  $D_{\text{CDF}} = \sup_z |F_{\text{MGH}}(z) - F_{\text{ref}}(z)|$ , is evaluated at the left and right limits of all grid points. The 95% quantile and CDF are theoretical comparisons and are not presented as published values from Table 5.1 of the original paper.

Table 12: Published values and MGH backward–forward results for the external benchmark (finest condition)

| Target | Policy | Mean        | Standard deviation | Median      | $q_{05}$    | Lower 5% mean |
|--------|--------|-------------|--------------------|-------------|-------------|---------------|
| 125    | PCMV   | 125/124.999 | 9/9.136            | 127/127.512 | 113/113.227 | 97/97.374     |
|        | DOMV   | 125/124.975 | 16/15.997          | 125/125.000 | 99/98.688   | 92/91.978     |
|        | cTCMV  | 125/124.837 | 15/14.421          | 125/124.813 | 101/101.125 | 95/95.092     |
|        | dTCMV  | 125/125.073 | 16/16.051          | 124/124.120 | 101/100.484 | 96/95.410     |
| 250    | PCMV   | 250/249.991 | 71/70.628          | 269/269.419 | 159/159.013 | 37/36.434     |
|        | DOMV   | 250/249.810 | 124/123.471        | 250/249.813 | 47/46.750   | −5/−4.875     |
|        | cTCMV  | 250/250.015 | 112/112.239        | 250/250.000 | 65/65.375   | 19/18.499     |
|        | dTCMV  | 250/250.877 | 444/447.275        | 123/122.732 | 17/17.159   | 11/11.294     |

Across all published output metrics, the maximum relative errors over both target means are 8.145% for PCMV, 2.508% for DOMV, 3.863% for cTCMV, and 10.844% for dTCMV. The metrics at which these maxima occur are, respectively, the probability of falling below the risk-free terminal value, the lower-5% mean, standard deviation, and excess kurtosis. The comparison includes the effect of integer and decimal rounding in the published values; relative error is not defined for published zeros, for which absolute error is reported. For example, at dTCMV target 125, excess kurtosis is 0.26747 versus the published value 0.3. Differences due to rounding of input coefficients are separated from these output errors. All published metrics and absolute errors are stored at `results/validation/van_staden_2021/published_vs_mgh.csv`.

Policies themselves are compared at nine points: the initial, intermediate, and final decision dates crossed with wealth values 50, 100, and 250. Across both target means, the maximum relative errors are 0.080% for PCMV, 0.231% for DOMV, 0.879% for cTCMV, and 9.522% for dTCMV. The largest dTCMV error remains at target 125. Terminal-CDF differences and policy errors at the nine evaluated points are therefore interpreted as separate metrics. They do not guarantee policy accuracy across all states.

### E.6.4 Grid convergence and boundary/control-domain diagnostics

The baseline condition is  $N_t = 320$ , seven-point GH quadrature, and log spacing 0.005 or uniform width 0.25. For target 250, 48 one-factor conditions vary the time step, state-grid width, GH order (3, 7, 15), state domain, control domain, and optimisation tolerance; these are stored together with eight baseline conditions across the two targets. To account for accumulation of interpolation error, we additionally use a separate coherent refinement path of 16 conditions in which the number of time intervals is doubled and quadrupled while state-grid widths are simultaneously halved and quartered. This is not a substitute for the one-factor checks, but a diagnostic of combined discretisation error. All 72 conditions are retained; no grid is selected ex post for being closest to the published values.

Table 13: Maximum CDF differences along the joint-refinement path and policy errors under the finest condition

| Target | Policy | $N_t = 320$ | 640      | 1280     | Maximum policy relative error |
|--------|--------|-------------|----------|----------|-------------------------------|
| 125    | PCMV   | 0.003702    | 0.003675 | 0.001092 | 0.080%                        |
|        | DOMV   | 0.006358    | 0.003559 | 0.001395 | 0.231%                        |
|        | cTCMV  | 0.007539    | 0.009329 | 0.005936 | 0.879%                        |
|        | dTCMV  | 0.023085    | 0.013041 | 0.004608 | 9.522%                        |
| 250    | PCMV   | 0.003702    | 0.003675 | 0.001092 | 0.080%                        |
|        | DOMV   | 0.004542    | 0.002726 | 0.000876 | 0.231%                        |
|        | cTCMV  | 0.003233    | 0.001955 | 0.000203 | 0.010%                        |
|        | dTCMV  | 0.007803    | 0.003837 | 0.002027 | 0.947%                        |

Table 13 shows that the CDF differences generally decline, but cTCMV at target 125 is non-monotone, and the dTCMV policy error at the same target increases from 7.95% under the intermediate condition to 9.52% under the finest condition. For PCMV at target 250, refining time alone increases the CDF difference to 0.01854. These results indicate sensitivity to the interaction between linear-interpolation grid phase and time discretisation, and do not support a claim of globally monotone convergence. Values for all conditions, including the mean, standard deviation,  $q_{05}$ , and  $q_{95}$ , are retained in the convergence CSV.

The state domain is an absolute centred-wealth range of  $10^7$  for PCMV, wealth  $[10^{-8}, 10^7]$  for dTCMV, and discounted wealth  $[-5000, 5000]$  for DOMV and cTCMV; outer boundaries are also halved and doubled. Across all conditions there is no negative mass, the maximum total-mass error is  $5.49 \times 10^{-14}$ , and the maximum cumulative probability of boundary escape is  $2.65 \times 10^{-19}$ . The largest standard-deviation change from altering the state domain is  $2.54 \times 10^{-9}$ , and the largest mean or standard-deviation change from halving or doubling the control domain is  $1.98 \times 10^{-6}$ ; no solution binds at the control-domain endpoints. For a fixed policy, backward–forward differences are below  $1.77 \times 10^{-11}$  for the mean and  $2.58 \times 10^{-9}$  for the standard deviation. Over this range, time discretisation, interpolation, control search, and formulation differences relative to the original paper are more plausible residual sources than boundary truncation. Small boundary probability alone does not guarantee small error in arbitrary higher moments.

### E.6.5 Scope of conclusions from the external validation

For PCMV and DOMV, errors in backward policies and terminal CDFs obtained without inserting analytical policies decrease along the present refinement path. cTCMV also agrees well at target 250, but target 125 retains non-monotonicity and a residual discrepancy. dTCMV is independently solved as an equilibrium problem without using published outputs as inputs,

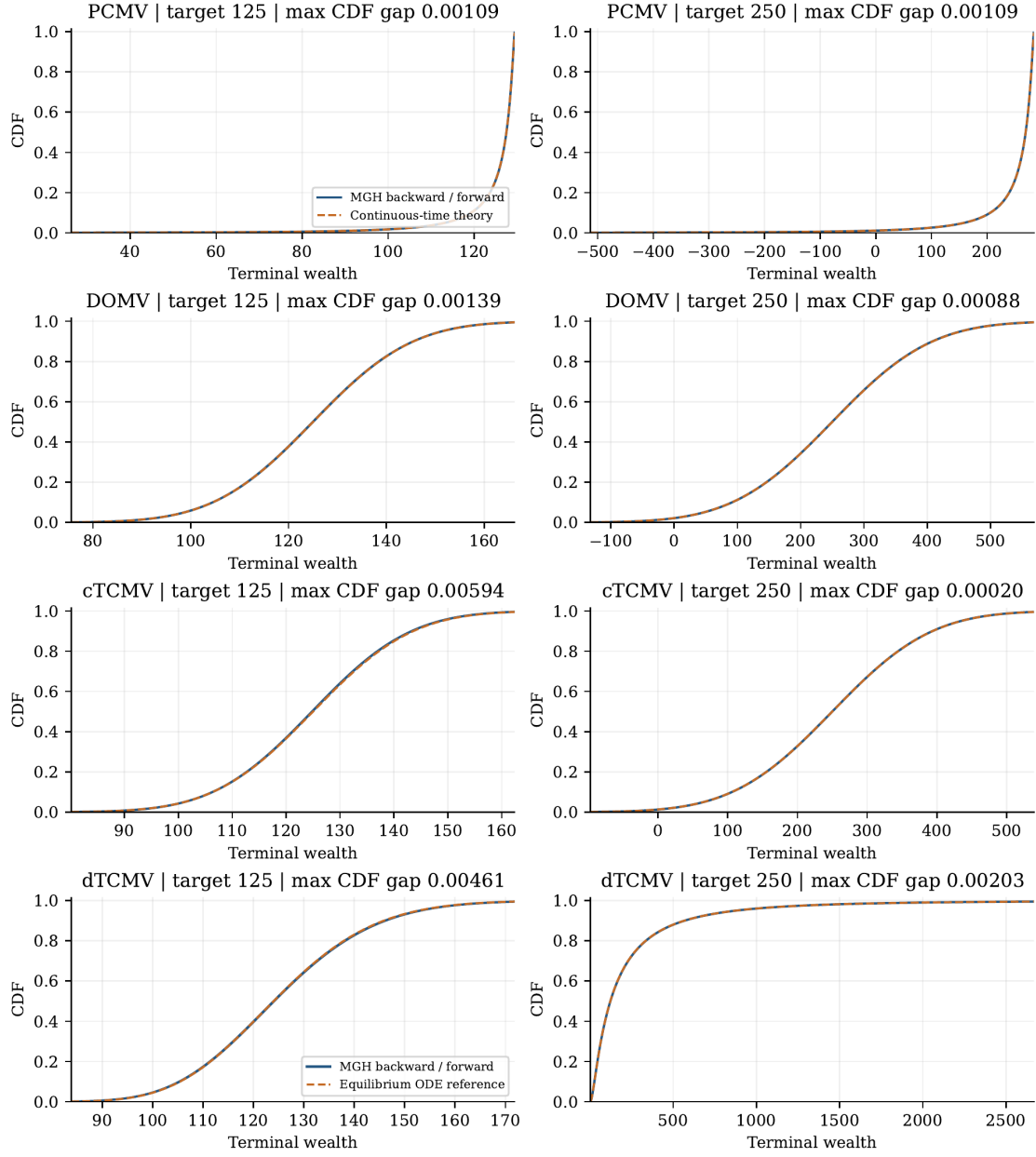


Figure 14: Terminal CDF comparison for the external problem. Solid lines are forward distributions of policies obtained by MGH backward induction; dashed lines are continuous-time reference distributions. The dTCMV reference is the equilibrium ODE derived from the objective function and is not a complete replication of the formula as printed or of the published table. Maximum CDF differences are calculated from the full discrete mass, including values outside the displayed range.

but because a difference from the printed formula and policy error remain, it is treated as a partial external validation of the original paper.

These results numerically assess the MGH backward–forward implementation for the evaluated external problem. They do not prove continuous-time optimality, general convergence, accuracy at every state under the DC constraint, or validity of exploratory MVS results.

## F Code, Numerical Data, and Validation Materials Supporting Reproducibility

### F.1 Public repository and current computational assets

The numerical implementation, principal CSV and NPZ data, validation outputs, and reproduction instructions used in this paper are available in the following public repository.

<https://github.com/yoshi123027-del/dc-glidepath-ica2026-en>

### F.2 Principal results in the current manuscript

The saved policies and independent evaluation values underlying Table 5 are in `results/current/`. Calibration files are `results/current/PCMV.json`, `results/current/DOMV.json`, `results/current/cTCMV.json`, `results/current/dTCMV.json`, and `results/current/CP.json`; saved policies are the corresponding `results/current/PCMV.npz`, `results/current/DOMV.npz`, `results/current/cTCMV.npz`, `results/current/dTCMV.npz`, and `results/current/CP.npz`. Independent-MC terminal statistics and samples/mean glide paths are in `results/current/independent_mc.csv` and `results/current/independent_mc.npz`; common settings are in `results/current/config.json`. Solvers/evaluators are under `recalibration/`; validation decisions are in `results/validation/recalibration/acceptance.json` and `results/validation/recalibration/checks.json`; evaluator-comparison outputs are under `results/validation/evaluator_comparison/`.

Section 7 reads `results/current/PCMV.npz`, `results/current/DOMV.npz`, `results/current/cTCMV.npz`, `results/current/dTCMV.npz`, and `results/current/CP.npz`, and linearly interpolates risky-asset investment in the state dimension. Coefficients remain at their initial calibration values, and one million paths per state are generated using the monthly Euler equation, independent standard-normal draws, no upper truncation, and common random numbers. The seeds for the states 20 years/balance 20, 30 years/balance 45, and 35 years/balance 30 are 20260915, 20360915, and 20460915, respectively. q05, median, and q95 are empirical quantiles; the lower-5% mean is the arithmetic mean of the lowest 5% of each sample. Table 14 summarises all five policies using the same evaluator as the detailed cTCMV and dTCMV comparison in the main text.

Table 14: Conditional outcomes from common states for all five policies (one million paths per state)

| State                | Policy | Initial proportion | Mean  | Standard deviation | q05   | q95    | Lower 5% mean |
|----------------------|--------|--------------------|-------|--------------------|-------|--------|---------------|
| 20 years, balance 20 | PCMV   | 1.000              | 73.45 | 22.15              | 28.69 | 97.51  | 22.84         |
|                      | DOMV   | 0.797              | 66.03 | 16.36              | 39.68 | 93.55  | 33.72         |
|                      | cTCMV  | 0.750              | 67.98 | 18.04              | 38.40 | 97.85  | 31.37         |
|                      | dTCMV  | 0.391              | 69.78 | 25.19              | 36.86 | 116.62 | 32.08         |
|                      | CP     | 0.469              | 67.76 | 21.58              | 39.37 | 107.81 | 35.17         |
| 30 years, balance 45 | PCMV   | 0.719              | 75.97 | 17.11              | 38.95 | 95.04  | 31.53         |
|                      | DOMV   | 0.406              | 70.24 | 10.32              | 53.32 | 87.28  | 49.06         |
|                      | cTCMV  | 0.438              | 72.19 | 13.00              | 50.83 | 93.59  | 45.40         |
|                      | dTCMV  | 0.453              | 77.95 | 24.63              | 44.70 | 123.43 | 39.53         |
|                      | CP     | 0.469              | 74.91 | 18.78              | 48.62 | 109.15 | 44.15         |
| 35 years, balance 30 | PCMV   | 1.000              | 44.81 | 15.76              | 22.80 | 74.10  | 19.73         |
|                      | DOMV   | 0.523              | 40.72 | 6.44               | 30.14 | 51.32  | 27.48         |
|                      | cTCMV  | 0.719              | 42.07 | 9.15               | 26.98 | 57.13  | 23.31         |
|                      | dTCMV  | 0.617              | 42.52 | 10.91              | 27.13 | 62.33  | 24.49         |
|                      | CP     | 0.469              | 40.95 | 7.26               | 30.21 | 53.86  | 28.15         |

### F.3 Exploratory analyses and supplementary figures

Sensitivity of policy shape to market and contribution assumptions and the MVS extensions are treated separately from the principal results in the main text. Corresponding sensitivity and MVS images used in this appendix are retained under `archive/papers/v14_7/figs/`, with related exploratory calculation assets under `archive/legacy_v5/`. These materials are historical and exploratory resources under `archive/`, distinct from the current principal results in `results/current/`. The sensitivity analysis corresponds to Appendix C, and the MVS extensions to Appendix D.

## G Existence and Verification Theorems for PCMV and DOMV Based on Viscosity Solutions and Relaxed Controls

Here we establish the existence results for PCMV and DOMV within a viscosity-solution and relaxed-control framework without assuming classical solutions or smooth free boundaries. For DOMV, joint measurable selection of the target and optimal control, together with appropriate solvability of the closed-loop equation after diagonal pasting, is imposed as an additional condition. We then present the classical-solution verification arguments used in Section A and the free-boundary and projection structure common to the four solution concepts. Local feedback representations and verification theorems assume  $C^{1,2}$  candidate functions and sufficient integrability.

### G.1 Existence for PCMV and DOMV Based on Viscosity Solutions and Relaxed Controls

In what follows, let  $c : [0, T] \rightarrow [0, \infty)$  be bounded and continuous, and let  $D_T \geq 0$  be constant. For  $x > 0$ , set  $u_s = \pi_s/X_s$ , and for  $x = 0$ , set  $u_s = 0$ ; the normalised control  $u$  is a progressively measurable process taking values in  $[0, 1]$ . The state equation then becomes

$$dX_s = \{c_s + (r + \beta u_s)X_s\} ds + \sigma u_s X_s dW_s, \quad X_t = x \geq 0 \quad (122)$$

For a fixed target  $m$ , define the value function by

$$L^m(t, x) = \inf_u \mathbb{E}_{t,x}[(X_T^u + D_T - m)^2] \quad (123)$$

**Proposition 3** (Existence for PCMV and DOMV Based on Viscosity Solutions and Relaxed Controls). *Under the conditions above, the following statements hold.*

- (i) *For every normalised control  $u$ , equation (122) admits a unique non-negative strong solution. Moreover, for every  $p \geq 2$  and every compact set of initial states,  $\mathbb{E}[\sup_{t \leq s \leq T} |X_s^u|^p]$  is uniformly bounded over controls  $u$ .*
- (ii)  *$L^m$  is finite, continuous in  $(t, x, m)$ , and has at most quadratic growth. It satisfies the dynamic programming principle and is a continuous viscosity solution of HJB (25). If a comparison principle holds in the class of continuous viscosity solutions with polynomial growth, then  $L^m$  is unique in that class. In addition, an optimal Markov control exists in the compactified class of relaxed controls. If a Borel-measurable strict selector attaining the minimum in the HJB exists and the resulting closed-loop SDE is appropriately solvable, an optimal strict Markov control may be selected.*
- (iii) *For  $\gamma > 0$ , define*

$$\Psi_\gamma(t, x; m) = m - \frac{1}{2\gamma} - \frac{\gamma}{2} L^m(t, x) \quad (124)$$

*Then  $m \mapsto \Psi_\gamma(t, x; m)$  is continuous and converges to  $-\infty$  as  $|m| \rightarrow \infty$ . Hence the outer maximisation in the scalarised PCMV problem admits at least one target  $m_P^*(t, x)$ .*

- (iv) *The remaining-horizon DOMV problem at each  $(t, x)$  also has a non-empty compact set of optimal targets,*

$$\mathcal{M}_D^*(t, x) = \arg \max_{m \in \mathbb{R}} \Psi_{\gamma_D}(t, x; m)$$

*From this correspondence, select a Borel-measurable target. If, in addition, the target and control can be selected jointly measurably and the closed-loop equation after diagonal pasting is appropriately solvable, the DOMV feedback can be constructed.*

The following four properties require additional assumptions:

- (a) *existence, for every prescribed target mean  $d$ , of an embedded target satisfying the mean-consistency condition;*
- (b)  *$C^{1,2}$  regularity of the value function;*
- (c) *smoothness of the free boundary; and*
- (d) *uniqueness and continuity of the optimal strict feedback.*

*Proof.* Normalisation makes the control set the fixed compact interval  $[0, 1]$ . The drift and diffusion coefficients in equation (122) are uniformly Lipschitz in  $x$  over the controls and have linear growth. Hence a unique strong solution exists for every progressively measurable control. Moreover, using the stochastic exponential

$$Z_s = \exp\left(\int_t^s \left(r + \beta u_q - \frac{1}{2} \sigma^2 u_q^2\right) dq + \int_t^s \sigma u_q dW_q\right)$$

we obtain

$$X_s = Z_s \left( x + \int_t^s Z_q^{-1} c_q dq \right) \geq 0 \quad (125)$$

Applying the Burkholder–Davis–Gundy and Gronwall inequalities yields the uniform moment estimate in (i).

The terminal loss is continuous with quadratic growth, and the control set is compact. Standard finite-horizon stochastic-control dynamic programming therefore gives finiteness and continuity of the value function, the dynamic programming principle, and the viscosity-solution property (Fleming and Soner, 2006). Compactifying a sequence of strict controls into relaxed controls yields an optimal relaxed Markov control. If a Borel selector of HJB-Hamiltonian minimisers exists and the selected closed-loop equation is appropriately solvable, the standard verification and selection argument recovers a strict Markov control. This proves (ii).

The moment estimate implies that the first moment of attainable terminal wealth  $W_T^u = X_T^u + D_T$  is uniformly bounded over controls. For some constant  $C_{t,x} < \infty$ ,  $\sup_u |\mathbb{E}_{t,x}[W_T^u]| \leq C_{t,x}$ ; Jensen's inequality then gives

$$L^m(t, x) \geq \inf_u \{\mathbb{E}_{t,x}[W_T^u] - m\}^2 \geq (|m| - C_{t,x})_+^2$$

Thus equation (124) tends to  $-\infty$  as  $|m| \rightarrow \infty$ . Together with continuity of  $L^m$ , this establishes existence of a maximiser and proves (iii). The same estimate applies to every remaining-horizon problem. Joint continuity of  $\Psi_{\gamma_D}$  and local uniform coercivity imply that the maximiser correspondence has non-empty compact values and is measurable, so the measurable maximum theorem yields a Borel selector  $m_D^*$ . Pasting the current optimal control obtained from each target problem state by state gives (iv).  $\square$

This proposition separates existence of PCMV and DOMV from existence of classical solutions and smooth free boundaries. Li et al. (2002) and Li and Xu (2016) provide the theoretical background on viscosity solutions and optimal strategies for constrained mean–variance problems, while Guan et al. (2022) gives a related classical-solution and regularity result for a free-boundary problem under borrowing constraints. Because the latter concerns a CRRA utility problem, the quadratic-loss HJB in this paper requires its own existence argument.

## G.2 Classical-Solution-Based Verification for PCMV and DOMV

For a fixed target  $m$  and any admissible control  $\pi$ , localise using the minimum  $\tau_k$  of the exit time of  $X^\pi$  from a bounded region and a stopping time controlling quadratic variation. Itô's formula gives

$$\mathbb{E}[v^m(\tau_k, X_{\tau_k}^\pi)] - v^m(t, x) = \mathbb{E} \int_t^{\tau_k} (v_s^m + \mathcal{L}^{\pi_s} v^m)(s, X_s^\pi) ds. \quad (126)$$

By HJB (25), the integrand on the right-hand side is non-negative for every  $\pi$  and is zero for the minimising control  $\pi^{m,*}$ . Polynomial growth and uniform integrability allow  $k \rightarrow \infty$ ; using the terminal condition then yields

$$v^m(t, x) \leq \mathbb{E}_{t,x}[(X_T^\pi + D_T - m)^2], \quad v^m(t, x) = \mathbb{E}_{t,x}[(X_T^{\pi^{m,*}} + D_T - m)^2]$$

. The part of the Hamiltonian depending on  $\pi$  is

$$q^m(\pi) = \beta v_x^m \pi + \frac{1}{2} \sigma^2 v_{xx}^m \pi^2$$

and is convex when  $v_{xx}^m > 0$ , so the constrained minimiser is given by equation (26). Mean-constrained PCMV reduces to this quadratic-loss minimisation through the identity shown in the main text. DOMV applies the same argument to the remaining-horizon problem at every  $(t, x)$  and pastes together the first control obtained at each date and state.

## Acknowledgements and Disclosure

Generative AI tools were used for TeX editing, code implementation of the Markov–Gauss–Hermite engine, drafting and refinement of the paper structure, background research, and checks of logical consistency. The author remains responsible for the theory, numerical results, and final content.

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